Data Technology and Market Efficiency*

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Abstract

How will advances in data technology affect market equilibrium and informational efficiency when adoption by investors is uneven? I show that data technology development and dissemination improve asset managers' stock-picking ability, especially for those with larger shares of data-science labor. Using U.S. institutional holdings and employment records, I estimate information production across investors and a demand system with heterogeneity in private information and learning from prices. I estimate investors' private information from holdings' forecasting power for future payoffs beyond public information and prices. Data technology developments are captured by increases in data scientist productivity. Estimates show that better-informed investors have more elastic demand, while learning from prices makes demand 61 percent less elastic at the median informative stock. A counterfactual advance calibrated to more than a decade of data scientist productivity growth raises price informativeness by 26 percent, more for small stocks. New technology adopters crowd out other investors, increasing market segmentation as stocks' investor bases diverge. Aggregate demand becomes more inelastic for 47 percent of stocks.

Keywords: data technology, institutional investors, price informativeness, demand system, investor bases, demand elasticity, learning from prices
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1 Introduction

Over the past two decades, rapid advances in data technology have changed how institutional investors process information (Goldstein et al., 2021). Asset managers have hired more data scientists and expanded their use of new technologies.[1] These teams use the tools to forecast sales, earnings, and other firm fundamentals that guide investment decisions.[2] The spread of these methods is changing information production across asset management, with market-wide effects on stock prices and the information they convey. Adoption is uneven, however, and investors differ greatly in their capacity to use new tools. As technology improves further, these differences will shape who gains information and how information is transmitted through financial markets. How will a further advance in data technology affect market equilibrium and informational efficiency?

I find moderate and uneven gains in price informativeness, the share of firm payoff uncertainty resolved by prices.[3] An advance equal to the estimated data scientist productivity growth over more than one decade raises average informativeness by 3.1 percentage points from 11.8 percent. Small stocks gain more: the increase averages 4.4 points in the smallest quintile and 2.3 points in the largest. Adopters' improved private signals strengthen their demand and crowd out other institutions, increasing market segmentation. Adopters' positions rise by 15.4 percent, while other institutions' positions fall by 1.85 percent. The average distance between stocks' investor bases rises from 53.12 to 53.60 percent. This distance measures the share of ownership that must change hands for two stocks' investor bases to match. The technology advance also makes aggregate demand less elastic for 46.5 percent of stocks.

To quantify these effects, I construct a framework in which investors differ in their capac-

[1] Data technology consists of the algorithms and computing infrastructure that let investors process a wider range of data at lower cost to produce information and signals for investing. Abis and Veldkamp (2024) document growth in the use of machine learning (ML) and AI by data analysts in investment management. Industry reports describe adoption of new data technologies (CFA Institute, *AI Pioneers in Investment Management*, 2019; AIMA, *Casting the Net*, 2020). Investor reports, prospectuses, Form ADV brochures and other communications describe the use of ML/NLP and alternative data (*Two Sigma, 2026; Citadel, 2026; BlackRock, 2024; AQR, 2025, 2026; etc.*).

[2] Applications to fundamental forecasting are described in the CFA Institute's *Handbook of Artificial Intelligence and Big Data Applications in Investments* (2023) and investor communications (*AllianceBernstein, 2019; T. Rowe Price, 2019; Allspring, 2024; Fidelity Investments, 2025; etc.*). Data technology also helps investors use trading data to extract information from prices (Farboodi and Veldkamp, 2020). I focus on technology used for forecasting firm fundamentals. Section 2.1, Appendix A.3, and Online Appendix OA2 give further examples.

[3] I focus on how well prices forecast future asset payoffs (Bond et al., 2012). Following Dávila and Parlatore (2026), I measure informativeness as the proportional reduction in forecast-error variance for firm payoffs when prices are added to other public information. Related work also regresses future cash flows on current valuations and public-information controls (Bai et al., 2016; Farboodi et al., 2022).

ity to process data into private forecasts of firm fundamentals and learn from prices when choosing portfolios. I measure investors' private information from the additional forecasting power of an investor's holdings for future firm payoffs, beyond public information and information contained in prices. Based on the model, I develop a demand system in which private information and price informativeness shape investor demand elasticity. I estimate information production and the demand function using detailed institutional holdings in the U.S. stock market and employment data, then calculate counterfactuals to quantify the effects of a further data technology advance on market equilibrium and informational efficiency.

Two facts motivate the analysis. First, data scientist employment has grown rapidly and unevenly across asset managers. Matching institutional holdings to employment histories, I find that data-science labor grew 8.5-fold, compared with 2.3-fold growth in investment analysts and portfolio managers. At the end of 2024, managers employing data scientists held 80.4 percent of all asset managers' equity assets but accounted for only 11.7 percent of those managers. Data scientist employment is concentrated among larger managers. Only 4.4 percent of managers in the bottom half by AUM employed data scientists, compared with 86.5 percent in the top one percent.

Second, asset managers with larger shares of data-science labor show greater gains in stock-picking ability around technology breakthroughs. Stock-picking ability is higher when a manager overweights firms relative to market weights before favorable earnings surprises. I focus on natural language processing (NLP) and computer vision (CV), which expand investors' ability to extract fundamental information from text and images. Extensive corporate disclosures give NLP models more material from which to extract information about firm fundamentals, while images of store-based retailers reveal customer traffic and can predict sales and earnings.[4] For each breakthrough, I compare changes in stock-picking ability on these stocks across managers with different pre-breakthrough shares of data-science labor in total employment. A one-standard-deviation higher pre-breakthrough share of data-science labor is associated with an additional gain in stock-picking ability of 0.91×10^{-3} after NLP breakthroughs and 0.55×10^{-3} after CV breakthroughs.

I develop a model in which investors with heterogeneous data scientist teams produce private information about firms, choose portfolios, and learn from equilibrium prices. Data scientists turn data into signals about firms' idiosyncratic payoffs. Investors combine these signals with the information conveyed by prices when evaluating expected payoffs and risk. Their trades reveal some of their information through market prices. A technology advance

[4]Textual analysis of corporate filings extracts information about future earnings, profitability, and cash flows (Li, 2010; Cohen et al., 2020; Azimi and Agrawal, 2021). Satellite images of store parking lots provide timely signals of retailers' sales and earnings (Zhu, 2019; Kang et al., 2021; Katona et al., 2025).

raises information productivity at investors with data scientist teams, giving them more precise private signals.

Who gains information, by how much, and which stocks they hold determine the effects of a technology advance. Data technology adopters' better forecasts raise their demand, and their demand makes prices more informative. This erodes other investors' informational advantage and reduces the expected payoff from trading on their existing private signals. Competition raises prices and lowers expected excess payoffs. In the counterfactual, other institutions are crowded out most where adopters contribute more information. They shift portfolio weight toward stocks with less information from adopters, increasing market segmentation. At the same time, learning from prices can reduce their uncertainty and partly cushion their aggregate ownership loss.

Uneven adoption also shapes aggregate demand elasticity. Adopters' better private information, by itself, makes their demand more elastic. By contrast, greater price informativeness strengthens learning from prices, weakening all holders' responses to price changes. Other investors receive no private-information gain. A price increase signals better fundamentals, so they raise their payoff forecasts and sell less. Demand elasticity is therefore an endogenous equilibrium outcome. For each stock, the distribution of ownership across investors with different data scientist teams shapes the gains in private information and how informative prices become. The model jointly determines demand elasticity and price informativeness. In the counterfactual, learning from prices dominates the direct effect of adopters' new information for many stocks, making aggregate demand less elastic.

To measure private information, I separate stock-picking ability from the management of exposure to common risk factors. Investors choose stocks for their expected payoffs relative to those of stocks with similar risk and for their contribution to factor exposures, following the stock-picking and factor-management logic of Haddad et al. (2026). The model decomposes holdings into demand based on public information, private signals, and factor exposure adjustments. I focus on private information about firms' idiosyncratic payoffs, measured using subsequent earnings conditional on stock characteristics, analyst forecasts, and prices. Investors who systematically hold larger positions in firms with favorable idiosyncratic earnings surprises have a positive holdings–surprise covariance. Correcting this covariance for information already revealed by prices and factor exposure management yields an investor-level information measure scaled by risk tolerance.[5] The measure captures holdings' additional forecasting power for future payoffs beyond public information and prices.

[5] Kacperczyk et al. (2016) measure fundamental stock-picking ability using market-relative portfolio weights and subsequent earnings shocks. I use positions and earnings news conditional on public information and prices and correct for learning from prices.

I derive the information estimator's sampling variance and asymptotic normal distribution. The variance depends on the amount and concentration of information.

Measured information is positive at asset managers, persistent within managers, and concentrated in a small group of investors. Managers in the highest quintile of past information still have more information than those in the lowest quintile over the following three years, after a one-year gap. Information is also concentrated. In the average quarter, the top decile of institutions accounts for 84 percent of positive measured information. These findings support a model in which persistent differences in data processing capacity generate substantial heterogeneity in the information investors have.

I then estimate how data scientist teams produce information for active asset managers. Following Abis and Veldkamp (2024), the production function has two layers: data-management workers turn raw data into a stock of clean data, and data-science teams use that data to produce information. Investor fixed effects account for persistent data processing efficiency and risk tolerance, while year fixed effects measure common changes in productivity. Estimated productivity in 2024 is 3.28 times its 2011 level. This increase in data scientist productivity captures the effects of data technology developments on information production. The estimated data-science labor exponent is 0.143: at given productivity and other staffing, doubling data scientist employment raises information production by about 10.5 percent.

The demand system estimates how private information and price informativeness affect investors' demand elasticity. I extend the granular instrumental variable (GIV) framework of Gabaix and Koijen (2024) by using measured information to account for correlated trading on payoff news. Investors receive signals about the same firm's fundamentals, so favorable news can make several holders buy together. The model determines how strongly their demand responds to this shared news and to prices. Within each stock-quarter, I project other holders' demand on these model-implied sensitivities, excluding the focal investor. The residual demand, weighted by the holders' assets under management, supplies the instrument. These residual demand shocks generate price variation that identifies the focal investor's demand elasticity. I use them in GMM estimation, building on the asset-market applications of Chaudhry and Li (2026) and Chaudhary et al. (2026).

The demand estimates support both information channels. Investors with more predicted private information respond more strongly to price changes, while investors respond less strongly where prices are more informative. Institutional demand is inelastic at the median, with an implied elasticity of -0.51. At the median informative stock, an uninformed investor's demand is 60.6 percent less elastic when she learns from prices. Learning from prices therefore accounts for a substantial part of the estimated demand elasticity.

I construct the counterfactual by combining the estimated information production function and demand system in the market of 2024Q4. I raise data scientist productivity by an increment equal to the estimated increase from 2011 to 2024. The technology advance applies to active managers employing data scientist teams, whom I call adopters. Each adopter gains 69.5 percent more private information at fixed data scientist employment, distributed across its stocks according to its initial information allocation. The model aggregates these information gains stock by stock, updates the information conveyed by prices, and determines beliefs and holdings. Market clearing gives counterfactual prices. Average price informativeness rises by 3.1 percentage points from 11.8 percent. Small stocks gain 4.4 points, compared with 2.3 for large stocks. Because adopters supply a larger share of small stocks' information, the same proportional advance brings these stocks a larger information gain.

Uneven information gains crowd out other institutions and increase market segmentation. Adopters' positions rise by 15.4 percent, while other institutions' positions fall by 1.85 percent. A decomposition separates the contribution of private-information gains from learning from prices. Holding information in prices fixed, adopters' private-information gains reduce other institutions' positions by 1.95 percent. Allowing prices to become more informative offsets 0.11 percentage points of that loss, as other holders learn more from prices. Crowding out is strongest where adopters' private information accounts for a larger share of information in the price. Other institutions' positions fall by 2.3 percent in the highest stock quintile of this share and 0.7 percent in the lowest. This uneven reallocation makes stocks' investor bases more distinct. Their average distance rises from 53.12 to 53.60 percent. Larger counterfactual technology advances further increase this separation.

The data technology advance also changes aggregate demand elasticity. Holding information in prices fixed, the private-information gains make aggregate demand 4.5 percent more elastic at the median among stocks with informative prices. Once investors learn from the more informative prices, aggregate demand is 2.5 percent less elastic at the median. It becomes less elastic for 46.5 percent of all stocks.

Related literature. This paper contributes to the economics of data by estimating how data technology developments and investors' information production shape market equilibrium. Research on data and technology explains how scale, data availability, and processing shape learning (Begenau et al., 2018; Farboodi and Veldkamp, 2020; Dugast and Foucault, 2018, 2025). Empirical studies show that information production responds to easier access to corporate disclosures and to changes in publicly supplied information (Gao and Huang, 2020; Chen et al., 2020). Abis and Veldkamp (2024) estimate a two-layer production function and

analyze how the growth of knowledge production affects price informativeness and the cost of capital. I build on their technology, measure investor-level information output from holdings, and estimate the productivity of data scientist teams. Combining these estimates with heterogeneous asset demand, I study how a common technology advance affects investors' demand and price informativeness. This also contributes to market macrostructure, which studies how institutions' organization and characteristics determine asset prices (Haddad and Muir, 2025). In related work, Bonelli and Foucault (2026) document ownership reallocation from long-term investors toward hedge funds as alternative-data coverage expands.

A second strand studies information acquisition and portfolio choice. Studies with rational expectations equilibrium models explain how heterogeneous private signals determine portfolio choices and become partly revealed through prices (Grossman and Stiglitz, 1980; Hellwig, 1980; Admati, 1985). Van Nieuwerburgh and Veldkamp (2010) jointly determine learning and portfolio choice, showing how information acquisition can lead investors to specialize. Kacperczyk et al. (2016) connect heterogeneous processing capacity to mutual funds' attention allocation between fundamental stock-picking and market-timing abilities. Institutional incentives also shape information acquisition through delegation and benchmarking (Gârleanu and Pedersen, 2018; Breugem and Buss, 2019). Goldstein and Yang (2015) show that investors learning about different components of a payoff can complement one another: information conveyed by prices reduces uncertainty about the other component and encourages trading and acquisition. Kacperczyk et al. (2025) show how investor size and market power affect both learning and the transmission of private information into prices. I estimate the information produced by data-science labor and evaluate a productivity increase at observed levels of data scientist employment.

The measurement contribution is to recover investor-level information from holdings and future firm payoffs. Forecasting price efficiency evaluates the information incorporated into market prices by how accurately those prices predict firms' future fundamentals (Bond et al., 2012). Farboodi et al. (2022) recover aggregate data precision from prices and fundamentals, separating private information from information conveyed by prices. Farboodi et al. (2025) value observed financial data from their incremental predictive content for returns, accounting for information revealed through prices. Stock-level measures quantify how much prices reveal about fundamentals (Bai et al., 2016; Dávila and Parlatore, 2026). My estimator builds on the portfolio-covariance logic of Kacperczyk et al. (2016). It attributes private information to individual investors by conditioning earnings news on public information and prices and correcting for information contained in prices and factor management. I derive its sampling variance and asymptotic distribution.

I build on the literature on asset demand systems (Koijen and Yogo, 2019, 2026; Huebner,

2023; Haddad et al., 2025; Jansen et al., 2024; Koijen et al., 2024; Li et al., 2026; Bretscher et al., 2026; Chaudhry and Li, 2026; Chaudhary et al., 2026). Koijen et al. (2024) use demand to measure investor informativeness and study how capital reallocations affect prices and price informativeness. My framework provides a microfoundation for measuring investors' private information when they learn from prices and links this information to data-science productivity. I also develop counterfactuals that map heterogeneous information gains into changes in price informativeness and investors' demand. Haddad et al. (2026) show how homogeneous substitution conditional on observables supports identification of relative demand elasticities when investors substitute across many assets. For each investor, my model extends the assumption by requiring homogeneous substitution conditional on observables, private information, and price informativeness, and I estimate how those information channels shape demand elasticities. Dou et al. (forthcoming) show that active funds tilt toward stocks with low exposure to common fund flows to hedge fund-size risk.

The analysis also builds on work explaining why demand elasticities vary. Haddad et al. (2025) show that investors' responses to one another matter for aggregate elasticity and that information acquisition can be one of the forces driving these responses. Chaudhry and Li (2026) document variation in elasticity with the size of demand shocks and accumulated price dislocations. Davis (2025) shows that quantitative investment strategies need not generate highly elastic demand. Bastianello and Fontanier (2025) show how mistaken inference from prices can make demand less elastic. I estimate how private information and rational learning from prices affect investor demand. Combining these estimates with heterogeneous productivity gains quantifies when a technology advance makes aggregate demand for a stock less elastic.

The paper connects asset managers' skill to its sources and market-wide effects. Holdings and trades reveal skill among some mutual fund managers (Kacperczyk et al., 2005; Kacperczyk and Seru, 2007; Kacperczyk et al., 2008; Cremers and Petajisto, 2009; Baker et al., 2010), and Kacperczyk et al. (2014) relate variation in skill to information processing and attention allocation. Recent evidence connects AI skills and data scientist staffing to investment performance (Cen et al., 2025; Aragon et al., 2026). Sheng et al. (forthcoming) find larger gains from ChatGPT's publication at hedge funds with pre-existing AI staff. Bonelli and Foucault (forthcoming) show that alternative data erode traditional experts' advantage in stock-picking ability and induce reallocation toward uncovered stocks. I estimate the productivity of data scientist teams from the earnings-based information measure and quantify how a productivity advance changes ownership and informational efficiency.

Feedback between prices and fundamentals could extend these effects to firms' decisions. Bond et al. (2012) and Goldstein (2023) review this feedback, and Chen et al. (2007) test

whether managers learn from prices. Goldstein et al. (2023) find that wider dissemination of corporate filings through EDGAR raised stock liquidity, lowered the cost of equity capital, and increased equity financing and investment. The counterfactual in this paper holds firm characteristics and fundamentals fixed. In a setting with feedback, the quantified changes in prices and information could also affect firms' financing and investment.

Section 2 presents the data and motivating evidence. Section 3 develops the model. Sections 4–6 measure information and estimate production and demand. Section 7 quantifies the technology advance.

2 Data and Motivating Facts

Institutional holdings and employment histories establish two facts that motivate the model. Data-science employment has grown rapidly and unevenly across asset managers. Managers with larger shares of data-science labor also show greater gains in stock-picking ability around breakthroughs in data technology.

2.1 Data Technology in Asset Management

Asset managers use data technology to turn data about firm fundamentals into forecasts for investment decisions. They combine alternative data, drawn from sources other than financial statements and market prices, with statistical and machine-learning methods to forecast sales, earnings, and other fundamentals (Green and Zhang, 2024). Natural-language processing extracts information from written and spoken language. Investors apply it to corporate filings and earnings-call transcripts to interpret disclosures and forecast fundamentals. Forward-looking tone and changes in financial reports predict future earnings (Li, 2010; Cohen et al., 2020); neural-network measures of annual-report sentiment predict future profitability and operating cash flows (Azimi and Agrawal, 2021). FinBERT, a version of Google's BERT language model adapted to financial text, improves information extraction from earnings calls (Huang et al., 2023). Goldman Sachs's quantitative equity funds began using natural-language processing on these calls in 2014.[6] Longer or more complex filings give these tools more text to process.

Computer vision, which identifies objects and patterns in images, provides one example. A model can count cars in satellite images of retailers' parking lots and turn those images into measures of customer traffic before firms report sales (Kang et al., 2021). Growth in

[6] Goldman Sachs Trust, *Tax-Advantaged Equity Funds Annual Report* (2014).

parking-lot fill rates predicts same-store sales and returns around earnings announcements (Katona et al., 2025).

Data-science teams develop and apply these methods, combining the tools with datasets about firm fundamentals to produce information for investment decisions. In an Alternative Investment Management Association survey, 53 percent of hedge-fund managers use alternative data, and 85 percent of those using it for more than five years are looking to hire data scientists.[7] Newly published language and computer-vision models expand these teams' forecasting tools. These applications motivate the stock-picking ability measure in Section 2.4 and the information-production model in Section 3. Appendix A.3 and Online Appendix OA2 give further institutional evidence.

2.2 Data

I combine quarterly institutional holdings from Thomson Reuters, 1984Q1–2024Q4, with CRSP prices and Compustat fundamentals. The holdings are quarter-end long positions in U.S.-listed equities reported on Form 13F by investors with discretion over at least $100 million in reportable securities. A 13F filer is the unit of observation, and its AUM is the market value of its reported equity holdings. I use institutional investor, institution and investor to refer to a filer.

Each filer receives a fixed investor type describing the business behind its reported portfolio (Appendix A.1). Asset managers are the subgroup whose primary activity is investment: hedge funds, mutual funds, investment advisers, and public pension and sovereign funds. Banks, insurers, and broker-dealers form a separate comparison group because their equity demand also reflects financial intermediation and insurance objectives. Banks and broker-dealers may hold shares as market-making inventories or as hedges against exposures from customer trades. Insurers' equity allocations are shaped by policyholder liabilities and regulatory capital requirements.

Stock characteristics come from the WRDS data library of Jensen et al. (2023), which constructs them from CRSP and Compustat. Size is log market capitalization, and book-to-market is book equity divided by market equity. The demand system uses the five nonprice characteristics in Koijen and Yogo (2019): log book equity, operating profitability, investment, dividends to book equity, and market beta. Profitability is earnings before interest, taxes, depreciation, and amortization, less interest expense, divided by book equity. Investment is the annual proportional change in total assets. The dividend ratio is total dividends over the preceding twelve months divided by book equity. Market beta is estimated from

[7] Noble and Balint, *Casting the Net* (AIMA, 2020), a survey of 100 hedge-fund managers.

a 60-month rolling regression of excess stock returns on excess market returns, requiring at least 36 observations. Filing length comes from the Loughran–McDonald summaries of firms' 10-K and 10-Q filings. The measures of stock-picking ability and information use quarterly EBIT from Compustat (Section 4). Online Appendix OA1 gives the variable definitions.

I/B/E/S provides revisions to median quarterly earnings-per-share forecasts, forecast dispersion, and analyst coverage. Along with stock characteristics, these variables control for public information when I estimate investors' private information.

Individual employment histories from Revelio Labs cover 2009Q1–2024Q4 and draw on publicly available professional profiles, including LinkedIn.[8] I match filers to the companies that employ their staff, allowing the match to change when a filer's name changes. Matches use company identifiers and standardized names, with websites and corporate affiliations used to resolve ambiguous cases. Data-science (DS) labor is the weighted headcount of data scientists and quantitative researchers in the investment process. It includes data-science work in statistical and machine-learning forecasting, text, computer vision, and alternative-data signals, as well as all quantitative-research titles. This measure captures computational research capacity. Appendix A.2 summarizes the classification, and Online Appendix OA1.2 gives the details of the classification.

The share of DS labor is DS employment divided by total employment, expressed as a percentage. A DS employer has positive DS employment in the quarter. When filers share an employer, I allocate its workforce in proportion to their 13F assets, or equally if none has assets, so each employee counts once. The employment panel includes all 13F institutions matched to Revelio in the corresponding quarter.

I combine institutional holdings with stock characteristics and analyst forecasts to construct a quarterly panel of 13F institutions. I use this full 13F panel to measure investors' information and estimate stock demand.

I match the 13F panel to the employment data to estimate how research teams produce information. Table 1 reports summary statistics for the matched employment panel, the full 13F panel, and the quarterly stock panel. Online Appendix OA1.5 describes the construction of the estimation samples.

2.3 Fact 1: DS Employment Has Grown Rapidly and Unevenly

DS employment at asset managers grew far faster than employment of investment analysts and portfolio managers. From 2009Q4 to 2024Q4, aggregate DS labor rose 8.5-fold, from 788 to 6,732 (Figure 1(a)), while employment of investment analysts and portfolio managers

[8] Revelio's sampling weights use government employment statistics to adjust for differences in profile coverage across occupations and locations. See Revelio Labs, *Data Dictionary* and *RPLS Methodology*.

Table 1: Descriptive Statistics

(a) Quarterly Investor Panel, with Employment Data

	N	Mean	SD	25%	50%	75%
1(DS Labor > 0) (%)	263,987	12.0	32.4	0.0	0.0	0.0
# DS Labor	263,987	1.8	21.0	0.0	0.0	0.0
Share of DS Labor (%)	263,987	0.67	4.15	0.00	0.00	0.00
# Investment Staff	263,987	41.8	410.5	1.0	3.1	9.6
# Data Engineering	263,987	2.7	34.6	0.0	0.0	0.0
# Admin. and Operations	263,987	29.7	321.7	0.0	1.1	4.1
# Finance and Accounting	263,987	4.7	32.5	0.0	0.0	2.1
# Human Resources	263,987	6.7	71.6	0.0	0.0	0.0
# IT Support	263,987	10.4	118.1	0.0	0.0	0.0
# Employees	263,987	316.5	3,283.5	5.1	12.5	37.6
AUM ($bn)	263,987	4.59	56.88	0.07	0.20	0.81
# Stocks Held	263,987	174	371	20	59	143
Years since First 13F	263,987	9.89	9.35	2.50	6.75	14.75

(b) Quarterly Investor Panel, Whole Sample

	N	Mean	SD	25%	50%	75%
AUM ($bn)	449,561	3.84	44.96	0.08	0.23	0.87
# Stocks Held	449,561	186	390	25	68	159
Years since First 13F	449,561	8.70	8.42	2.25	6.00	12.50

Table 1: Descriptive Statistics

(c) Quarterly Stock Panel

	N	Mean	SD	25%	50%	75%
Size (ln ME)	910,743	5.19	2.32	3.50	5.05	6.76
B/M	910,743	0.79	0.93	0.29	0.56	0.95
Op. Profitability	910,743	0.04	0.76	−0.01	0.17	0.29
Asset Growth	910,743	0.27	1.08	−0.03	0.07	0.23
Market Beta	910,743	1.11	0.74	0.61	1.03	1.50
Idio. Vol. (Daily, %)	910,743	3.56	2.47	1.86	2.87	4.49
13F Ownership (%)	910,743	37.9	31.7	8.5	31.0	64.7
# 13F Holders	910,743	100	208	8	31	111
# Analysts	910,743	3.7	5.4	0.0	1.0	5.0
Earnings Surprise (SUE)	910,743	0.83	8.88	−1.80	0.73	3.46

Notes. Panel (a) describes institutions matched to Revelio employment records; panels (b) and (c) describe the full investor and stock panels defined in Section 2.2. DS Labor counts data scientists and quantitative researchers under the definitions in Section 2.2; its share is DS labor per 100 employees. Employee counts use Revelio's sampling weights. Investment staff includes investment analysts and portfolio managers. AUM is the market value of holdings in the U.S. equity panel, in billions of dollars. Investor age is years since the first 13F report in the data. ME is market equity in millions of dollars; B/M is book-to-market equity. Operating profitability is operating profits over book equity. Asset growth is annual total-asset growth. Market beta uses 60 months. Idiosyncratic volatility is the annual-window daily standard deviation of market-model residuals, in percent. Ownership statistics use stock-quarters covered by the 13F data; ownership is institutions' percentage of outstanding shares. SUE is the standardized earnings surprise defined in Online Appendix OA1.3. Online Appendix OA1 gives the construction details.

grew 2.3-fold. The DS share of investment staff rose from 2.39 to 8.41 percent (Figure 1(b)). The DS share of investment staff divides DS labor by DS labor plus investment analysts and portfolio managers. Growth also occurred within managers: among those present in 2009Q1, DS labor grew 5.3-fold by 2024Q4.



Figure 1: DS Labor at U.S. Asset Managers

Notes. Asset managers and DS labor follow Section 2.2. Headcounts use Revelio's sampling weights. Panels (a) and (b) plot fourth-quarter DS employment and its share of investment staff, respectively. The share uses the orange dashed line. The denominator comprises DS labor, investment analysts and portfolio managers. The share is the ratio of aggregate headcounts. Panel (c) reports the percentage of managers with positive DS employment in each AUM percentile group in 2024Q4. Every manager has equal weight. The red dashed line gives the overall percentage calculated across all managers.

DS employers are a small share of managers and hold most assets. At the end of the employment panel, 11.7 percent of managers employed DS labor and held 80.4 percent

of AUM, up from 57.3 percent in 2009Q4. The AUM share reached 92.6 percent among mutual funds and 68.5 percent among hedge funds (Online Appendix Table OA1). Entry by managers without DS staff lowered the overall fraction of DS employers. Among managers observed in both 2009Q4 and 2024Q4, however, that fraction rose from 14.7 to 24.8 percent.

DS employment is concentrated to larger asset managers. Only 4.4 percent of managers in the bottom half by AUM had DS staff, compared with 13.0 percent between the 50th and 90th percentiles, 38.4 percent between the 90th and 99th, and 86.5 percent in the top one percent (Figure 1(c)). Employment also differs across manager types. Using each type's aggregate employment, the DS labor share was 7.45 percent at hedge funds, 1.26 percent at investment advisers, 0.90 percent at mutual funds, and 0.36 percent at public pension and sovereign funds.

2.4 Fact 2: DS Labor and Gains in Stock-Picking Ability after Technology Breakthroughs

I study asset managers' stock-picking ability around major breakthroughs in natural language processing (NLP) and computer vision (CV).[9] Each technology helps forecast the fundamentals of particular firms, whose stocks I call exposed stocks. For NLP, exposed stocks are text-heavy stocks, the top decile of filing length before a breakthrough after adjusting for size and industry. Their longer reports give the tools more material to process. For CV, they are store-based retailers and restaurants: satellite images of their parking lots reveal customer traffic (Katona et al., 2025; Bonelli and Foucault, forthcoming). The classification uses retail SIC codes, including restaurants and excluding nonstore retailers (Appendix A). If DS teams help managers use new tools, stock-picking ability on exposed stocks should improve more at managers with larger pre-event shares of DS labor.

Following Kacperczyk et al. (2016), I measure the fundamental stock-picking ability of asset manager i on the exposed stocks in quarter t as

$$\mathrm{SP}_{i,t} = N_{i,t}^{-1} \sum_{n \in E_{i,t}} \left(w_{i,n,t} - w_{n,t}^{m} \right) \mathrm{SUE}_{n,t+4}, \tag{1}$$

where $E_{i,t}$ is the set of exposed stocks the asset manager holds with an observed earnings surprise, and $N_{i,t}$ is the number of those stocks. The portfolio weight $w_{i,n,t}$ uses the asset manager's whole portfolio; $w_{n,t}^{m}$ is the stock's market weight. The earnings surprise $\mathrm{SUE}_{n,t+4}$ is operating earnings over the next four quarters minus operating earnings over the past four,

[9] The NLP events focus on information extraction from disclosures. Recent generative large language models also support coding and autonomous workflows and fall outside this event set.

scaled by the standard deviation of the eight preceding year-over-year changes in quarterly earnings. The measure is high when the asset manager overweights firms before favorable earnings surprises. It is defined when the asset manager holds at least ten exposed stocks with an observed surprise.

The breakthroughs come from a registry of new NLP and CV methods and models that appear in the Epoch AI database and rank in the top 0.01 percent of their field and year by citations (Appendix A; Online Appendix OA2.5). I group nearby developments into waves. Each event falls in the first quarter ending at least two months after the highest-ranked model's code or trained models became public, allowing time for use before holdings are observed. The NLP events are Word2Vec and BERT; the CV events are AlexNet, ResNet, Mask R-CNN, ViT, and SAM (Appendix Table A1).

I build a separate dataset around each event and stack them for estimation (Cengiz et al., 2019). Each dataset includes asset managers with at least one employment observation in the four quarters before the event and follows their stock-picking ability on exposed stocks from six quarters before the event to six after. DS employers have DS staff in any of those four pre-event quarters. Comparison asset managers have none during those quarters.

The pre-event share of DS labor, $\mathrm{DS}_{i,s}$, is asset manager i's mean DS labor per 100 employees over the four quarters before event s, standardized across asset managers in the event's dataset. Asset managers without DS staff have a zero share before standardization.

The event study is

$$\mathrm{SP}_{i,s,t} = \sum_{k \neq -1} \beta_k \, \mathbf{1}\{t - t_s = k\} \times \mathrm{DS}_{i,s} + \alpha_{i,s} + \lambda_{s,t} + u_{i,s,t}, \tag{2}$$

where t_s is the event quarter, and $k = -1$, the quarter before the event, is the reference. Asset-manager-by-event effects $\alpha_{i,s}$ absorb each asset manager's average stock-picking ability within the window, and event-by-quarter effects $\lambda_{s,t}$ absorb common movements. Thus, β_k measures the change from quarter -1 to quarter k per standard deviation of the pre-event share of DS labor.

Managers with larger shares of DS labor show greater gains in stock-picking ability on exposed stocks. Figure 2 plots the NLP coefficients in panel (a) and CV coefficients in panel (b). Both increase at the event quarter; the NLP gains are largest in the first four post-event quarters. For CV, the pooled gain per standard deviation is 0.55×10^{-3}, about 13 percent of mean stock-picking ability on store-based retailers in the estimation sample. Appendix Tables A2 and A3 report pooled estimates, specifications adding shares of six other types of employment, an employer-indicator specification, and results for complex-text stocks and satellite-covered retailers. Online Appendix Table OA2 gives all staffing

coefficients from column (2). Appendix B describes the pooled specification, weights, and pre-event tests.

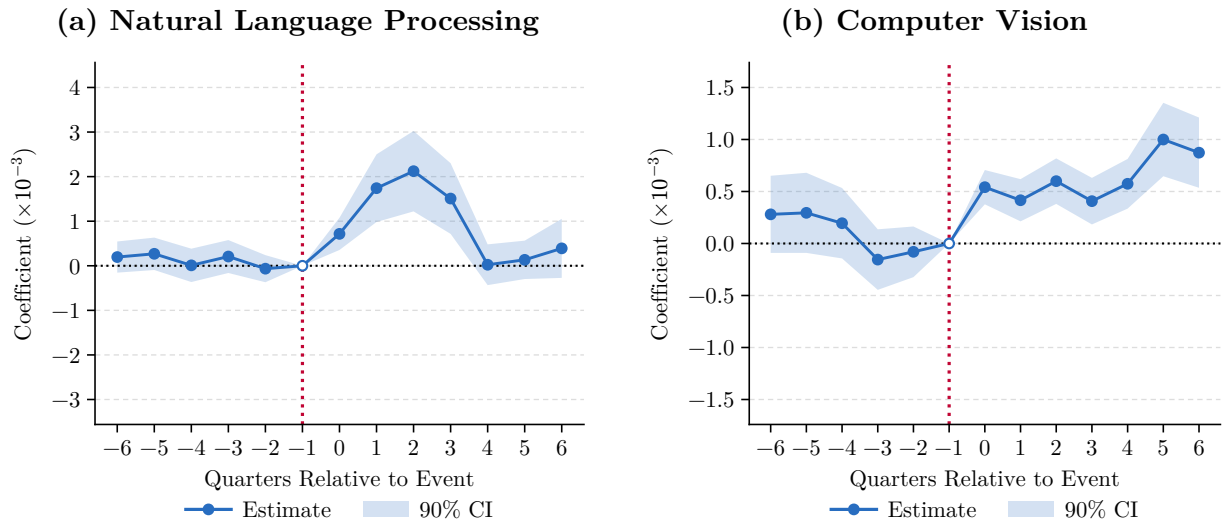


Figure 2: Stock-Picking Ability around Data-Technology Breakthroughs

Notes. Changes in stock-picking ability around the two language events in panel (a) and five computer-vision events in panel (b), listed in Appendix Table A1. Each panel pools its events in one regression. Points are the event-time coefficients, β_k, in equation (2), multiplied by 10^3; bands give 90 percent confidence intervals. Each coefficient measures the change per standard deviation of the pre-event DS share of total employment, relative to quarter −1. Stock-picking ability, defined in equation (1), uses text-heavy stocks in panel (a) and store-based retailers in panel (b). The hollow marker and vertical line mark the omitted quarter. The sample and fixed effects follow column (1) of Appendix Tables A2 and A3. DS employers have weight one; comparison investors are reweighted to give each event the same ratio of comparison investors to DS employers (Wing et al., 2024). Standard errors are clustered by investor.

2.5 From Facts to Model

The facts connect data scientist employment to investment outcomes. DS labor has grown rapidly and remains concentrated, while managers with larger DS employment shares show greater gains in stock-picking ability around breakthroughs. The model gives research teams a role in producing private signals about firm payoffs and lets their trades convey some of that information to others through prices. The production function in Section 5 estimates how much more information a given team produces as productivity improves. This estimated change sets the size of the counterfactual technology advance.

3 Model

Data technology improves the information of investors with data scientist teams. As these investors trade, they compete with other holders for shares and convey some of their information through prices. I study these effects in a noisy rational-expectations economy with heterogeneous investors and many stocks (Grossman and Stiglitz, 1980; Hellwig, 1980; Admati, 1985). The model links an investor's information to its holdings and explains how learning from prices transmits an uneven technology advance across investors. Appendix C gives the proofs.

3.1 Setup

Timing. There are three dates. At date 0, data scientist teams use their existing data stock to produce signals and allocate them across stocks. At date 1, investors observe their signals and choose portfolios. At date 2, payoffs are realized. Data scientist teams and data stocks are in place at the beginning of date 0.

Assets. There are N stocks, each with supply normalized to one share, and a riskless asset with gross return R_f. Stock n trades at price $p(n)$, and investor i holds the fraction $q_i(n)$ of its shares. I assume that payoffs follow a one-factor model.[10] The payoff of stock n is

$$f(n) = \mu(n) + \beta(n)\, F + \varepsilon(n), \qquad F \sim N(0,1), \qquad \varepsilon(n) \sim N\Big(0, 1/\tau_\varepsilon(n)\Big), \tag{3}$$

where $\mu(n)$ is the expected payoff, $\beta(n)$ the loading on the common factor F, and $\varepsilon(n)$ the idiosyncratic payoff innovation. The factor and the innovations are mutually independent. The payoff precision $\tau_\varepsilon(n)$ is the inverse of the innovation's prior variance. Following Koijen et al. (2024), I allow expected payoffs and factor loadings to depend on observable characteristics $x(n)$:

$$\mu(n) = \Phi_\mu^\top x(n) + \xi_\mu(n), \qquad \beta(n) = \Phi_\beta^\top x(n) + \xi_\beta(n). \tag{4}$$

Φ_μ and Φ_β are vectors of coefficients on firm characteristics. Known residuals $\xi_\mu(n)$ and $\xi_\beta(n)$ allow expected payoffs and factor loadings to differ among stocks with the same characteristics.

Investors. There is a continuum of investors, indexed by i.[11] Investor i has initial wealth

[10] With multiple factors and no mandates, Proposition C.2 gives the corresponding factor-management term in demand. The information estimator keeps its form, and price informativeness is unchanged in the benchmark economy.

[11] Appendix H derives the information moment with finitely many investors. Under the benchmark inference rule and the restrictions on price impact stated there, demand also keeps its form with effective risk

W_{1i} and chooses a portfolio q_i, whose elements are $q_i(n)$, to maximize U_i subject to her budget constraint:

$$U_i = \gamma_i \,\mathbb{E}[W_{2i} \mid \mathcal{I}_i] - \frac{\gamma_i^2}{2} \operatorname{Var}(W_{2i} \mid \mathcal{I}_i) + \frac{\delta_i^\phi}{2}\Big(\phi^\top q_i\Big)^2, \qquad W_{2i} = R_f W_{1i} + q_i^\top\Big(f - R_f p\Big), \quad (5)$$

where W_{2i} is wealth at date 2, f and p are vectors of payoffs and prices, $\mathcal{I}_i$ is her information set, and $\gamma_i > 0$ is absolute risk aversion.[12] Risk tolerance is γ_i^{-1}. Investment mandates and priorities also shape portfolio choice (Haddad et al., 2026). Some institutions must maintain a prescribed mix of stocks; others follow broader objectives stated in their prospectuses. The last term captures a soft investment mandate or priorities that the investor can trade off against expected return and risk: $\phi(n) \geq 0$ is a stock attribute, such as carbon emissions, and $\phi^\top q_i$ is total portfolio exposure to it. A negative coefficient δ_i^ϕ penalizes the magnitude of this exposure; a positive one rewards it.[13] Without a mandate, $\delta_i^\phi = 0$. Because the objective depends on total exposure, changing one position affects the incentive to hold other stocks with the same attribute.

Information. Investors use data to learn about payoffs. I measure their information in units of signal precision: one data point on stock n is a signal about its payoff innovation that is exactly as informative as the prior. The count therefore measures how much the data teach the investor about payoffs.[14] Investor i's jth data point on the stock is

$$\eta_i^j(n) = \varepsilon(n) + u_i^j(n), \qquad u_i^j(n) \sim N\Big(0, 1/\tau_\varepsilon(n)\Big), \qquad j = 1, \ldots, k_i(n), \quad (6)$$

where $k_i(n) \geq 0$ counts her data points on the stock. I assume that the noise $u_i^j(n)$ of a data point is normal, independent across data points, stocks and investors, and independent of the factor and the innovations. The average of her data points, $\eta_i(n) = k_i(n)^{-1} \sum_j \eta_i^j(n)$, is a sufficient statistic for them, and I call it her signal on the stock. Given $\varepsilon(n)$, it is normal with mean $\varepsilon(n)$ and precision $k_i(n)\tau_\varepsilon(n)$, so her private precision about the stock and her total private information are

$$\tau_i(n) = k_i(n)\,\tau_\varepsilon(n), \qquad K_i = \sum_n k_i(n). \quad (7)$$

tolerance. The appendix gives the conditions required for production estimation and counterfactuals.

[12] Under normality, the first two terms equal $-\ln \mathbb{E}[\exp(-\gamma_i W_{2i}) \mid \mathcal{I}_i]$ and give the CARA portfolio (Veldkamp, 2011; Kacperczyk et al., 2016).

[13] A hard mandate fixes exposure at $\phi^\top q_i = \bar{c}_i$, as with a required allocation to an asset category. Online Appendix Proposition OA3.1 derives optimal demand under this constraint.

[14] This count expresses the information in a dataset in precision units; raw records may have different information content. It can take noninteger values because signal precision is continuous.

She can also learn from prices p, so her information set is $\mathcal{I}_i = \{\eta_i(1), \ldots, \eta_i(N), p\}$. Signals reveal information about idiosyncratic payoffs, and investors keep their prior about the common factor.[15]

The count $k_i(n)$ measures private precision relative to payoff precision. It is unchanged by the units of payoffs or shares, and the first data point halves the prior variance. Thus K_i sums information in a common unit across stocks.[16]

Allocation of data. A predetermined rule allocates an investor's total information across stocks:

$$k_i(n) = K_i \, \Psi_i\Big(x(n)\Big), \qquad \Psi_i\Big(x(n)\Big) \geq 0, \qquad \sum_{n \in \mathcal{N}_i} \Psi_i\Big(x(n)\Big) = 1, \tag{8}$$

where $x(n)$ is a vector of observable stock characteristics known at date 0, and $\mathcal{N}_i$ is the investor's stock universe. Ψ_i is zero outside it. She holds no stocks outside this universe. The predetermined rule depends on stocks through these variables and separates scale from mix: a proportional rise in K_i raises every $k_i(n)$ in proportion. It can reflect the coverage of a data source. Since the variance of the payoff innovation can be included in $x(n)$, this allocation rule can also summarize an information choice, including specialization in stocks on which information is most valuable (Van Nieuwerburgh and Veldkamp, 2009, 2010).

Noise traders. Besides investors, noise traders trade every stock, buying $z(n)$ shares of stock n. $z(n)$ is normal with mean zero and variance $\sigma_z^2(n)$, independent across stocks and of every other shock. Together, the factor, the payoff innovations, the noise of every data point and the noise trades are therefore jointly normal and mutually independent.[17]

3.2 Equilibrium

Investors choose optimal portfolios given their information sets, and prices clear markets.

Definition 1 (Equilibrium)**.** Given the data counts $k_i(n)$, an equilibrium is a set of functions for prices p, and portfolio choices $\{q_i\}$, such that three conditions hold. (i) Given $\mathcal{I}_i$, each q_i maximizes the objective (5). (ii) Markets clear,

$$\int q_i \, di + z = \mathbf{1}, \tag{9}$$

where z stacks the noise trades and $\mathbf{1}$ is the vector of supplies. (iii) Each investor rationally

[15] This stock-level information complements the attention allocation between aggregate and stock-specific shocks studied by Kacperczyk et al. (2016).

[16] A budget on these counts is a scale-free precision budget. An entropy budget instead charges the same for each halving of remaining variance(Van Nieuwerburgh and Veldkamp, 2010; Veldkamp, 2011). Here each successive halving requires twice as many new data points.

[17] Assumption C.1 states the boundedness and factor-pervasiveness conditions used in the proofs.

updates her beliefs by Bayes' rule given the price function.

Beliefs. The price conveys part of what other investors know. I conjecture, and verify in Proposition 1, that when $\kappa(n) > 0$, stock n's price reveals the signal

$$\eta_p(n) = \varepsilon(n) + \frac{z(n)}{\kappa(n)}, \qquad \kappa(n) = \int \gamma_i^{-1}\tau_i(n)\, di, \qquad \tau_p(n) = \frac{\kappa(n)^2}{\sigma_z^2(n)}, \tag{10}$$

where $\kappa(n)$ is aggregate information weighted by risk tolerance and $\tau_p(n)$ is price-signal precision. Demand responds to each private signal with coefficient $\gamma_i^{-1}\tau_i(n)$. Adding these coefficients gives $\kappa(n)$. Noise trading prevents full revelation. Doubling $\kappa(n)$ divides price-signal noise by two and multiplies its precision by four. When $\kappa(n) = 0$, investors learn nothing from the price and omit its signal from their beliefs.

I write $\tilde{\tau}(n) = \tau_\varepsilon(n) + \tau_p(n)$ for the public precision of stock n and $\hat{\tau}_i(n) = \tilde{\tau}(n) + \tau_i(n)$ for investor i's posterior precision.

Lemma 1 (Beliefs)**.** *Under the conjecture, investor i's beliefs about the payoff innovations are normal and independent across stocks, with*

$$\mathbb{E}[\varepsilon(n) \mid \mathcal{I}_i] = \frac{\tau_i(n)\,\eta_i(n) + \tau_p(n)\,\eta_p(n)}{\hat{\tau}_i(n)}, \qquad \mathrm{Var}\big(\varepsilon(n) \mid \mathcal{I}_i\big) = \frac{1}{\hat{\tau}_i(n)}. \tag{11}$$

Her beliefs about the factor are its prior.

Conditional independence makes precisions add. The price contributes the same precision to every investor, while private signals distinguish their beliefs. No signal reveals the factor. Price informativeness, $\tau^{\mathrm{R}}(n) = \tau_p(n)/\tilde{\tau}(n)$, is the share of prior uncertainty about the innovation that the price resolves (Dávila and Parlatore, 2026).

Optimal demand. At a given expected excess payoff, a better-informed investor takes a larger position because she faces less uncertainty per share. Buying the stock may also increase her portfolio's factor exposure or move it away from a mandate. She adjusts other positions to account for these portfolio effects.

Write $m_i(n) = \mathbb{E}[f(n) \mid \mathcal{I}_i] - R_f p(n)$ for expected excess payoff, and $M_i = \hat{\Omega}_i + \beta\beta^\top - \delta_i^\phi \gamma_i^{-2}\phi\phi^\top$ for the curvature matrix of the objective. Here $\hat{\Omega}_i$ is the diagonal posterior covariance matrix of idiosyncratic innovations, with entries $1/\hat{\tau}_i(n)$. I assume M_i is positive definite, which makes the objective strictly concave and the optimal portfolio unique. This holds automatically for $\delta_i^\phi \leq 0$.

In the demand rule below, $\gamma_i^{-1}\hat{\tau}_i(n)m_i(n)$ is the position the investor would choose based on the stock's expected excess payoff and idiosyncratic uncertainty. The other two terms

adjust this position for factor and mandate exposures, following the portfolio logic of Haddad et al. (2026). Their shadow prices, g_i^β and g_i^ϕ, are common to all stocks in her portfolio. Each stock's loadings and posterior precision determine how much its position changes.

Lemma 2 (Optimal demand)**.** *Investor i's problem has the unique solution $q_i = \gamma_i^{-1} M_i^{-1} m_i$. Stock by stock,*

$$q_i(n) = \gamma_i^{-1} \hat{\tau}_i(n) \Big[m_i(n) - \beta(n) g_i^\beta - \phi(n) g_i^\phi \Big], \tag{12}$$

where the factor and mandate adjustments g_i^β and g_i^ϕ solve the two-by-two system

$$\begin{pmatrix} g_i^\beta \\ g_i^\phi \end{pmatrix} = G_i^{-1} \begin{pmatrix} \beta^\top \hat{\Omega}_i^{-1} m_i \\ \phi^\top \hat{\Omega}_i^{-1} m_i \end{pmatrix}, \qquad G_i = \begin{pmatrix} 1 + \beta^\top \hat{\Omega}_i^{-1} \beta & \beta^\top \hat{\Omega}_i^{-1} \phi \\ \beta^\top \hat{\Omega}_i^{-1} \phi & -\gamma_i^2 / \delta_i^\phi + \phi^\top \hat{\Omega}_i^{-1} \phi \end{pmatrix}.$$

Without a mandate, $\delta_i^\phi = 0$, the system reduces to its first equation, so that $g_i^\phi = 0$ and $g_i^\beta = \beta^\top \hat{\Omega}_i^{-1} m_i / (1 + \beta^\top \hat{\Omega}_i^{-1} \beta)$. The portfolio's exposures are $\beta^\top q_i = \gamma_i^{-1} g_i^\beta$ and, for $\delta_i^\phi \neq 0$, $\phi^\top q_i = -\gamma_i g_i^\phi / \delta_i^\phi$.

A favorable signal can induce the investor to buy one stock and adjust other positions to manage the resulting exposure. The two-by-two system makes these choices consistent across her portfolio. Its shadow prices depend on the signals and prices observed when trading, even though preferences and loadings are fixed.

Equilibrium price function. Aggregating these demands and imposing market clearing gives the following price function. The benchmark replaces realized factor and mandate adjustments by their means in aggregate demand. Individual portfolios keep their realized adjustments.[18]

Proposition 1 (Equilibrium price function)**.** *Under the benchmark market-clearing approximation, stock n has the equilibrium price function*

$$R_f \, p(n) = \mu(n) - \frac{1 + \beta(n) H_\beta(n) + \phi(n) H_\phi(n)}{\mathcal{T}(n)} + \frac{\kappa(n) \, \varepsilon(n) + z(n)}{\mathcal{T}_p(n)}. \tag{13}$$

The coefficient $\mathcal{T}(n)$ determines the expected excess payoff needed to absorb supply; $\mathcal{T}_p(n)$ determines the price impact of trading. The terms $H_\beta(n)$ and $H_\phi(n)$ aggregate the mean factor and mandate adjustments. These quantities are defined below. If $\kappa(n) > 0$, the price is a strictly increasing affine function of $\eta_p(n)$, which verifies the conjecture. If $\kappa(n) = 0$, the price reveals nothing and moves with $z(n)$ alone.

[18] Under the diversification and market-depth conditions in Appendix C, the approximation error shrinks as investors' stock universes expand; the same discussion gives finite-universe bounds.

Consider first a stock with $\beta(n) = \phi(n) = 0$. Before signals and noise trades are realized, its expected excess payoff is $1/\mathcal{T}(n)$, the compensation investors require to hold its unit supply. Greater risk tolerance or more precise information lowers this compensation and raises the expected price. Let $\bar{\gamma}^{-1} = \int \gamma_i^{-1}\, di$ be the total risk tolerance of investors who can hold the stock, suppressing its dependence on n. Then

$$\mathcal{T}(n) = \int \gamma_i^{-1}\hat{\tau}_i(n)\, di = \bar{\gamma}^{-1}\Big[\tau_\varepsilon(n) + \tau_p(n)\Big] + \kappa(n).$$

Holding payoff beliefs fixed, a purchase of one share from these investors raises the price by $1/[R_f\mathcal{T}(n)]$. When the purchase comes from noise traders, investors also learn from the resulting price rise. They interpret part of it as favorable payoff news and require a larger price increase to sell the same number of shares.

For an investor with no private signal, the demand slope based on the prior alone is $-R_f\gamma_i^{-1}\tau_\varepsilon(n)$. Learning from prices multiplies this slope by

$$d_p(n) = \frac{\kappa(n)}{\kappa(n) + \bar{\gamma}^{-1}\tau_p(n)} = \frac{1}{1 + \bar{\gamma}^{-1}\kappa(n)/\sigma_z^2(n)} \in (0, 1], \tag{14}$$

with $d_p(n) = 1$ at $\kappa(n) = 0$. As aggregate private information $\kappa(n)$ rises or noise-trading variance $\sigma_z^2(n)$ falls, prices convey more information and $d_p(n)$ falls: the investor sells less in response to a price increase.

For an investor with a private signal, private precision adds $-R_f\gamma_i^{-1}\tau_i(n)$ to this slope. Aggregating across investors gives the demand slope $-R_f\mathcal{T}_p(n)$, where

$$\mathcal{T}_p(n) = \kappa(n) + \bar{\gamma}^{-1}\tau_\varepsilon(n)\, d_p(n) = \mathcal{T}(n)\, d_p(n). \tag{15}$$

A noise-trader purchase changes the price by $1/[R_f\mathcal{T}_p(n)]$ per share, $1/d_p(n)$ times the impact with payoff beliefs fixed. At a given price, a unit payoff innovation raises informed demand by $\kappa(n)$; the equilibrium price rises by $\kappa(n)/[R_f\mathcal{T}_p(n)]$. Independent private-signal errors average out, leaving the payoff innovation and noise trading as the two sources of price fluctuations.[19]

Factor risk and mandates enter expected excess payoffs through $\beta(n)H_\beta(n)/\mathcal{T}(n)$ and $\phi(n)H_\phi(n)/\mathcal{T}(n)$. The first term compensates holders for factor exposure; the second reflects how holding the stock serves their investment objectives. The aggregates are

$$H_\beta(n) = \int \gamma_i^{-1}\hat{\tau}_i(n)\, \bar{g}_i^{\beta}\, di, \qquad H_\phi(n) = \int \gamma_i^{-1}\hat{\tau}_i(n)\, \bar{g}_i^{\phi}\, di.$$

[19] With finitely many investors, individual signal errors also enter prices. An investor then learns from the part of the price not accounted for by her own signal; Appendix H.2 gives the inference rule.

Here $\bar{g}_i^a = \mathbb{E}_0[g_i^a]$ for $a \in \{\beta, \phi\}$ is the mean adjustment given date-0 primitives and information allocations.

Relative demand slopes. Price changes also induce trades to adjust factor and mandate exposures. These adjustments cancel from the difference in holdings of two stocks whose positions respond equally to changes in either shadow price. For distinct stocks $n, n' \in \mathcal{N}_i$, equation (12) gives the conditions

$$\hat{\tau}_i(n')\beta(n') = \hat{\tau}_i(n)\beta(n), \qquad \hat{\tau}_i(n')\phi(n') = \hat{\tau}_i(n)\phi(n).$$

Precision enters these conditions because a given change in either shadow price induces a larger adjustment when the investor has a more precise payoff forecast. Because posterior precision varies across investors, the matching conditions are investor specific.[20]

Proposition 2 (Relative demand slopes)**.** *Under the benchmark equilibrium price function, hold information allocations, private signals and all other prices fixed. For the two stocks defined above,*

$$\frac{\partial\left[q_i(n) - q_i(n')\right]}{\partial p(n)} = -R_f\,\gamma_i^{-1}\Big[\tau_i(n) + \tau_\varepsilon(n)\,d_p(n)\Big]. \tag{16}$$

The coefficients on the right-hand side sum to $-R_f\mathcal{T}_p(n)$ across the stock's eligible holders.

A more precise private signal makes the demand more elastic. Learning from prices makes the demand less elastic through $d_p(n)$. Section 6 estimates these two information effects. Slopes are measured in shares per dollar of price.

3.3 Data Processing and Information Production

Following Abis and Veldkamp (2024), I separate data preparation from analysis. Data managers acquire and organize raw records into usable datasets. Data scientist teams analyze these data to produce information about firms' payoffs. For an investor i with a data scientist team, the two stages in quarter t are

$$D_{it} = (1-\delta_D)D_{i,t-1} + \left(L^{\mathrm{DM}}_{i,t-1}\right)^{\rho},$$
$$K_{it} = \tilde{a}_i A_t D_{it}^{\nu}\left(L^{\mathrm{DS}}_{i,t-1}\right)^{1-\nu}.$$

Here K_{it} is total private information, D_{it} is the usable data stock, L^{DM} is data-management labor, and L^{DS} includes data scientists and quantitative researchers. Both labor inputs

[20] Corollary C.1 gives the own-price response, including the portfolio-exposure adjustment.

are measured at the previous quarter-end, before signals arrive. The depreciation rate is $\delta_D \in [0,1]$. The exponents $0 < \rho, \nu < 1$ describe diminishing returns to data-management labor and to data in analysis. Investor efficiency $\tilde{a}_i$ and common productivity A_t determine how much information a team produces with given inputs.

The information measure uses quarterly earnings innovations. I treat usable data as specific to forecasting the next earnings announcement and assume full depreciation within one quarter, $\delta_D = 1$. Each announcement reveals the outcome that these data help forecast. Because the sample has limited coverage of data-management labor, I also assume $L^{\mathrm{DM}}_{i,t-1} = c_i L^{\mathrm{DS}}_{i,t-1}$, where $c_i > 0$ is constant over time. Substituting these assumptions gives

$$K_{it} = a_i A_t \left(L^{\mathrm{DS}}_{i,t-1}\right)^{\alpha}, \tag{17}$$

where $\alpha = 1 - \nu + \rho\nu$ and $a_i = \tilde{a}_i c_i^{\rho\nu}$. The labor elasticity α combines the gains from preparing more data and from analyzing them as the team expands. Section 5 estimates this relationship using the information expressed in investors' holdings.

3.4 What a Data-Technology Advance Does

A data technology advance raises information productivity at investors with DS teams. Holding employment fixed, each adopter's private information grows in proportion to the increase in A_t. The gains reach stocks unevenly because these investors hold different portfolios and contribute different amounts of information. Their demand affects other investors through more informative prices and competition for the fixed supply of stocks.

Price informativeness. The effect differs across stocks because adopters account for different shares of aggregate information. For $\kappa(n) > 0$, write $\Delta_\kappa(n) = \kappa'(n)/\kappa(n) - 1$ for the proportional increase in aggregate information, where a prime denotes the economy after the data technology advance. This increase weights each adopter's gain by her initial contribution to $\kappa(n)$. Price signal precision then rises by the factor $[1 + \Delta_\kappa(n)]^2$. For a given proportional gain, price informativeness rises most at an intermediate initial level: nearly uninformative prices convey little new knowledge, while highly informative prices leave little uncertainty to resolve.[21]

Crowding out. For a clean example, take a stock with $\beta(n) = \phi(n) = 0$, so it has no factor or mandate exposure. The expected trading payoff attributable to investor i's private

[21] The gain is largest at initial informativeness $1/[2 + \Delta_\kappa(n)]$, just below one half for a small advance (Corollary C.5).

information is

$$\Pi_i(n) = \gamma_i^{-1} \frac{\tau_i(n)}{\tilde{\tau}(n)} = \gamma_i^{-1} k_i(n)[1 - \tau^{\mathrm{R}}(n)]. \tag{18}$$

A more informative price reduces this informational rent from a given private signal.[22] Bonelli and Foucault (forthcoming) show how this erosion of informational advantage lowers the trading returns of investors whose expertise improves less with data technology.

Expected ownership depends on the investor's willingness to bear risk relative to that of other holders. Even an investor with no private signal holds shares in expectation because the stock compensates her for bearing payoff risk. For the same stock, the expected excess payoff $1/\mathcal{T}(n)$ clears the unit supply, giving

$$\mathbb{E}[q_i(n)] = \frac{\gamma_i^{-1} \hat{\tau}_i(n)}{\mathcal{T}(n)}. \tag{19}$$

Investors with more precise forecasts or greater risk tolerance demand more. A technology advance raises posterior precision but lowers the expected excess payoff as investors compete for the fixed supply. At unchanged risk tolerance,

$$\frac{\mathbb{E}[q_i'(n)]}{\mathbb{E}[q_i(n)]} = \frac{\hat{\tau}_i'(n)}{\hat{\tau}_i(n)} \frac{\mathcal{T}(n)}{\mathcal{T}'(n)}. \tag{20}$$

An investor's expected demand falls when her precision grows less proportionally than holders' average precision, weighted by risk tolerance. Learning from prices can improve her forecast even as better-informed adopters crowd her out.[23]

Proposition 3 (Crowding out)**.** *Let $\beta(n) = \phi(n) = 0$, and let some investors' private precision on stock n rise, holding $\sigma_z(n)$, $\tau_\varepsilon(n)$, $\bar{\gamma}^{-1}$ and all other investors' private precisions fixed. If $\tau^{\mathrm{R}}(n) < 1/2$, a small increase in aggregate private information $\kappa(n)$ lowers the expected demand of every investor whose private precision is unchanged. A finite increase in $\kappa(n)$ by the proportion $\Delta_\kappa(n) > 0$ lowers their expected demand whenever $\tau^{\mathrm{R}}(n) < 1/[2 + \Delta_\kappa(n)]$.*

Investors whose private precision is unchanged gain information through more informative prices, while adopters' stronger demand lowers the stock's expected excess payoff. For a small increase in $\kappa(n)$ with initial $\tau^{\mathrm{R}}(n) < 1/2$, this payoff falls proportionally more than any of these investors' posterior precisions rise, so all of their expected holdings fall.[24]

[22] For nonzero factor loadings, Proposition C.1 gives this rent for the stock-selection position after removing the realized factor adjustment.

[23] With zero mandate penalties, the two propositions extend to the conditional stock-selection demand of stocks with nonzero factor loadings.

[24] Easley and O'Hara (2004) find lower average holdings of uninformed investors when a larger share of

Corollary 1 (Crowding out across investors)**.** *Under the conditions for crowding out in Proposition 3, consider investors i and j whose private precisions on stock n remain unchanged. If $\tau_i(n) > \tau_j(n)$ and the price becomes more informative, investor i experiences a larger percentage decline in expected holdings:*

$$\frac{\mathbb{E}[q_i(n)] - \mathbb{E}[q_i'(n)]}{\mathbb{E}[q_i(n)]} > \frac{\mathbb{E}[q_j(n)] - \mathbb{E}[q_j'(n)]}{\mathbb{E}[q_j(n)]}.$$

More informative prices add the same precision to both investors' forecasts. This is a larger proportional improvement for the investor with less private information. Her expected holdings therefore fall by a smaller percentage when competition from adopters lowers the stock's expected excess payoff.

The effect of learning from prices on expected demand depends on the investor's private precision. Holding private precisions fixed, greater public precision raises expected demand among investors whose private precision is below the risk-tolerance-weighted average, $\kappa(n)/\bar{\gamma}^{-1}$, and lowers it among those above (Proposition C.1). It cushions crowding out for relatively uninformed investors and reinforces it for better-informed investors whose private information is unchanged.

Aggregate demand elasticity. A technology advance affects aggregate demand elasticity through better private information and learning from prices. For a stock with $\beta(n) = \phi(n) = 0$, the magnitude of aggregate demand elasticity at a fixed positive price and quantity is proportional to $\mathcal{T}_p(n)$. A fall in $\mathcal{T}_p(n)$ therefore makes aggregate demand less elastic.

Proposition 4 (Decrease in Aggregate Elasticity)**.** *For a stock with $\beta(n) = \phi(n) = 0$, fix $\sigma_z(n)$, $\tau_\varepsilon(n)$ and $\bar{\gamma}^{-1}$. At $\kappa(n) > 0$, a small rise in $\kappa(n)$ lowers the magnitude of aggregate demand elasticity, evaluated at a fixed positive price and quantity, if and only if*

$$\kappa(n) < \bar{\gamma}^{-1}\tau_\varepsilon(n)d_p(n)[1 - d_p(n)].$$

Better private information reduces adopters' uncertainty about payoffs and makes their demand more elastic. Their informed trading also makes prices more informative, strengthening learning from prices for all investors. A higher price then leads investors to revise their payoff forecasts upward, making their demand less elastic. Since aggregate demand is the sum of individual demand curves, the change in aggregate demand elasticity depends on which force dominates.

fixed total information is private. Here some investors' private information increases; Corollary C.2 gives the finite-change condition.

Under the condition in Proposition 4, learning from prices dominates. To interpret the condition, consider a small percentage increase in aggregate private information. In equation (15), the private-information component of the magnitude of aggregate demand elasticity is proportional to $\kappa(n)$ and grows by that percentage. The component based on investors' common prior is proportional to $\bar{\gamma}^{-1}\tau_\varepsilon(n)d_p(n)$; stronger price learning reduces it by $1 - d_p(n)$ times that percentage. The condition holds when the component based on the common prior is large enough that its fall exceeds the rise in the private-information component in absolute size.[25] Section 7 measures how these forces change prices, demand elasticity and portfolios when information gains differ across stocks.

4 Measuring Investors' Information

Private information gives investors' holdings forecasting power for future payoffs beyond public information and prices. Following the portfolio-covariance logic of Kacperczyk et al. (2016), I measure information by asking whether an investor holds more of stocks with subsequently favorable payoff news. The model supplies corrections for information already revealed by prices and for portfolio adjustments that offset the investor's stock-level demand.

Consider an investor's position in a stock before its next earnings announcement. I remove the earnings news predicted by public variables and the price, then ask whether the investor's position predicts the remaining earnings surprise. A holding above the amount predicted by public controls contributes positively when the earnings surprise is favorable; a holding below that amount contributes positively when the earnings surprise is unfavorable. The investor may have learned from public information, such as analyst forecasts. What matters is whether her forecast contains information beyond that already in public variables and prices.

4.1 From Holdings to Information

Let $\tilde{\varepsilon}(n) = \varepsilon(n) - \mathbb{E}[\varepsilon(n) \mid p]$ denote payoff news left unrevealed by prices, measured empirically by the residual earnings surprise. The investor's risk-tolerance-weighted information is $\theta_{it} = \gamma_{it}^{-1} K_{it}$. Risk tolerance matters because an investor willing to bear more risk takes larger positions on the same signal.

Two adjustments recover information from the covariance of demand with residual payoff news. First, prices reveal the fraction $\tau^{\mathrm{R}}(n)$ of payoff uncertainty, reducing the covariance

[25] From equation (14), $\partial \log d_p(n) / \partial \log \kappa(n) = -[1 - d_p(n)]$. Lemma C.1 and Corollary C.3 give the local and finite-change conditions.

even when private information is unchanged. Second, buying on favorable private information changes the portfolio's factor and mandate exposures. Managing these exposures offsets part of the purchase. The share $h_i(n)$ measures this reduction in the stock's response to its private signal; dividing by $1 - h_i(n)$ recovers the information behind that response.[26]

Theorem 1 (Information identity)**.** *The covariance of investor i's demand for stock n with its residual payoff news is $\mathbb{E}[q_i(n)\tilde{\varepsilon}(n)] = \gamma_i^{-1} k_i(n)[1 - \tau^{\mathrm{R}}(n)][1 - h_i(n)]$. Hence*

$$\gamma_i^{-1} K_i = \sum_n \frac{\mathbb{E}\Big[q_i(n)\,\tilde{\varepsilon}(n)\Big]}{\Big[1 - \tau^{\mathrm{R}}(n)\Big]\Big[1 - h_i(n)\Big]}. \tag{21}$$

Demand based on public information and trades on private signal noise have zero covariance with residual payoff news. With full positions and known model objects, realized products in equation (21) therefore give an unbiased estimate of risk-tolerance-weighted information.[27]

An investor without private information has a zero population moment. The identity holds for raw demand: removing a public component leaves its population covariance unchanged (Corollary C.6). Public-information trades nevertheless add sampling noise because they can coincide with realized payoff news by chance. I residualize positions to reduce this noise and correct below for the effect of fitting the position controls in the same sample.

The stock-picking ability measure in equation (1) uses portfolio weights and earnings surprises. The information measure instead uses positions as fractions of shares outstanding, removes the news predicted by public information and prices, and corrects for the two sources of attenuation.[28] I implement it in four steps; Appendix D.1 specifies the forecasting regressions and estimator corrections.

(1) I use operating earnings announced in the quarter after holdings are observed, subtract earnings for the same fiscal quarter a year earlier, and divide the difference by book equity at the holdings date. This comparison accounts for recurring seasonal patterns. I forecast this earnings change using stock characteristics and analyst-forecast revisions, coverage and dispersion. The augmented forecast adds price, its interactions with

[26] Appendix C derives $h_i(n) = \hat{\tau}_i(n) v_n^\top G_i^{-1} v_n$, with $v_n = (\beta(n), \phi(n))^\top$. For $\delta_i^\phi \leq 0$, $0 \leq h_i(n) < 1$. If $\phi(n)$ is spanned by the included characteristics, controls containing $\tilde{\tau}(n)\phi(n)$ remove the public-precision component of the mandate adjustment. The private-precision component remains. Online Appendix Proposition OA3.1 covers hard mandates; Proposition C.2 covers multiple factors.

[27] With finitely many investors, the moment measures incremental precision beyond the price, weighted by effective risk tolerance when price impact is present (Appendix H).

[28] The forecasting stage adapts Dávila and Parlatore (2026). Attributing risk-tolerance-weighted information to investors complements the latent-demand attribution of Koijen et al. (2024) and the aggregate price-informativeness trends in Bai et al. (2016).

characteristics, and industry effects. Its error, $\tilde{\varepsilon}_{t+1}(n)$, is the residual earnings surprise.

(2) The inverse conditional variances of the two forecast errors give prior and public precisions, $\hat{\tau}_\varepsilon(n)$ and $\hat{\tau}_\varepsilon(n)+\hat{\tau}_p(n)$. The baseline variance divided by the augmented variance gives the correction $1/[1-\hat{\tau}^{\mathrm{R}}(n)]$. I anchor this correction to variance ratios within size and market-to-book bins and bound it between one and 100. The lower bound assigns zero measured price informativeness to stocks whose fitted variance does not fall.

(3) I express positions in book-equity dollars: $Q_{it}(n)$ is the fraction of outstanding shares held times the firm's book equity. Residualizing these positions on public controls, including every forecasting regressor, gives $\tilde{Q}_{it}(n)$. The estimation universe $\mathcal{N}_{it}$ contains stocks held in any of the previous four quarters; a current position is zero if the stock is no longer held. Since 13F reports long positions only, the measure captures information expressed through those positions.[29]

(4) I sum the corrected products of residual positions and earnings surprises:

$$\hat{\theta}_{it} = \sum_{n\in\mathcal{N}_{it}} \frac{\tilde{Q}_{it}(n)\,\tilde{\varepsilon}_{t+1}(n)}{\left[1-\hat{\tau}^{\mathrm{R}}(n)\right]\left[1-\hat{h}(n)\right]\left[1-P_{nn}\right]}. \tag{22}$$

To estimate $h_i(n)$, I weight each stock's squared estimated market beta by its estimated public precision and divide by the sum across the investor's stock universe. Market betas are estimated using 60 months of monthly returns. This feasible $\hat{h}(n)$ replaces unobserved investor-specific precisions with public precision (Corollary C.7). The diagonal P_{nn} of the position regression's projection matrix is the weight on stock n's own position in its fitted value. Subtracting this fitted value removes part of the information covariance; dividing by $1-P_{nn}$ restores that part. Investor and quarter indices are suppressed.[30]

4.2 Statistical Properties

The estimator's precision depends on how much information an investor has and how she allocates it across stocks. The results condition on public information and the predetermined

[29] Long positions yield a lower bound in expectation on risk-tolerance-weighted information in the Gaussian benchmark, up to the exposure and projection approximations. Censoring multiplies each stock's expected information contribution by its probability of being held, which is at most one (Lemma D.2).

[30] I cap $\hat{h}(n)$ at 0.5 and floor $1-P_{nn}$ at 0.05. An observed mandate attribute outside the span of market beta must also enter the exposure correction. Appendix D.1 gives the specification.

information allocation. Write V_{it} for the leading sampling variance. Appendix D.2 states the regularity conditions and gives the proofs.

Proposition 5 (Bias and precision)**.** *Under the conditions in Appendix D.2,*

$$\mathbb{E}[\hat{\theta}_{it}] - \theta_{it} = O(\theta_{it}/|\mathcal{N}_{it}|), \qquad \mathrm{Var}(\hat{\theta}_{it}) = V_{it}\Big[1 + O(|\mathcal{N}_{it}|^{-1})\Big].$$

For $K_{it} > 0$, the leading relative variance is

$$\frac{V_{it}}{\theta_{it}^2} = 2\,\mathrm{HHI}_{it} + \frac{\bar{w}_{it}}{K_{it}} + \frac{\sigma^2_{\mathrm{mis},it}}{\theta_{it}^2},$$

where $\mathrm{HHI}_{it} = \sum_n [k_{it}(n)/K_{it}]^2$ measures information concentration, $\bar{w}_{it} = \sum_n \frac{k_{it}(n)}{K_{it}[1-\tau^{\mathrm{R}}(n)]}$ averages the price-informativeness correction, and $\sigma^2_{\mathrm{mis},it}$ is the variance contributed by public demand left in residual positions.

Relative bias vanishes as the stock universe grows. Spreading information across stocks reduces dependence on payoff news from a few stocks, lowering the concentration term. Reducing the signal-noise term $\bar{w}_{it}/K_{it}$ also requires more information, at given allocation shares and price informativeness. More informative prices amplify sampling noise because a larger correction is needed to recover information already revealed by prices. Public demand left after residualization adds noise without contributing information.

Corollary 2 (Asymptotic normality)**.** *Under the conditions of Proposition 5, let $V_{it}(n)$ denote stock n's contribution to $V_{it} > 0$. If $\max_n V_{it}(n)/V_{it} \to 0$ as $|\mathcal{N}_{it}| \to \infty$, then*

$$\frac{\hat{\theta}_{it} - \theta_{it}}{\sqrt{V_{it}}} \xrightarrow{d} N(0,1).$$

The normal approximation applies when no stock accounts for a nonvanishing share of the estimator's sampling variance. If information also becomes diffuse, K_{it} grows and residual public-demand variance becomes negligible relative to θ_{it}^2, the estimate recovers the investor's information with a vanishing proportional error.

In the data, I sum the squared information contributions of twelve industries to measure sampling uncertainty, allowing stocks within each industry to move together.[31]

I report two normalizations. *Information per dollar* divides $\hat{\theta}_{it}$ by lagged AUM, the value of the entire 13F portfolio at the previous quarter-end, and is reported per billion dollars.

[31] These industry contributions are uncentered, so the resulting scale is not the variance V_{it} in Corollary 2. Appendix D.2 gives its expectation and shows that standardized information is bounded in absolute value by $\sqrt{12}$. Standard errors for group statistics use their quarterly time series.

This normalization removes the size effect on the estimated total information.[32] *Standardized information* divides by the estimated standard error and measures the estimated information relative to sampling noise. Multiplying every position by the same positive factor scales the estimate and its standard error proportionally. Standardization therefore removes the proportional effect of position size and risk tolerance. Larger positions alone do not raise standardized information (Lemma D.1). In the data, I sum the squared information contributions of twelve industries to measure sampling uncertainty, allowing stocks within each industry to move together.[33]

4.3 Properties of Measured Information

Measured information is higher in the asset-management group than in the comparison group of banks, insurers and broker-dealers (Table 2; Section 2.2). Mean standardized information is 0.059 for the asset-management group and 0.010 for the comparison group. The difference is 0.049; the difference in information per dollar is 0.123. Both are significant. The comparison group's mean is statistically indistinguishable from zero.

Hedge funds rank first on both measures, consistent with holdings-based evidence on hedge-fund skill (Griffin and Xu, 2009; Agarwal et al., 2013). Their standardized information of 0.205 is statistically significant. The median hedge fund's information per dollar, averaged over quarters, is 0.951 per billion dollars of lagged AUM, compared with 0.169 for mutual funds and 0.108 for investment advisers.

Measured information is persistent. Each quarter, I sort investors within each group into quintiles by their mean standardized information over the previous twelve quarters, leave a four-quarter gap, and follow them for twelve quarters. In the asset-management group, the top quintile exceeds the bottom by 0.114, with monotonically increasing quintile means (Table 2, Panel (b)). The comparison group's spread is 0.003 and statistically indistinguishable from zero. The gap prevents the formation and evaluation windows from using the same quarterly earnings observation. The persistence is not driven by investor size. Controlling for size and breadth leaves a spread of 0.107; sorting these investors on size instead gives an imprecisely estimated spread of -0.033 (Online Appendix Table OA4).

Information is also concentrated. In the average quarter, the top decile of institutions accounts for 84 percent of positive measured information. Figure 3 compares the distribution of investors' standardized information. For investors with at least twelve quarterly estimates,

[32] If risk tolerance is proportional to assets, the ratio measures expressed information times relative risk tolerance, $1/\gamma_{it}^{R}$ (Assumption F.1).

[33] Appendix D.2 gives its expectation and shows that standardized information is bounded in absolute value by $\sqrt{12}$. Standard errors for group statistics use their quarterly time series.

Table 2: Measured Information of Institutional Investors

(a) By Institution Type

	N(Inst.)	13F Share (%)	Info. per $bn	Information (Standardized)
Asset Managers	3,549	71.1	0.167***	0.0595***
			(0.042)	(0.0120)
Hedge Funds	261	2.4	0.951***	0.2049***
			(0.347)	(0.0423)
Mutual Funds	562	50.5	0.169***	0.0632***
			(0.050)	(0.0159)
Inv. Advisers	2,665	13.2	0.108**	0.0388***
			(0.050)	(0.0118)
Public Pension & Sovereign	61	5.0	0.041	0.0285
			(0.108)	(0.0494)
Banks, Insurers & Broker-Dealers	724	28.0	0.044	0.0103
			(0.065)	(0.0216)
Difference			0.123***	0.0492***
			(0.036)	(0.0154)

(b) Persistence: Quintile Means Relative to the Group's Mean

	Low	2	3	4	High	High − Low
Asset Managers	−0.0465	−0.0177	−0.0086	0.0051	0.0677	0.1142***
						(0.0187)
Banks, Insurers & Broker-Dealers	0.0081	−0.0142	−0.0164	0.0112	0.0112	0.0031
						(0.0275)

Notes. Information estimates follow Section 4; the sample and institution groups are defined in Section 2.2. Panel (a) reports each group's share of the market value of positions, summed over quarters. Info. per $bn is the time average of each group's quarterly median information per billion dollars of lagged AUM. Standardized information divides the estimate by its industry-based noise scale. The column averages quarterly group means. Difference is asset managers minus banks, insurers and broker-dealers. Panel (b) sorts institutions by mean standardized information over the preceding twelve quarters, leaves a four-quarter gap, and reports means over the next twelve quarters relative to the mean of the five quintiles. Newey–West standard errors use 8 lags in panel (a) and 24 in panel (b). *, **, and *** denote significance at the 10, 5, and 1 percent levels.

I average their quarterly information estimates in levels using inverse sampling variances as weights and compute the implied standard error. The asset-manager distribution has a fatter right tail than the distribution for banks, insurers and broker-dealers. The orange area marks the excess density in this tail.

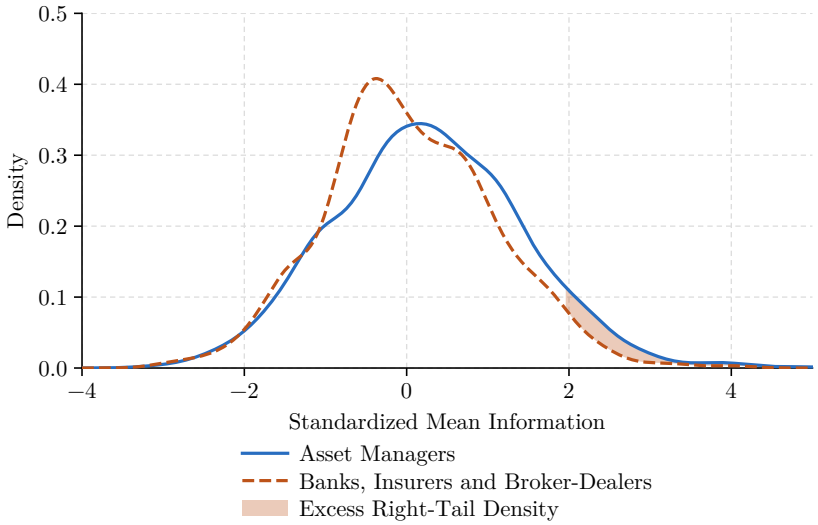


Figure 3: The Distribution of Measured Information across Investors

Notes. Distributions of standardized mean information for institutions with at least twelve quarterly estimates. Every institution has equal weight within its group. The solid blue and dashed dark-orange curves are Gaussian kernel densities for asset managers and for banks, insurers and broker-dealers, using a common pooled Silverman bandwidth. Orange shading marks the excess density in the asset managers' fatter right tail.

5 The Information Production Function

I estimate how data scientist teams process data and produce information about firm fundamentals. The output of the information production function is the measured total information of investors from the previous section. Higher productivity allows a given data scientist team to generate more precise signals about firm payoffs.

5.1 Specification

Following the model in Section 3.3, I estimate the information production function in logs, allowing productivity to vary by calendar year and controlling for employment in other occupations:

$$\log \hat{\theta}_{it} = \omega_i + \omega_{y(t)} + \alpha \log L^{\mathrm{DS}}_{i,t-1} + X^{\top}_{i,t-1} b + u_{it}. \tag{23}$$

Here $\hat{\theta}_{it}$ is investor i's measured total information in quarter t, and $L^{\mathrm{DS}}_{i,t-1}$ is DS employment. The two controls in $X_{i,t-1}$ are log combined employment of investment analysts and portfolio managers, and log employment in all remaining occupations, excluding DS. They account for the size of the investor's other investment staff and the broader organization. All employment inputs use headcounts from the preceding quarter. I assume that risk tolerance is constant within each investor. Investor fixed effects ω_i then absorb these factors along with persistent differences in efficiency. The calendar-year effects $\omega_{y(t)}$ measure common changes in information productivity at given employment. The disturbance u_{it} includes unobserved changes in inputs and productivity, together with sampling error in measured information.

The main sample consists of active asset managers. Active share is half the absolute distance between portfolio and market weights (Cremers and Petajisto, 2009). Managers above the 20th percentile of full-sample mean active share form the active group. They employ 60 percent of DS labor and their holdings are more likely to express stock-level research than those of managers close to the market portfolio. I require positive DS employment in at least four of the preceding eight quarters to focus on investors with an established DS workforce. The log specification uses observations with positive measured information and positive employment in each input.

I estimate the coefficients jointly by least squares. Changes in DS employment within investors identify the labor elasticity α, controlling for other employment and common annual changes. Appendix E gives the formal conditions and estimation details.

5.2 Annual Information Productivity

I normalize the 2011 year effect to zero and measure productivity relative to that year:

$$\frac{\widehat{A}_y}{\widehat{A}_{2011}} = \exp\bigl(\hat{\omega}_y - \hat{\omega}_{2011}\bigr), \qquad \hat{\omega}_{2011} = 0. \tag{24}$$

Holding DS and other employment fixed, the estimated production function yields 3.28 times as much information in 2024 as in 2011 (Figure 4). Productivity reaches 1.62 times its initial level in 2016 and 2.53 in 2021, and reaches its highest estimated value in 2024.

The estimated DS labor elasticity is 0.143. At unchanged productivity and other employment, doubling DS employment raises fitted information by 10.5 percent. With data-management labor proportional to DS employment, this elasticity combines the gains from preparing more data and from analyzing them. This estimate is consistent with Abis and Veldkamp (2024).

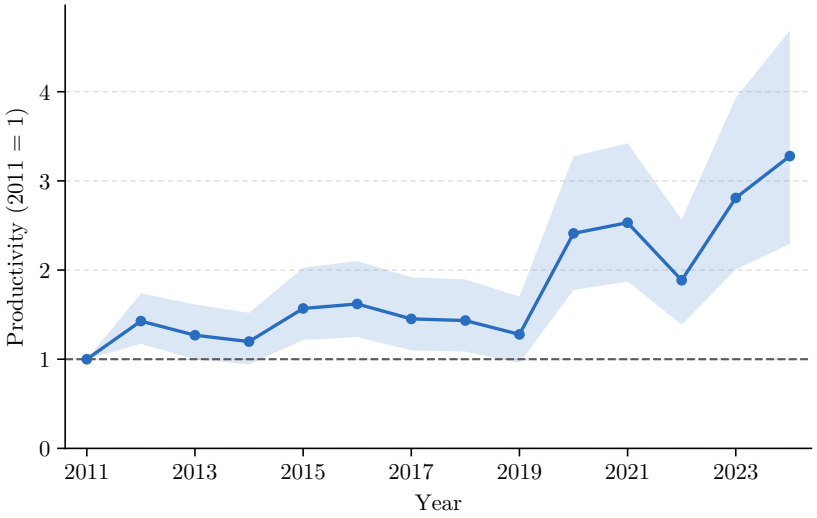


Figure 4: Annual Information Productivity

Notes. Annual productivity from equation (23), normalized to one in 2011. The active-manager sample follows Section 2.2. The line plots $\exp(\hat{\omega}_y - \hat{\omega}_{2011})$. Shading gives pointwise 90 percent confidence intervals.

6 Demand Estimation

I estimate a demand system in which private information and learning from prices shape investors' portfolio choices. The estimation uses quarterly changes in 13F holdings over 1984Q3–2024Q4.

6.1 The Demand System

Private information and learning from prices affect demand through investors' uncertainty about firm payoffs and their interpretation of price changes. More precise private signals and more informative prices both reduce posterior uncertainty. Holding the payoff forecast fixed, this reduction makes demand more sensitive to price. Learning from prices also changes the forecast itself: a higher price signals a higher payoff, offsetting part of the demand response to its higher cost. The estimated price slope combines these effects.

I estimate the relative demand slope to separate stock selection from factor management and adjustments to mandate exposures. Equation (16) compares two stocks for which managing these exposures induces equal position adjustments when one stock's price changes. The difference between their demand responses removes this common adjustment. Following Haddad et al. (2026), I use comparisons across stocks within an investor's portfolio, controlling for stock characteristics and their interactions with information. These interactions matter because stocks with the same factor loading can require different adjustments

to factor exposure when their posterior precisions differ.

I measure demand in book-equity dollars, $Q_{it}(n) = \mathrm{BE}_t(n) q_{it}(n)$, where $q_{it}(n)$ is investor i's fraction of outstanding shares and $\mathrm{BE}_t(n)$ is the firm's book equity (Koijen et al., 2024). I construct the quarterly demand flow $\Delta Q_{it}(n)$ from the change in split-adjusted shares, valued at the previous quarter's book equity per share.[34] Valuing both quarters' shares at the same book equity per share separates trading from changes in valuation. Price is the market-to-book ratio. Its change in the same fixed book-equity units is $\Delta \mathrm{MB}_t(n) = \mathrm{MB}_{t-1}(n) r_t(n)$, where $r_t(n)$ is the ex-dividend return.

Taking first differences of the model's demand schedule gives a price-change term whose coefficient depends on private information and learning from prices. The remaining terms guide the controls used to isolate these information effects. Appendix F gives the derivation. I estimate

$$\begin{aligned} \Delta Q_{it}(n) &= \zeta_{it}(n) \tau_\varepsilon(n) \Delta \mathrm{MB}_t(n) + \Theta_{it}^\top X_{it}(n) + e_{it}(n), \\ \zeta_{it}(n) &= \zeta_0 + \zeta_{\mathrm{type}(i)} + \zeta_{\mathrm{age}} \ln(1 + \mathrm{age}_{i,t-1}) \\ &\quad + \underbrace{\zeta_k \bar{k}_{i,t-1} / N_{i,t-1}}_{\text{private information}} + \underbrace{\zeta_\ell \ell_{t-1}(n)}_{\text{learning from prices}} \\ &\quad + \zeta_b \bar{\ell}_{t-1} \mathbf{1}\{\tau^{\mathrm{R}}_{t-1}(n) > 0\} + \zeta_c \mathbf{1}\{\tau^{\mathrm{R}}_{t-1}(n) = 0\}. \end{aligned} \tag{25}$$

The first term is the demand response to a price change. $\tau_\varepsilon(n)$ is the inverse of prior payoff variance. The payoff-precision term increases the magnitude of the price response for stocks with less payoff risk. The coefficient $\zeta_{it}(n)$ captures variation in the slope, with the two underbraced terms measuring the effects of private information and learning from prices. The baseline ζ_0 and the type and age terms account for other heterogeneity across investors. The remaining terms control for changes in relative price signal precision over time and stocks with uninformative prices. Our estimation uses the cross-sectional variation in price informativeness. The controls $\Theta_{it}^\top X_{it}(n)$ account for other demand shifts, and $e_{it}(n)$ is residual demand.

The model predicts a negative coefficient on private information and a positive coefficient on learning from prices, $\zeta_k < 0$ and $\zeta_\ell > 0$. Better private information reduces payoff uncertainty. Investors then buy more when prices fall and sell more when prices rise. Learning from prices also reduces uncertainty, but a higher price raises investors' payoff forecasts. This revision outweighs the effect of lower uncertainty, so investors sell less as prices rise. These information effects extend the characteristics-based demand system of Koijen and

[34] Here $\Delta Q_{it}(n)$ denotes the dollar trading flow divided by lagged assets under management (AUM). Assumption F.1 sets $\gamma_i^{-1} = \mathrm{AUM}_i / \gamma_i^{\mathrm{R}}$, matching the local risk tolerance of CRRA preferences evaluated at AUM. Dividing by AUM removes the component of the relative slope proportional to investor size. Relative risk aversion γ_i^{R} is constant over time within each investor and may differ across investors.

Yogo (2019); Koijen et al. (2024).

For the private-information interaction in demand, I assume that investors allocate information evenly across the $N_{i,t-1}$ stocks they hold in the sample used to estimate information.[35] The prediction $\bar{k}_{i,t-1}$ measures information per dollar of that portfolio's value. Dividing by $N_{i,t-1}$ gives predicted information per dollar per position. Because both the information estimate and demand come from holdings, I predict information using investor type, lagged size and a historical information score. This captures persistent differences in information and reduces the mechanical link with current trading. The score divides average past information by its estimated standard error and uses estimates through $t-5$. The persistence documented in Section 4 supports this use of past information. I fit the prediction coefficients using other investors' information estimates to reduce overfitting to the investor's own measurement noise. I standardize $\bar{k}_{i,t-1}/N_{i,t-1}$ across investor–stock observations, so its coefficient measures the slope difference associated with a one-standard-deviation increase in predicted information per dollar per position.

To measure learning from prices, I use the ratio of aggregate private information to noise-trading variance. I construct this ratio from estimated price informativeness and information aggregated across the stock's holders. A higher ratio makes demand less responsive to price in the model, holding other stock and investor characteristics fixed. For informative prices, $\ell_{t-1}(n)$ is the log of the previous quarter's ratio minus that quarter's median across investor–stock observations. This is the Relative Price Signal Precision measure in Table 3. The term $\bar{\ell}_{t-1}$ measures the quarterly median's deviation from the median over the full sample, so the coefficient on $\ell_{t-1}(n)$ compares stocks within a quarter. I also allow a separate slope for stocks with uninformative prices. Appendix F gives the construction details.

The controls $X_{it}(n)$ account for demand differences associated with observable stock characteristics, factor management and mandate exposures. Stock characteristics and their changes, together with public payoff forecasts, control for expected-payoff variation based on information other than the contemporaneous stock price. Interactions of factor loadings with information allow exposure adjustments to depend on the precision of payoff forecasts, with coefficients that vary across investor-quarters. Appendix F derives their connection to the relative demand slope.

[35] In an alternative specification, I assume a parametric form for investors' information allocation across investors, based on the interaction of investor features and stock characteristics. Estimating that demand system gives similar results but with lower power.

6.2 Identification with a GIV

Prices and investors' holdings are jointly determined in equilibrium, making prices endogenous in the estimation of demand elasticities. The granular instrumental variable (GIV) uses residual demand from other investors to isolate price variation (Gabaix and Koijen, 2024). The model supplies a natural source of this variation: investor-specific noise in private signals. These signal errors are independent across investors, and, with finitely many investors, they need not cancel in aggregate demand. Noise in other investors' signals can therefore move the market price while the focal investor's private signal stays fixed.

Constructing the instrument removes two components of observed demand. First, I account for investors' heterogeneous price elasticities of demand. This separates changes along their demand schedules from residual demand, which includes trading on private signals. Second, I remove the payoff innovation shared by those signals. Each private signal contains the same firm-level payoff innovation plus investor-specific noise. Responses to the shared innovation are stronger for investors with more private information. Removing this component isolates the variation associated with their signal noise.

I remove both components in a single projection of flows within each stock-quarter. The projection includes the investor variables governing price responses and the sensitivities to common payoff news implied by predicted information and payoff precision. Because flows difference two quarterly positions, the projection includes payoff-news sensitivities at both dates.

The heterogeneous price response can be removed before its coefficients are known. Within a stock-quarter, $\tau_\varepsilon(n)\Delta\mathrm{MB}_t(n)$ is common to all investors, so the price-response term in equation (25) is a linear combination of the observed variables in $\zeta_{it}(n)$. Projecting on these variables removes that combination for any values of its coefficients.[36] The instrument can therefore be constructed before estimating the demand coefficients, and the GMM moments are linear in those coefficients. This procedure follows Gabaix and Koijen (2024) and builds on the asset-market applications of Chaudhary et al. (2026); Chaudhry and Li (2026).

For investor i, I exclude her own flow from the projection and aggregate the other investors' residual flows:

$$Z_{it}(n) = \sum_{j \neq i} \mathrm{AUM}_{j,t-1} \Delta Q_{jt}^{\perp,(-i)}(n). \tag{26}$$

Here $\Delta Q_{jt}^{\perp,(-i)}(n)$ is investor j's residual flow from the projection excluding i. Multiplication

[36] For example, if the price coefficient varies only by investor type, subtracting each type's mean flow within that stock-quarter removes its price-response component without knowing the type coefficients. Projecting on all the variables in $\zeta_{it}(n)$ extends this argument to the specified heterogeneity in demand slopes.

by lagged AUM converts it back to book-equity dollars, giving larger investors' demand shocks more weight. Excluding investor i throughout the construction keeps her own signal noise out of the instrument. The resulting price variation traces her demand response, including learning from prices (Haddad et al., 2026).

The GIV instruments the baseline price term. To estimate heterogeneity, I interact it with each variable in $\zeta_{it}(n)$, including type indicators, log age, predicted information and the price-informativeness terms. Let $\widetilde{Z}_{l,it}(n)$ denote each instrument after removing the included controls, where l indexes the baseline and its interactions. I estimate the coefficients by linear GMM using

$$\mathbb{E}\Big[\widetilde{Z}_{l,it}(n)e_{it}(n)\Big] = 0 \qquad \text{for every slope term } l. \tag{27}$$

Controls serve as their own instruments. Appendix F derives the instrument and estimating moments.

6.3 Implementation

The sample follows positions held in consecutive quarters and includes stocks with at least sixty investors and investors with at least twenty stocks. I allow the coefficients on twenty controls to vary across investor-quarters. The investor-quarter intercept makes the comparison across stocks held by the same investor in the same quarter. The stock characteristics include log book equity, profitability, investment, dividends to book equity and market beta, entered in levels and changes. Together with changes in public payoff forecasts net of the information conveyed by prices, these characteristics control for expected-payoff differences. The information and factor-exposure interactions control for factor management. Additional expected-payoff controls use the payoff precision measured at each date and have separate coefficients common across investors and quarters, estimated jointly with the price coefficients.[37]

Observations receive equal weight in the estimating moments. The GIV projection weights flows by inverse residual variance estimated from other quarters. I calculate standard errors following Chaudhary et al. (2026) to account for each investor's shock entering other investors' instruments.

For each position, I calculate elasticity by multiplying the fitted demand slope by lagged market-to-book and dividing by lagged book-equity holdings per dollar of AUM. I average these elasticities within each investor-quarter using book-equity weights.

[37] These controls accompany the baseline, type and age terms and the terms for relative price signal precision. Appendix F gives the complete control set.

Table 3: Demand Estimates

	(1) Main Specification
Price (ζ_0)	−0.51
	(0.06)
Price × Information (ζ_k)	−0.05
	(0.01)
Price × Relative Price Signal Precision (ζ_ℓ)	0.25
	(0.04)
Implied Elasticity, Median	
All Institutions	−0.51
Hedge Funds	−1.10
Mutual Funds	−0.39
Inv. Advisers	−0.51
Banks & Broker-Dealers	−0.91
Pension Funds	0.21
Insurers	−0.21
Other	−1.27
Controls	Y
Price × Type, Age	Y
N(Inst.-Quarters)	106,144
N(Stock-Quarters)	197,128

Notes. Section 6 describes estimation. Information is standardized predicted private information per dollar per position. Relative Price Signal Precision is the log ratio of aggregate private information to noise-trading variance, centered at the quarterly median. Both measures are lagged one quarter. Elasticities equal fitted slopes times lagged market-to-book divided by lagged book-equity holdings per dollar of AUM. I average them within each investor-quarter using book-equity weights, then report medians across investor-quarters for all institutions and within each type. All price coefficients and their standard errors are reported on a common scale of the median investor-quarter elasticity. This sets the reported coefficient ζ_0 to −0.51 and expresses the interactions as slope differences on a common scale. Standard errors appear in parentheses and treat the scaling factor as fixed.

6.4 Estimates

Table 3 shows that investor demand is inelastic at the median, with an implied elasticity of −0.51. Hedge funds have a median elasticity of −1.10, banks and broker-dealers −0.91, investment advisers −0.51, and mutual funds −0.39. This ordering accords with Koijen and Yogo (2019); Koijen et al. (2024).

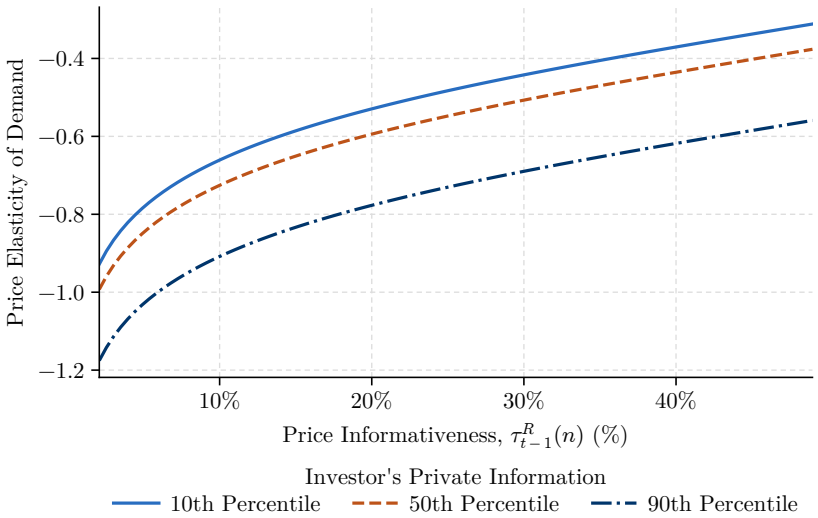


Figure 5: Demand Elasticity and Information

Notes. Fitted demand elasticities for a hypothetical position at the 10th, 50th and 90th percentiles of predicted private information per dollar per position. Price informativeness, $\tau^{\mathrm{R}}_{t-1}(n)$, is the percentage of idiosyncratic payoff uncertainty resolved by prices. The axis spans its 5th–95th percentiles across distinct informative stock-quarters. Age uses the investor-quarter median, and type is fixed at investment adviser. Holdings use the median across investor-quarters of each portfolio's median informative holding. Reference prices and quantities stay fixed along the curves. More negative values mean more elastic demand. Appendix F gives the calculation.

Figure 5 shows how fitted elasticity varies with private information and price informativeness for a hypothetical position. Its three curves compare the 10th, 50th and 90th percentiles of predicted information per dollar per position among investors with positive estimated private information. The horizontal axis is price informativeness, $\tau^{\mathrm{R}}_{t-1}(n)$: the fraction of payoff uncertainty resolved by prices. Other stock inputs, investor age and position size are fixed at their medians, with type fixed at investment adviser. The term for movements in the quarterly median of the price-learning measure is zero. Price and quantity stay fixed along each curve.

Better-informed investors have more negative demand slopes. A one-standard-deviation increase in predicted information per dollar per position changes the slope coefficient by −0.05, about a tenth of the baseline coefficient. This estimate agrees with the model: more precise private signals reduce uncertainty and increase the willingness to trade against a

price change. At the median price informativeness of 19.0 percent, Figure 5's elasticities at the 10th, 50th and 90th percentiles of private information are -0.54, -0.61 and -0.79. Investors with more private information have a larger percentage demand response to a one percent price change.

Learning from prices makes demand less elastic. A one-log-unit increase in the ratio of aggregate private information to noise-trading variance within a quarter raises the slope coefficient by 0.25. Holding other stock inputs fixed, a more informative price raises this ratio. Each curve in Figure 5 therefore rises with price informativeness: the elasticity becomes less negative as investors learn more from price changes.

Setting private information to zero at the median informative stock gives an AUM-weighted average slope coefficient of -0.42. Combined with the price-learning coefficient, Lemma F.1 gives a discount of $1 + 0.25/(-0.42) \simeq 0.4$. Learning from prices therefore reduces the price-response magnitude to about two fifths of its value under the common prior alone. This discount summarizes the effect on trading; price informativeness τ^{R} measures the share of payoff uncertainty resolved by prices.

7 Counterfactual: A Data-Technology Advance

I combine the estimated information production function and demand system to study the effects of a future advance in data technology. The experiment takes 2024Q4, the final quarter of the sample, as its baseline and repeats the historical increase in data-science productivity. I analyze the effects on price informativeness, the composition of stocks' investor bases and market segmentation, and aggregate demand elasticity.

An information gain changes both the level of demand and its slope. Better private signals and learning from prices reduce posterior payoff variance. Lower variance raises demand at a given expected excess payoff and increases the magnitude of the demand slope when payoff forecasts are held fixed. When investors revise their forecasts in response to prices, a higher price also signals a higher expected payoff, offsetting part of its effect on demand. The model combines the reduction in risk and the revision of forecasts to determine the net change in demand elasticity.

7.1 The Experiment

I combine 2024Q4 13F holdings and price informativeness with lagged information inputs. Employment of data scientists, investors' risk preferences and mandates, and the set and characteristics of stocks available to each investor remain fixed. Each stock's supply to

investors in the sample equals their initial aggregate holdings. The outside sector keeps its demand fixed and supplies no private information.

The advance raises private information at active asset managers with DS staff in 2024Q3, whom I call active adopters. They initially hold 10.7 percent of the sample's book-equity positions. Active status follows the production sample's classification in Section 5. The productivity gains estimated for active managers motivate this choice. Private information is unchanged for other investors, including low-active adopters; the advance affects them through competition from adopters and learning from prices.

I repeat the estimated increase in annual productivity from 2011 to 2024. For adopter i, the proportional gain in the baseline information measure θ_i is

$$\frac{\Delta\theta_i}{\theta_i} = \delta \frac{\widehat{A}_{2024} - \widehat{A}_{2011}}{\widehat{A}_{2024}} = \delta \left(1 - \frac{\widehat{A}_{2011}}{\widehat{A}_{2024}}\right). \tag{28}$$

The parameter δ measures the advance in multiples of the historical level increase, $\widehat{A}_{2024} - \widehat{A}_{2011}$. The estimated productivity ratio is 3.28 (Figure 4), so $\delta = 1$ raises each adopter's private information by 69.5 percent. The experiments with $\delta = 2$ and 3 double and triple this increment. The production function is multiplicative. At fixed DS employment, its common productivity factor scales each adopter's information in the same proportion. This mapping gives every initially held stock the same amount of information in units of prior precision.[38] The data technology gain raises every stock's private precision proportionally within an adopter. Appendix G gives this mapping and the restriction on factor management.

The technology advance changes price informativeness in general equilibrium. Adopters' stronger demand raises prices, moving other investors along their demand curves. As adopters trade on more precise signals, prices also reveal more information about payoffs. This changes how all investors learn from prices. The resulting changes in payoff uncertainty and in the response of beliefs to prices shift investors' demand curves and change their slopes.

Aggregating adopters' gains determines how much additional payoff information enters each stock's price. Investors combine this more precise price signal with their private signals. I add the resulting changes in model expected holdings to observed positions, set negative positions to zero and rescale within each stock to preserve its aggregate institutional holdings.

I aggregate the shifts in investors' demand caused by information gains to calculate the expected price change that clears each stock's fixed supply. Reported 90 percent confidence intervals are the 5th and 95th percentiles of 200 joint draws of demand coefficients using their estimated covariance, conditional on the production estimates, technology gain and

[38] Private-signal precision is the inverse variance of signal noise. Each data point has the stock's prior precision $\tau_\varepsilon(n)$, so $k_i(n)$ data points give signal precision $\tau_i(n) = k_i(n)\tau_\varepsilon(n)$.

other inputs.

7.2 Price Informativeness

Table 4 shows that average price informativeness rises by 3.1 percentage points from 11.8 percent. The gain is 5.2 points among stocks with initially informative prices. Doubling and tripling the advance raise the all-stock average by 6.1 and 8.9 points.

Table 4: Price Informativeness after a Data-Technology Advance

	(1)	(2)	(3)
	Productivity Increase (δ)		
	$\delta = 1$	$\delta = 2$	$\delta = 3$
Initial Informativeness (%)	11.8	11.8	11.8
Final Informativeness (%)	14.9	17.9	20.7
Gain (pp)	3.1	6.1	8.9

Notes. Equal-weighted price informativeness across stocks in the 2024Q4 counterfactual. Informativeness is the fraction of idiosyncratic payoff uncertainty resolved by prices. Initial and Final Informativeness are percentages. Gain is in percentage points. The productivity increase, δ, is expressed in multiples of the historical increment. The columns repeat, double and triple that increment at fixed research employment. Section 7.1 defines the advance and adopters.

The gains depend on how much new information reaches each stock and how informative its price is initially. Let $\Delta_\kappa(n)$ be the proportional rise in aggregate information, weighting adopters' gains by their initial contributions to the stock. Price-signal precision then rises by the factor $[1 + \Delta_\kappa(n)]^2$: stronger informed demand reduces the noise in the information conveyed by prices. When a price is initially very noisy, a proportional improvement reveals little additional information. When it is already highly informative, little uncertainty remains to resolve. The gain is therefore largest at an intermediate initial level. In the cross-section, gains rise up to initial informativeness of about 45 percent and then decline (Figure 6; Corollary C.5). Larger advances move this peak toward lower initial informativeness.

Small stocks gain more. The smallest size quintile gains 4.4 points on average, compared with 2.3 for the largest. Adopters supply 32.8 percent of small firms' aggregate private information and 20.5 percent of large firms', so their advances add proportionally more information to small firms. Table 5 relates the gains to stock characteristics. Conditional on the other characteristics, gains are larger for stocks with greater idiosyncratic volatility, analyst coverage and market beta.

Table 5: Price Informativeness and Crowding Out across Stock Characteristics

	(1) Change in Price Informativeness (pp)	(2) Crowding Out (%)
Size (ln ME)	−0.10	0.08***
	(0.14)	(0.02)
B/M (ln)	0.33***	−0.09***
	(0.09)	(0.02)
Op. Profitability	−0.39***	0.09***
	(0.09)	(0.02)
Asset Growth	0.23***	−0.11***
	(0.06)	(0.01)
Dividends / BE	−0.27***	0.00
	(0.09)	(0.02)
Market Beta	1.16***	−0.18***
	(0.08)	(0.01)
Idio. Vol.	1.25***	−0.25***
	(0.10)	(0.02)
Turnover (ln)	−0.29***	0.16***
	(0.09)	(0.01)
Firm Age (ln)	−0.01	0.00
	(0.08)	(0.01)
Analyst Coverage (ln)	1.39***	−0.03*
	(0.12)	(0.02)
Mean of Dependent Variable	5.25	1.53
Adjusted R^2	0.49	0.43
Stocks	1147	1933

Notes. Cross-sectional regressions after a productivity increase equal to the historical increment ($\delta = 1$). Column (1) uses stocks with positive initial price informativeness. The outcome is the gain in informativeness in percentage points. Column (2) uses the full stock sample. The outcome is the percentage decline in other institutions' total positions. A positive coefficient indicates a larger percentage reduction in positions. Characteristics follow Table 1. B/M and six-month average daily turnover enter in logs. Firm age in months and analyst coverage enter as log one plus their values. Dividends / BE is twelve-month dividends over book equity. Regressors are winsorized at the 1st and 99th percentiles and standardized across the full stock sample. Parentheses report standard errors. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

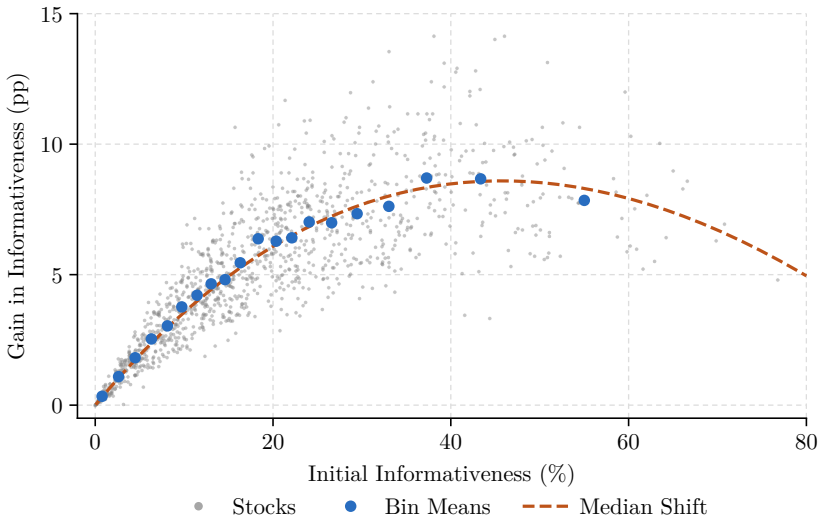


Figure 6: The Gain in Price Informativeness across Stocks

Notes. Gains in price informativeness after a productivity increase equal to its 2011–2024 increment ($\delta = 1$), for stocks with positive initial informativeness in 2024Q4. The horizontal axis is its initial level in percent. The vertical axis is its change in percentage points. Gray points are stocks. Bin Means averages twenty equal-count bins of initial informativeness. Section 7.1 defines the advance and adopters.

7.3 Crowding Out and Market Segmentation

7.3.1 Portfolio Reallocation

I measure portfolio reallocation by the percentage change in each group's total positions. A position is the fraction of a firm's outstanding shares held by an institution, measured in units of the firm's initial book equity.

Active adopters expand their positions by 15.4 percent, while other institutions reduce theirs by 1.85 percent (Table 6). Adopters' share of total positions rises from 10.7 to 12.3 percent.

Competition from adopters' private-information gains drives this reallocation. Lower payoff uncertainty raises their demand. Prices rise as investors compete for the fixed supply, reducing the expected excess payoff available to every holder. An investor is crowded out when the improvement in its forecast precision is too small to offset this decline (equation (20)).

To separate the two sources of information gains, I first hold price informativeness at its initial value. Adopters' new private information then reduces other institutions' positions by 1.95 percent. Allowing prices to become more informative offsets 0.11 percentage points of that decline through learning from prices. Adopters gain 16.3 percent in the first step; learning from prices reduces their gain by 0.90 points (Table 7, Panel (a)). Expected excess

Table 6: Crowding Out of Other Institutions

Panel (a) Position Changes (%)

		Adopters' Coverage					
	All	Low	2	3	4	High	High − Low
Adopters	15.4	4.9	9.4	12.9	13.7	20.2	15.4
	[11.3, 19.9]						[11.5, 19.3]
Other Institutions	−1.8	−0.7	−1.2	−1.5	−1.8	−2.3	−1.5
	[−2.4, −1.4]						[−1.9, −1.1]

Panel (b) Portfolio-Weight Changes (pp)

		Adopters' Coverage					
	All	Low	2	3	4	High	High − Low
Other Institutions		0.02	0.03	0.06	0.07	−0.17	−0.19
							[−0.23, −0.15]
N(Stocks)	1933	387	386	387	386	387	

Notes. Changes after a productivity increase equal to the historical increment ($\delta = 1$), starting from 2024Q4. Adopters are active asset managers with DS staff receiving new private information. Other Institutions are the remaining institutions. Within each coverage quintile, panel (a) reports each institution group's percentage change in total positions. Panel (b) reports changes in each stock quintile's share of the group's portfolio, in percentage points, valuing holdings at initial prices. Adopters' coverage is their private-signal precision divided by all holders' total posterior precision. Stocks are sorted into five bins. All pools the five groups. High minus Low compares the extreme quintiles. Brackets give 90 percent confidence intervals.

payoffs adjust to clear the fixed supply in both steps. Appendix G gives the decomposition, and Online Appendix Table OA5 reports alternative orders and larger advances.

Learning from prices adds the same precision to every investor's forecast, so it gives a larger proportional improvement to investors with less private information. It also erodes the trading value of existing private information as prices reveal more to competitors (Bonelli and Foucault, forthcoming). For other institutions as a group, the benefit is small relative to the stronger competing demand from adopters. Every institution type in this group loses positions in aggregate (Table 8).

Among adopters, hedge funds expand their positions by 35.3 percent, investment advisers by 12.0 percent and mutual funds by 6.5 percent. Hedge funds begin with more information, so the proportional technology gain adds more to their private precision.

Crowding out is strongest where adopters contribute more of the information supporting demand. I define adopter coverage as the share of the aggregate demand slope at fixed payoff forecasts attributable to adopters' private information. The denominator includes demand supported by the common prior and learning from prices as well as private signals. A given proportional gain in adopters' information shifts aggregate demand proportionally more where this share is high. Across coverage quintiles, adopters' positions rise by 4.9

Table 7: Private Information, Learning from Prices, and Institutional Demand

	(1) Private Information Gains	(2) More Informative Prices	(3) Total
Panel (a) Position Changes (%)			
Adopters	16.30	−0.90	15.41
	[11.94, 20.97]	[−1.13, −0.68]	[11.27, 19.86]
Other Institutions	−1.95	0.11	−1.85
	[−2.51, −1.43]	[0.08, 0.14]	[−2.38, −1.35]
Panel (b) Change in Elasticity Magnitude (%)			
Median across Informative Stocks	4.52	−7.01	−2.49
	[4.10, 4.96]	[−7.96, −6.15]	[−3.85, −1.19]

Notes. This table decomposes changes in positions and demand elasticity. Panel (a) reports percentage position changes: private-signal precision increases first at unchanged price informativeness, followed by the effect of more informative prices. Panel (b) reports median percentage changes in demand-elasticity magnitude across informative stocks, evaluated at initial prices and holdings. In the Private Information Gains column of Panel (b), the effect of learning from prices on the demand slope is held fixed. The learning contribution is the difference between the total and private-information median changes. Negative values mean less elastic demand. Brackets give 90 percent confidence intervals. Online Appendix Table OA5 reports alternative orders of the position decomposition and larger productivity increases.

Table 8: Crowding Out of Other Institutions by Institution Type

	Hedge Funds	Inv. Advisers	Mutual Funds	Pension Funds	Banks & Br.-Dealers	Insurers	Other
Adopters	35.3	12.0	6.5	–	–	–	–
Other Institutions	−7.2	−3.9	−1.2	−0.5	−2.2	−0.9	−2.8

Notes. Percentage changes in positions by institution type after a productivity increase equal to the historical increment ($\delta = 1$), starting from 2024Q4. Adopters, other institutions and the position adjustment follow Table 6. Each entry divides the indicated group's change in total positions across stocks by its initial total, multiplied by 100. Institution types follow Table 3; Other Types collects the remaining institutions. A dash indicates no adopters in that type.

percent in the least-covered stocks and 20.2 percent in the most-covered. Other institutions' positions fall by 0.7 and 2.3 percent, respectively (Figure 7(a)). Other institutions also shift portfolio weight away from the most-covered stocks. Valuing holdings at initial prices, their weight in the highest-coverage quintile falls by 0.17 percentage points, while their weight in the lowest rises by 0.02 points (Table 6, Panel (b)).

More-informed institutions receiving no private-information gain face larger losses. Figure 7(b) sorts these institutions by predicted information per dollar per stock, $\bar{k}_i/N_i$. The median decline in total positions rises from 1.8 percent in the lowest information quintile to 14.5 percent in the highest. The ordering is consistent with Corollary 1: for the same stock, a better-informed investor gains less proportionally from learning from prices and suffers a larger percentage loss when crowded out.

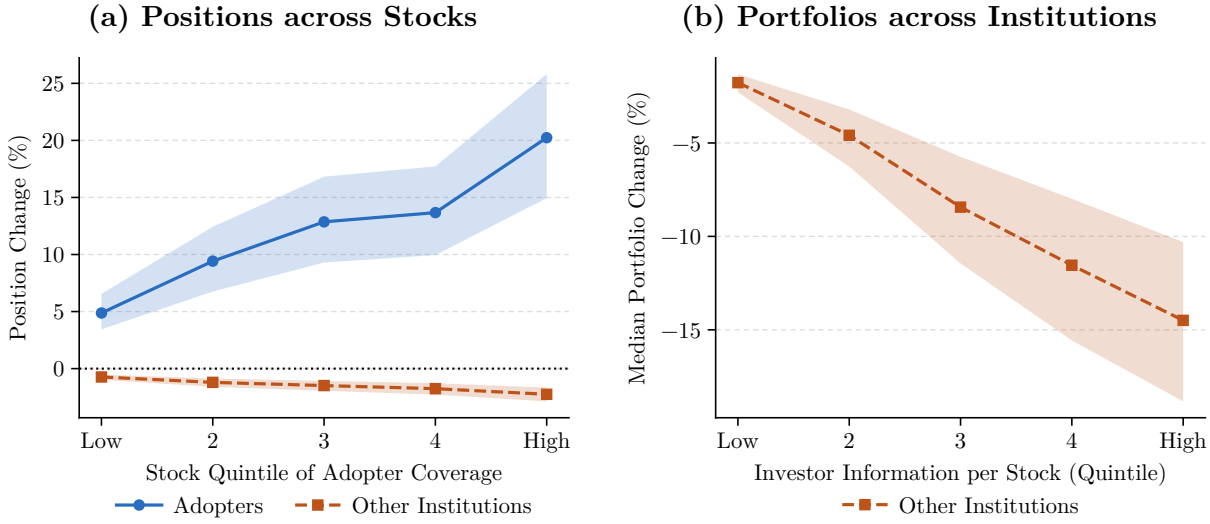


Figure 7: Crowding Out across Stocks and Institutions

Notes. Percentage changes in positions after a productivity increase equal to its 2011–2024 increment ($\delta = 1$). Panel (a) sorts stocks into five approximately equal-count groups by adopters' coverage and reports each institution group's percentage change in total positions within that stock group. Coverage and institution groups follow Table 6. Panel (b) sorts non-adopters by predicted information per dollar per stock, $\bar{k}_i/N_i$. For each investor, I divide the change in total positions across stocks by its initial total positions. Points give medians within information quintiles. Stock and investor groups stay fixed. Shading gives 90 percent confidence intervals.

Crowding out also varies with stock characteristics (Table 5). A one-standard-deviation increase in log firm size is associated with a 0.08 percentage-point greater reduction in other institutions' positions, and higher turnover with a 0.16-point greater reduction. Losses are smaller for stocks with greater idiosyncratic volatility, market beta and analyst coverage, conditional on the other characteristics. These stocks also gain more information from prices, consistent with learning cushioning crowding out.

7.3.2 Investor Bases across Stocks

A stock's investor base is the distribution of its holdings across institutions in the sample. I measure the distance between two bases as half the sum of the absolute differences in investors' ownership shares: the fraction of ownership that would have to change hands for the two bases to match. This is the dissimilarity index of Duncan and Duncan (1955), analogous to active share (Cremers and Petajisto, 2009).

The average distance rises from 53.12 to 53.60 percent, an increase of 0.48 percentage points (Table 9). Each pair of stocks receives equal weight. The increase reflects uneven reallocation toward adopters and crowding-out of non-adopters. Stocks where adopters expand more come to be owned by a different mix of institutions. Doubling and tripling the advance raise the distance by 0.90 and 1.28 points. The distance between the average investor bases of the highest- and lowest-coverage quintiles rises by 0.12 points from 22.63 percent, and by 0.41 and 0.75 points under the larger advances. Each stock receives equal weight within its quintile.

Table 9: Market Segmentation after the Advance

	(1)	(2)	(3)	(4)
	2024Q4	Change after the Advance (pp)		
Productivity Increase		$\delta = 1$	$\delta = 2$	$\delta = 3$
Panel (a) All Pairs of Stocks (%)				
Average Distance between Two Stocks	53.12	0.48	0.90	1.28
		[0.35, 0.62]	[0.66, 1.15]	[0.94, 1.62]
Panel (b) Average Investor Bases of the Most- and the Least-Covered Quintile (%)				
All 13F Institutions	22.63	0.12	0.41	0.75
		[0.08, 0.17]	[0.22, 0.60]	[0.47, 1.07]

Notes. Investor-base distances in the 2024Q4 counterfactual. The productivity increase, δ, is a multiple of its historical increment. The distance between two stocks is half the sum of absolute differences in investors' ownership shares, normalized by each stock's total holdings among sample institutions. Holdings are valued at initial prices. Panel (a) gives every pair of distinct stocks equal weight. Panel (b) compares the average investor bases of the highest- and lowest-coverage stock quintiles in Table 6, giving every stock equal weight within its quintile. Brackets give 90 percent confidence intervals for the changes after the advance.

7.4 Aggregate Demand and Equity Prices

Aggregate demand becomes less elastic for 46.5 percent of stocks (Table 10). These stocks represent 78.3 percent of those with informative prices and account for 19.3 percent of total book equity. Among informative stocks, aggregate elasticity magnitude falls by 2.5 percent at the median. The share of all stocks becoming less elastic falls to 43.3 percent under twice

the advance and 40.3 percent under three times the advance. I evaluate elasticities at initial prices and quantities to measure changes in the demand slope.

Table 10: Demand Elasticity and Equity Prices after the Advance

	(1)	(2)	(3)
	Change after the Advance		
Productivity Increase	$\delta = 1$	$\delta = 2$	$\delta = 3$
Panel (a) Demand Elasticity			
Uninformed Investor, Median Change (%)	−10.5	−19.1	−26.1
	[−11.7, −9.5]	[−20.9, −17.3]	[−28.4, −23.9]
Aggregate, Median Change (%)	−2.5	−3.7	−3.9
	[−3.9, −1.2]	[−6.2, −1.3]	[−7.1, −0.5]
Less Elastic, Share of Stocks (%)	46.5	43.3	40.3
	[38.4, 52.3]	[33.9, 49.9]	[30.7, 46.8]
Less Elastic, Share of BE (%)	19.3	17.5	15.8
	[9.3, 36.9]	[8.0, 33.4]	[7.0, 30.2]
Panel (b) Equity Prices			
Price Change, VW (%)	2.9	5.5	7.9
	[2.0, 4.0]	[3.8, 7.7]	[5.5, 11.0]
Price Change, EW (%)	1.4	2.8	4.1
	[1.0, 2.1]	[1.9, 4.0]	[2.8, 5.8]

Notes. Changes in elasticity magnitudes and equity prices from in the counterfactual. The productivity increase, δ, is a multiple of its historical increment. Uninformed Investor refers to an investor without private information. Aggregate refers to total institutional demand. Median Change is the median percentage change in elasticity magnitude across informative stocks, evaluated at initial prices and holdings. Negative values mean less elastic demand. Less Elastic, Share of Stocks is the percentage of all stocks with a decline in aggregate elasticity magnitude. Share of BE weights this indicator by book equity. Price Change follows the anticipated repricing calculation in Appendix G; VW and EW are market-equity-weighted and equal-weighted means across stocks. Brackets give 90 percent confidence intervals.

The decline reflects that the learning from prices effect outweighs the direct effect of better private information on the demand slope. Adopters' better private signals make them more willing to trade against a price change. More informative prices reduce uncertainty for all investors, but they also make a price increase stronger evidence of favorable payoff news. Investors raise their valuations and sell less in response. Aggregate demand becomes less elastic when learning from prices effect dominates.

Holding the contribution of learning from prices to the demand slope fixed, new private information raises aggregate elasticity magnitude by 4.5 percent at the median across informative stocks. Allowing prices to become more informative lowers this median change by 7.0 percentage points, giving a net decline of 2.5 percent (Table 7, Panel (b)). As the advance grows, the direct private-information gain dominates for more stocks, reducing the share that become less elastic (Corollary C.3).

Figure 8(a) shows larger elasticity declines where adopters supply a larger share of aggregate private information. Across the lowest and highest information-share groups, the mean decline in elasticity magnitude rises from 0.5 to 4.9 percent. Both the direct private-information effect and the effect of learning from prices grow with adopters' share, but the learning effect grows more.

Across groups sorted by initial price informativeness, the largest mean decline is 4.6 percent, in the group with average initial informativeness of 12.2 percent (Panel (b)). In the group with the most informative prices, the direct private-information effect dominates and demand becomes 3.0 percent more elastic on average.

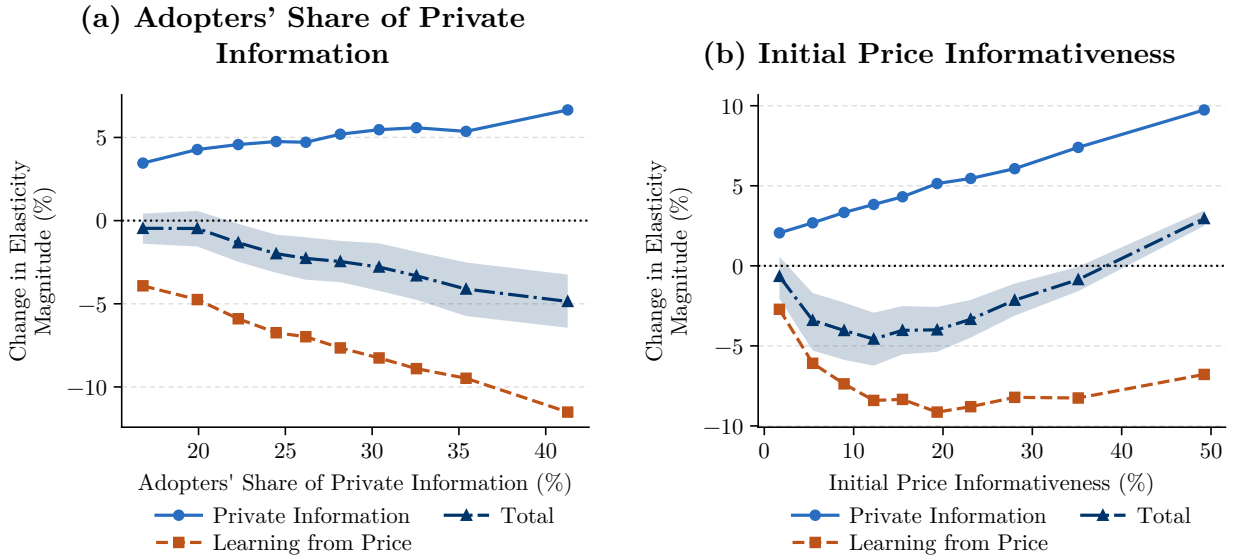


Figure 8: Private Information, Learning from Prices, and Aggregate Demand Elasticity

Notes. Percentage changes in aggregate demand-elasticity magnitude after a productivity increase equal to its 2011–2024 increment ($\delta = 1$), evaluated at initial prices and holdings for informative stocks. Panel (a) sorts stocks by adopters' share of aggregate private information. Panel (b) sorts by initial price informativeness. Both panels give equal-weighted means within ten bins. Private Information holds the contribution of learning from prices to the demand slope fixed. Learning from Price gives the additional effect of more informative prices. Total is their sum. Negative changes mean less elastic demand. Shading around Total gives 90 percent confidence intervals.

For an investor without private information, the median decline in demand-elasticity magnitude across informative stocks is 10.5 percent. The declines reach 19.1 and 26.1 percent under the larger advances.

The same information gain can raise demand at initial prices and reduce its price elasticity. Better private signals and learning from prices lower payoff uncertainty, so investors require less compensation to absorb the fixed supply. Expected equity prices rise by 2.9 percent value-weighted and 1.4 percent equal-weighted, lowering expected excess payoffs until desired holdings match that supply. Appendix G gives the price calculation. Doubling and

tripling the advance raise value-weighted prices by 5.5 and 7.9 percent.

8 Conclusion

Asset managers with larger shares of data-science labor show greater gains in stock-picking ability around technology breakthroughs. I measure investors' private information from holdings' forecasting power for future payoffs beyond public information and prices. Combining U.S. institutional holdings and employment records, I estimate how data scientist teams produce information and how private information and learning from prices shape investor demand.

A counterfactual advance equal to the estimated 2011–2024 increase in data-science productivity raises average price informativeness by 26 percent. Small stocks gain more because adopters supply a larger share of the private information about these firms. Adopters' better forecasts strengthen their demand, raising prices and crowding out other institutions. These institutions shift portfolio weight toward stocks less affected by the advance, making investor bases more distinct. Learning from prices reduces their payoff uncertainty and partly offsets their aggregate holdings losses.

Holding price informativeness fixed, better private forecasts reduce payoff uncertainty and make adopters' demand more elastic. More informative prices also lead all holders to revise their payoff forecasts more in response to price changes. A higher price signals better fundamentals, so investors sell less. This learning effect dominates for 46.5 percent of stocks, making aggregate demand less elastic. The distribution of information processing capacities therefore shapes both the information in prices and how investors respond to price changes.

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Appendix

A Data Construction and the Breakthrough Registry

A.1 Holdings and Institution Types

I use the holdings, prices, and fundamentals described in Section 2.2 and stock characteristics from Jensen et al. (2023). Each filer receives one fixed investor type based on its available regulatory history and the business behind its own 13F portfolio. Starting with Bushee's entity types and Thomson's codes, I use fund links and Form ADV to distinguish mutual funds and hedge funds from other advisers. Seven rules then use regulatory identifiers and the filer's business to refine the classification. Asset managers comprise investment advisers, hedge funds, mutual funds, and public pension and sovereign funds; banks, insurers, and broker-dealers form the comparison group. Online Appendix OA1.1 gives the source hierarchy, rules, and counts. Appendix F uses the alternative classification of Koijen and Yogo (2019) plus hedge funds.

A.2 Employment Measurement

Revelio Labs' 2024 release provides 56,785,614 U.S. positions at matched employers and companies in their corporate families for measuring employment over 2009Q1–2024Q4. Identifiers, names, and independent reviews link 12,080 filers to 11,033 Revelio companies. The reviews use no holdings, performance, or employee counts. A worker counts once per investor-quarter, with Revelio's sampling weight, when a position overlaps the quarter.

Job titles determine functions. DS labor includes data scientists working on statistical forecasts, machine learning, text, computer vision, and alternative-data signals, as well as all quantitative researchers, including execution, risk, and pricing research. Generic analytics and financial-data analysis fall outside the definition. Filers sharing an employer divide its staff in proportion to 13F assets, or equally when none has assets. The matched employment panel includes all 13F institutions linked to Revelio in the corresponding quarter. Online Appendix OA1.5 describes the samples for employment comparisons and estimation, and Online Appendix OA1.2 gives the matching, title, and counting protocols. Online Appendix Table OA1 reports employment trends by institution type.

Breakthrough waves and stock groups. A candidate must introduce a new method or model, appear in Epoch AI's notable-model database, and rank in the top 0.01 percent of its field and year by citations. Models made public within eight quarters of a wave's first model

belong to that wave. The wave is dated by its highest-ranked model: the event falls in the first calendar quarter ending at least two months after that model became public with code or trained models. Table A1 summarizes the seven events. Online Appendix OA2.5 lists all fifteen models and documents their dates.

Store-based retailers have SIC codes 5200–5999, excluding 5961, 5962, and 5963 (mail-order, vending-machine, and direct-selling businesses). The group includes restaurants. Each stock's latest nonmissing SIC in the cleaned stock panel fixes its classification across events. SIC comes from JKP, with CCM/CRSP codes filling missing values. Online Appendix OA1.4 describes the text-heavy, complex-text, and satellite-covered stock groups.

Table A1: Data-Technology Breakthrough Waves

Technology	Release	Public date	Event quarter	Window
Language processing	Word2Vec	2013-07-29	2013Q3	$[-6, 6]$
	BERT	2018-11-02	2019Q1	$[-6, 6]$
Computer vision	AlexNet	2012-10-13	2012Q4	$[-6, 6]$
	ResNet	2016-02-03	2016Q2	$[-6, 6]$
	Mask R-CNN	2018-01-22	2018Q1	$[-6, 6]$
	ViT	2020-10-22	2020Q4	$[-6, 6]$
	SAM	2023-04-05	2023Q2	$[-6, 6]$

Notes. The table lists the seven events in Section 2.4. Public date is the day the method became public with code, trained models or a demonstrated result (Online Appendix OA2.5); for BERT, it is the public announcement of its code and models. The event quarter is the first calendar quarter ending at least two months later. Models released within eight quarters of a wave's first model belong to that wave. Its most prominent model determines the wave's date. Windows are measured in quarters relative to the event quarter, which also begins the post period. Online Appendix Table OA3 lists all fifteen models.

A.3 Data Technology and Fundamental Forecasting

Data technology changes the cost of forecasting firms' payoffs. Investors combine alternative data with financial statements and disclosures, and DS teams develop methods to turn these inputs into investment signals (Green and Zhang, 2024).

Language models extract information from disclosures and earnings calls. Filing tone is informative about unexpected earnings (Loughran and McDonald, 2011), changes in filings predict earnings and profitability (Cohen et al., 2020), and FinBERT improves information extraction from calls (Huang et al., 2023). The language-processing analysis therefore uses firms with extensive disclosures, measured as in Section 2.4.

Satellite images make customer traffic observable before retailers announce earnings. Orbital Insight processes parking-lot images into car counts (Kang et al., 2021); parking-lot fill rates predict same-store sales and announcement returns (Katona et al., 2025). Real-

time sales measures also explain earnings surprises (Froot et al., 2017). These applications motivate the store-based retailers and restaurants used in the computer-vision analysis.

Asset managers describe the same forecasting goals for DS teams, including forecasts of earnings and profitability (Chen and Zhou, 2023; Cui and Shahrabani, 2023). These applications produce information about firm payoffs, the object of private signals in the model, and complement the extraction of other investors' information from prices studied by Farboodi and Veldkamp (2020).

Online Appendix OA2 documents the institutional applications and improvements in models and computing costs. Its breakthrough registry gives public-release dates and the grouping of fifteen models into seven events.

B Stock-Picking Ability around Breakthroughs: Specification and Estimates

Datasets. For event s, I follow asset managers with an employment observation in the four preceding quarters, using the sample in Online Appendix OA1.5. Investors without DS staff in those quarters form the comparison group, including those that hire DS staff later. BERT's exposed stocks and employment eligibility are evaluated at its public release in 2018Q4; DS labor shares and event time use 2019Q1, the event quarter. Investor-events observed in only one quarter are absorbed by the investor-by-event effects and dropped. Tables A2 and A3 report the resulting sample sizes.

Windows and overlap. Each window runs from six quarters before the event to six after and ends before the next event of the same technology and the sample endpoint in 2024Q4. The two NLP events are 22 quarters apart, with separate windows. Consecutive CV events are 7 to 14 quarters apart. Later pre-event quarters can therefore fall in an earlier event's post-event window: six quarters for Mask R-CNN, two for ViT, and three for SAM. The AlexNet and ResNet windows are separate. An investor-quarter in two windows enters both datasets with the corresponding investor-by-event effects. Clustering by investor accounts for dependence between these entries.

Public availability can precede the event quarter. BERT became public in its 2018Q4 reference quarter and ResNet in its 2016Q1 reference quarter. Moreover, the measure of stock-picking ability uses four subsequent earnings quarters, so leads -4 through -1 include earnings realized in or after the event quarter. The joint lead test restricted to -6 and -5, with the primary investor clustering, gives $p = 0.473$ for NLP and $p = 0.325$ for CV.

Pooled specification. The pooled regression replaces the quarter indicators of equation (2) with the indicator $\text{Post}_{s,t} = \mathbf{1}\{t \geq t_s\}$ for the event quarter and the six quarters after it:

$$\text{SP}_{i,s,t} = \beta \, \text{Post}_{s,t} \times \text{DS}_{i,s} + \alpha_{i,s} + \lambda_{s,t} + u_{i,s,t}. \tag{A1}$$

The coefficient β compares stock-picking ability on exposed stocks in the six pre-event quarters with the seven post-event quarters, per standard deviation of the pre-event DS labor share. The event-study coefficients compare each quarter with quarter -1. In columns (3) and (5) of Tables A2 and A3, $\text{DS}_{i,s}$ is an indicator for employing DS labor in the four pre-event quarters. Column (2) adds six other pre-event staffing shares, each standardized within event and interacted with $\text{Post}_{s,t}$: investment staff; data engineering; administration and operations; finance and accounting; human resources; and IT support (Online Appendix OA1.2).

Weights. Following Wing et al. (2024), DS employers receive weight one and event s's comparison investors receive $Q_s = (N_s^T/N^T)/(N_s^C/N^C)$. Here N_s^T and N_s^C count DS employers and comparison investors in event s, and N^T and N^C sum them across the technology's events. This gives every event the same ratio of comparison investors to DS employers. Weights are recomputed for each column's sample and also used in Figure 2. In a balanced panel, the employer-indicator coefficient averages event-specific coefficients using each event's share of all DS employers; the estimation samples approximate these weights.[39]

Estimates. Column (1) of Tables A2 and A3 gives the pooled counterpart of the event study: 0.906×10^{-3} (s.e. 0.279×10^{-3}) for NLP and 0.545×10^{-3} (s.e. 0.112×10^{-3}) for CV. The pre-event coefficients are jointly insignificant ($p = 0.48$ and 0.45; $p = 0.73$ in column (2) for both technologies). The employer-indicator estimates in column (3) are positive, with insignificant pre-event coefficients for NLP ($p = 0.39$) and jointly significant coefficients for CV ($p < 0.01$). Columns (4) and (5) use complex-text stocks and satellite-covered retailers. Their larger estimates accompany jointly significant pre-event coefficients in all four columns ($p < 0.01$).

Clustering by both investor and calendar quarter gives column (1) standard errors of 0.390×10^{-3} for NLP and 0.134×10^{-3} for CV. The pooled post-event coefficients remain significant at the 5 percent level. The joint pre-event p-values are 0.002 for NLP and 0.615 for CV, so the NLP pre-event test is sensitive to the clustering choice.

[39] An event's actual weight is its share of the weighted sum of squared regressors after removing fixed effects. Its difference from the DS-employer share is at most 0.05 for NLP and 0.011 for CV in columns (1) and (3).

Table A2: Stock-Picking Ability around Natural-Language-Processing Breakthroughs

Dep. Var.:	SP on Text-Heavy Stocks			SP on Complex-Text Stocks	
	(1)	(2)	(3)	(4)	(5)
Different Measures of DS Labor:	Share of DS Labor	Share of DS Labor (w. Other Staff)	Employs DS Labor (0/1)	Share of DS Labor	Employs DS Labor (0/1)
Post × DS Labor	0.906***	0.822**	3.113**	2.125***	9.388***
	(0.279)	(0.325)	(1.275)	(0.325)	(1.101)
Other Staff Controls		Y			
WFH Control Weights	Y	Y	Y	Y	Y
Investor × Event FE	Y	Y	Y	Y	Y
Event × Quarter FE	Y	Y	Y	Y	Y
Joint p, Other Staff		0.20			
Mean of Dep. Var.	−0.31	−0.31	−0.31	0.39	0.39
N(Observations)	19,971	19,971	19,971	27,456	27,456
N(Investors)	1,419	1,419	1,419	1,814	1,814

Notes. The table reports coefficients on Post × DS Labor in equation (A1) across the Word2Vec and BERT events. Coefficients, standard errors and dependent-variable means are multiplied by 10^3. The dependent variable is stock-picking ability on text-heavy stocks in columns (1)–(3) and complex-text stocks in columns (4)–(5). Text-heavy stocks are in the top decile of pre-event filing length; complex-text stocks are in the top decile of the share of Loughran–McDonald complexity words in 10-K filings. Both measures are adjusted for size and industry (Online Appendix OA1.4), and investor-quarters require at least ten such stocks. Column (2) adds pre-event employment shares for investment staff, data engineering, administration and operations, finance and accounting, human resources, and IT support, each standardized within event and interacted with Post. Online Appendix Table OA2 reports all six coefficients. Other definitions follow Table A3. DS employers have weight one, and comparison investors are reweighted to give every event the same ratio of comparison investors to DS employers (Wing et al., 2024). The pre-event coefficients of the event study are jointly insignificant in columns (1)–(3) and jointly significant at the 1 percent level in columns (4) and (5). Standard errors clustered by investor are in parentheses. *, **, and *** denote 10, 5, and 1 percent significance.

Table A3: Stock-Picking Ability around Computer-Vision Breakthroughs

Dep. Var.:	SP on Store-Based Retailers			SP on Satellite-Covered Retailers	
	(1)	(2)	(3)	(4)	(5)
Different Measures of DS Labor:	Share of DS Labor	Share of DS Labor (w. Other Staff)	Employs DS Labor (0/1)	Share of DS Labor	Employs DS Labor (0/1)
Post × DS Labor	0.545***	0.555***	1.827***	1.984***	8.538***
	(0.112)	(0.125)	(0.432)	(0.272)	(0.968)
Other Staff Controls		Y			
WFH Control Weights	Y	Y	Y	Y	Y
Investor × Event FE	Y	Y	Y	Y	Y
Event × Quarter FE	Y	Y	Y	Y	Y
Joint p, Other Staff		0.17			
Mean of Dep. Var.	4.07	4.07	4.07	13.39	13.39
N(Observations)	48,889	48,889	48,889	87,335	87,335
N(Investors)	1,939	1,939	1,939	3,216	3,216

Notes. The table reports coefficients on Post × DS Labor in equation (A1) across the five computer-vision events. Coefficients, standard errors and the dependent-variable mean are multiplied by 10^3. The dependent variable, SP, is stock-picking ability, defined in equation (1), on store-based retailers in columns (1)–(3) and on the satellite-covered retailers of Katona et al. (2025) in columns (4)–(5). Investor-quarters require at least ten such stocks, or three for the satellite-covered retailers. Post marks the event quarter and the six quarters after it; the window is $[-6, 6]$. In the row label, DS Labor refers to the measure named in the column header. Share of DS Labor is pre-event DS labor per 100 employees, standardized within event; Employs DS Labor indicates DS staff in the four pre-event quarters. Column (2) adds pre-event employment shares for investment staff, data engineering, administration and operations, finance and accounting, human resources, and IT support, each standardized within event and interacted with Post. Joint p reports the p-value for a joint test of these six coefficients; Online Appendix Table OA2 reports them individually. Comparison investors have no DS staff in the four pre-event quarters. WFH Control Weights follow Wing et al. (2024): DS employers have weight one, and comparison investors are reweighted to give every event the same ratio of comparison investors to DS employers (Appendix B). Means are unweighted. The five pre-event coefficients of the corresponding event study (quarters −6 to −2) are jointly insignificant in columns (1) and (2) and jointly significant at the 1 percent level in columns (3)–(5). Standard errors clustered by investor are in parentheses. *, **, and *** denote 10, 5, and 1 percent significance.

C Proofs and Further Theory

Unless stated otherwise, expectations condition on the public date-0 primitives and information allocations; $\mathbb{E}_0$ makes this conditioning explicit. Write $\tilde{\tau}(n) = \tau_\varepsilon(n) + \tau_p(n)$, $\hat{\tau}_i(n) = \tilde{\tau}(n)+\tau_i(n)$ and $\eta_i(n) = \varepsilon(n)+\bar{u}_i(n)$, where mean signal noise has precision $\tau_i(n)$. Matrices and portfolio adjustments follow Lemma 2. Let $B = [\beta, \phi]$, with row $v_n^\top = (\beta(n), \phi(n))$, and $\bar{g}_i^a = \mathbb{E}_0[g_i^a]$ for $a \in \{\beta, \phi\}$. Without a mandate, $B = \beta$, $v_n = \beta(n)$ and $G_i = 1+\beta^\top \hat{\Omega}_i^{-1}\beta$.

Investor-specific matrices run over $\mathcal{N}_i$, and N in their order bounds denotes $N_i = |\mathcal{N}_i|$. Cross-investor matrix integrals use zeros outside each investor's universe. Scalar primes mark counterfactual values; n' denotes another stock.

Assumption C.1 (Primitives)**.** (i) The factor, the payoff innovations, the noise trades and the noise of every data point are jointly normal and mutually independent, and $\tau_\varepsilon(n)$ and $\sigma_z^2(n)$ are bounded above and away from zero uniformly in n. The prices that Section 7 treats as uninformative are the limit $\sigma_z(n) \to \infty$ of this bound, in which a price reveals nothing whatever $\kappa(n)$.

(ii) Investors are price takers in a continuum, the noise of their signals averages out exactly, $\int \gamma_i^{-1}\tau_i(n)\bar{u}_i(n)\, di = 0$, aggregate risk tolerance and aggregate information are finite, and the counts $k_i(n)$ are bounded uniformly in i and n. Each stock has positive aggregate risk tolerance among its eligible holders, and the aggregate matrix integrals below are finite.

(iii) The loadings $\beta(n)$ and the attribute $\phi(n)$ are bounded. Within each investor's expanding universe, $N_i^{-1}\sum_{n\in\mathcal{N}_i}\beta(n)^2$ is bounded away from zero. With a mandate, the smallest eigenvalue of $N_i^{-1}\sum_{n\in\mathcal{N}_i} v_n v_n^\top$ is also bounded away from zero. These pervasiveness conditions keep each exposure direction represented as the universe expands. They also hold with weights bounded above and away from zero.

Proof of Lemma 1. Prices reveal $\eta_p(n)$ under the conjecture. Conditional on $\varepsilon(n)$, private and price signals have independent normal errors with precisions $\tau_i(n)$ and $\tau_p(n)$. Normal updating adds these to the prior precision and weights each signal by its share of the sum, giving equation (11). Independence across stocks gives diagonal posterior covariance. Neither signals nor prices load on F, so its posterior remains $N(0,1)$ and independent of the innovations. □

Proof of Lemma 2. By Lemma 1, the payoff vector is normal given $\mathcal{I}_i$, with mean $\mu+\mathbb{E}[\varepsilon \mid \mathcal{I}_i]$ and covariance $\beta\beta^\top + \hat{\Omega}_i$. The objective (5) therefore equals $\gamma_i R_f W_{1i} + \gamma_i q_i^\top m_i - \frac{\gamma_i^2}{2} q_i^\top M_i q_i$,

where m_i stacks the $m_i(n)$. With M_i positive definite it is strictly concave, and its first-order condition gives the unique solution $q_i = \gamma_i^{-1} M_i^{-1} m_i$. With a mandate, $M_i = \hat{\Omega}_i + B\ \mathrm{diag}(1, -\delta_i^{\phi}\gamma_i^{-2})\, B^{\top}$ and $G_i = \mathrm{diag}(1, -\gamma_i^2/\delta_i^{\phi}) + B^{\top}\hat{\Omega}_i^{-1}B$, so the Woodbury identity gives

$$M_i^{-1} = \hat{\Omega}_i^{-1} - \hat{\Omega}_i^{-1} B\, G_i^{-1} B^{\top} \hat{\Omega}_i^{-1}.$$

Row n of $\gamma_i^{-1} M_i^{-1} m_i$ is $\gamma_i^{-1}\hat{\tau}_i(n)[m_i(n) - v_n^{\top} G_i^{-1} B^{\top} \hat{\Omega}_i^{-1} m_i]$, which is equation (12). Without a mandate, $M_i = \hat{\Omega}_i + \beta\beta^{\top}$, and the Sherman–Morrison formula gives the same form with $B = \beta$ and $G_i = 1 + \beta^{\top}\hat{\Omega}_i^{-1}\beta$. For the exposures, $B^{\top}\hat{\Omega}_i^{-1}B = G_i - \mathrm{diag}(1, -\gamma_i^2/\delta_i^{\phi})$ gives

$$B^{\top} q_i = \gamma_i^{-1}\Big(I - B^{\top}\hat{\Omega}_i^{-1} B\, G_i^{-1}\Big) B^{\top}\hat{\Omega}_i^{-1} m_i = \gamma_i^{-1}\,\mathrm{diag}\left(1, -\gamma_i^2/\delta_i^{\phi}\right) \begin{pmatrix} g_i^{\beta} \\ g_i^{\phi} \end{pmatrix},$$

whose two rows are the exposure identities. Without a mandate the first row holds alone. □

The portfolio adjustments are realized at the trading date, so $\mathbb{E}[g_i^a \mid \mathcal{I}_i] = g_i^a$. Their aggregates and date-0 means are

$$H_a^{\mathrm{real}}(n) = \int \gamma_i^{-1}\hat{\tau}_i(n) g_i^a\, di, \qquad H_a(n) = \mathbb{E}_0[H_a^{\mathrm{real}}(n)] = \int \gamma_i^{-1}\hat{\tau}_i(n)\bar{g}_i^a\, di, \tag{A2}$$

for $a \in \{\beta, \phi\}$, integrating over eligible holders. Deterministic weights and integrability give the second equality. The benchmark substitutes these means for the realized aggregates in market clearing, subject to the finite-stock conditions below.

Proof of Proposition 1. Using $\hat{\tau}_i(n) m_i(n) = \hat{\tau}_i(n)[\mu(n) - R_f p(n)] + \tau_i(n)[\varepsilon(n) + \bar{u}_i(n)] + \tau_p(n)\eta_p(n)$ and the exact averaging of private signal noise, replace the realized aggregates in market clearing by their date-0 means. The resulting benchmark equation is

$$\mathcal{T}(n)\Big[\mu(n) - R_f p(n)\Big] + \kappa(n)\varepsilon(n) + z(n) + \bar{\gamma}^{-1}\tau_p(n)\eta_p(n) - \beta(n)H_{\beta}(n) - \phi(n)H_{\phi}(n) = 1.$$

Since $\kappa\varepsilon + z = \kappa\eta_p$, the coefficient on η_p is $(\kappa + \bar{\gamma}^{-1}\tau_p)/\mathcal{T} = \kappa/(\mathcal{T} d_p)$. Also,

$$\mathcal{T}\, d_p = \frac{\Big[\bar{\gamma}^{-1}\tau_{\varepsilon} + (\bar{\gamma}^{-1}\tau_p + \kappa)\Big]\kappa}{\kappa + \bar{\gamma}^{-1}\tau_p} = \bar{\gamma}^{-1}\tau_{\varepsilon} d_p + \kappa = \mathcal{T}_p.$$

This proves equation (13). At $\kappa > 0$, the positive coefficient makes price equivalent to η_p with precision κ^2/σ_z^2. At $\kappa = 0$, $\tau_p = 0$, $d_p = 1$ and $R_f p = \mu - [1 + \beta H_{\beta} + \phi H_{\phi} - z]/\mathcal{T}$.

To determine the mean adjustments, let $y(n) = \beta(n)H_{\beta}(n) + \phi(n)H_{\phi}(n)$. Substitute

$\mathbb{E}_0[m_i(n)] = [1 + y(n)]/\mathcal{T}(n)$ into their system and aggregate:

$$y = \Xi \operatorname{diag}(\mathcal{T})^{-1}(\mathbf{1} + y), \qquad \Xi = \int \gamma_j^{-1}\left(\hat{\Omega}_j^{-1} - M_j^{-1}\right) dj,$$

where Woodbury gives $\hat{\Omega}_j^{-1} B G_j^{-1} B^\top \hat{\Omega}_j^{-1} = \hat{\Omega}_j^{-1} - M_j^{-1}$. Positive definiteness of M_j implies $x^\top \Xi x < \sum_n \mathcal{T}(n) x(n)^2$ for nonzero x. Thus $\Xi \operatorname{diag}(\mathcal{T})^{-1}$ is similar to a symmetric matrix with all eigenvalues below one. The linear system and the mean portfolio adjustments are unique. □

Finite-stock approximation. Let $\mathbf{g}_i = (g_i^\beta, g_i^\phi)^\top$ and $\bar{\mathbf{g}}_i = \mathbb{E}_0[\mathbf{g}_i]$, dropping the mandate component when $\delta_i^\phi = 0$. The benchmark price of each stock depends only on its own shocks, making $m_i(n) - \mathbb{E}_0[m_i(n)]$ independent across stocks. Hence

$$\begin{aligned} \mathbf{g}_i - \bar{\mathbf{g}}_i &= G_i^{-1} \sum_{n \in \mathcal{N}_i} \hat{\tau}_i(n) v_n \Big(m_i(n) - \mathbb{E}_0[m_i(n)]\Big), \\ \operatorname{Cov}_0(\mathbf{g}_i) &= G_i^{-1} \left[\sum_{n \in \mathcal{N}_i} \hat{\tau}_i(n)^2 \operatorname{Var}_0(m_i(n)) v_n v_n^\top\right] G_i^{-1}. \end{aligned} \tag{A3}$$

Suppose posterior precisions, loadings and $\operatorname{Var}_0(m_i(n))$ are uniformly bounded over the expanding universe. For $\delta_i^\phi \le 0$, pervasiveness within that universe gives $\lambda_{\min}(G_i) \ge c_i N_i$ for some $c_i > 0$. The matrix in brackets is then $O(N_i)$ and $G_i^{-1} = O(N_i^{-1})$, so $\mathbb{E}_0\|\mathbf{g}_i - \bar{\mathbf{g}}_i\|^2 = O(N_i^{-1})$. Thus the centered adjustments are $O_p(N_i^{-1/2})$. Their means may themselves depend on the number of stocks.

The variance bound holds for a fixed population trading a common universe with fixed, finite, positive aggregate risk tolerance. With the primitive bounds, equation (15) bounds $\mathcal{T}_p(n)$ away from zero, keeping price variance and the variance of $m_i(n)$ bounded. For restricted universes, a sufficient additional condition is a uniform positive lower bound on each stock's participating risk tolerance, or directly on $\mathcal{T}_p(n)$. Pervasiveness must hold within each investor's universe.

At the benchmark price, the clearing error from keeping realized individual portfolios is exactly

$$\begin{aligned} \int q_i(n)\, di + z(n) - 1 = &- \beta(n)\Big[H_\beta^{\text{real}}(n) - H_\beta(n)\Big] \\ &- \phi(n)\Big[H_\phi^{\text{real}}(n) - H_\phi(n)\Big]. \end{aligned} \tag{A4}$$

It has date-0 mean zero. The dispersion of each aggregate is bounded by

$$\Big\|H_a^{\text{real}}(n) - H_a(n)\Big\|_{L^2} \le \int \gamma_i^{-1} \hat{\tau}_i(n) \Big\|g_i^a - \bar{g}_i^a\Big\|_{L^2} di.$$

Uniform rate bounds across eligible holders, growing minimum portfolio breadth and uniformly bounded aggregate risk tolerance make the clearing error vanish at the benchmark price. These are bounds on quantity errors. Convergence of exact finite-stock equilibrium prices also depends on market depth, since small quantity errors can require large price adjustments when demand is sufficiently inelastic.

In an exact finite-stock equilibrium, realized portfolio adjustments enter the full price vector and its signal covariance. An investor with $K_i = 0$ holds a public-information-measurable portfolio and has an exact zero information moment. Public-component invariance also follows from conditioning on that price vector. Under the benchmark's diagonal public posterior, the stock-by-stock zero and scalar attenuation of Theorem 1 apply. Appendix H treats finitely many investors, a separate issue.

Corollary C.1 (The own-price slope)**.** *Investor i's own-price slope for stock n is*

$$-R_f\gamma_i^{-1}[\tau_i(n) + \tau_\varepsilon(n)d_p(n)][1 - h_i(n)],$$

where $h_i(n) = \hat{\tau}_i(n)v_n^\top G_i^{-1}v_n$ is the exposure attenuation of Theorem 1. The exposure term vanishes when $\beta(n) = \phi(n) = 0$ and, for $\delta_i^\phi \le 0$, is of relative order $1/N$. Across stocks, exposure terms form a matrix of rank at most one, or two with a mandate.

Proof of Proposition 2 and Corollary C.1. Fix information allocations, private signals and other prices. At $\kappa(n) > 0$, a unit rise in $R_f p(n)$ raises the inferred price signal by $\mathcal{T}_p(n)/\kappa(n)$. Using $\tau_p\mathcal{T}_p/\kappa = \tau_p + \tau_\varepsilon(1 - d_p)$ gives

$$\frac{\partial m_i(n)}{\partial p(n')} = -\mathbf{1}\{n = n'\}R_f\frac{\tau_i(n) + \tau_\varepsilon(n)d_p(n)}{\hat{\tau}_i(n)}.$$

This also holds at $\kappa(n) = 0$, where $\tau_p(n) = 0$ and $d_p(n) = 1$. Precisions and M_i stay fixed, so Woodbury gives

$$\frac{\partial q_i(n)}{\partial p(n')} = -R_f\gamma_i^{-1}\Big[\mathbf{1}\{n = n'\} - \hat{\tau}_i(n)v_n^\top G_i^{-1}v_{n'}\Big]\Big[\tau_i(n') + \tau_\varepsilon(n')d_p(n')\Big].$$

For distinct stocks with $\hat{\tau}_i(n')v_{n'} = \hat{\tau}_i(n)v_n$, subtracting entry (n', n) from (n, n) cancels the exposure terms, proving equation (16). Appendix F gives the corresponding empirical comparison restrictions. Integrating its coefficients yields $\kappa + \bar{\gamma}^{-1}\tau_\varepsilon d_p = \mathcal{T}_p$. The own-price aggregate slope also includes the exposure terms. Benchmark prices, which clear using mean adjustments, give $\partial p/\partial z = 1/(R_f\mathcal{T}_p)$.

The diagonal gives the corollary. For $\delta_i^\phi \le 0$, precision bounds and pervasiveness imply $\lambda_{\min}(G_i) \ge c_i N_i$, hence $h_i(n) = O(1/N_i)$. The exposure matrix factors through B and

has rank at most its number of columns. Finally, the payoff-forecast response $\tau_p \mathcal{T}_p/(\kappa\hat{\tau}_i)$, averaged with weights $\gamma_i^{-1}\hat{\tau}_i/\mathcal{T}$, equals $1 - d_p$; it is zero at $\kappa = 0$. □

Under the conditions for crowding out in Proposition 3, the decline in expected holdings per unit of risk tolerance increases with initial private precision among investors whose private precision is unchanged.

Corollary C.2 (Crowding out over a finite rise)**.** *Let $\beta(n) = \phi(n) = 0$ and let κ rise to $\kappa' > \kappa$, fixing σ_z, τ_ε, $\bar{\gamma}^{-1}$ and each nonrecipient's private precision $\tau_i \geq 0$. That investor's expected holding falls if and only if*

$$\kappa\kappa' < \sigma_z^2(\tau_\varepsilon + \tau_i) + \tau_i\bar{\gamma}^{-1}(\kappa + \kappa').$$

At $\tau_i = 0$, the condition is $\tau^{\mathrm{R}} + \tau^{\mathrm{R}\prime} < 1$. If $\kappa > 0$ and $\kappa' = (1 + \Delta_\kappa)\kappa$, it is $\tau^{\mathrm{R}} < 1/(2 + \Delta_\kappa)$, which ensures that every nonrecipient's expected holding falls.

Proof of Proposition 3 and Corollaries 1 and C.2. With $\beta(n) = \phi(n) = 0$, demand is $\gamma_i^{-1}\hat{\tau}_i m_i$ and $\mathbb{E}[m_i] = 1/\mathcal{T}$, proving equation (19). Holdings sum to one because $\int \gamma_i^{-1}\hat{\tau}_i \, di = \mathcal{T}$. At fixed τ_i,

$$\frac{\partial \mathbb{E}[q_i]}{\partial \kappa} = \frac{\gamma_i^{-1}}{\mathcal{T}^2}\left[\frac{2\kappa}{\sigma_z^2}(\kappa - \bar{\gamma}^{-1}\tau_i) - (\tilde{\tau} + \tau_i)\right].$$

The bracket decreases in τ_i and equals $\tau_p - \tau_\varepsilon$ at zero, proving the local loss and its ordering per unit of risk tolerance. The proportional rate, $\partial \log \mathbb{E}[q_i]/\partial\kappa$, has derivative $-2\kappa/(\sigma_z^2\hat{\tau}_i^2)$ in τ_i, strictly negative at $\kappa > 0$.

For a finite rise, the sign of the holding change is the sign of

$$(\tilde{\tau}' + \tau_i)\mathcal{T} - (\tilde{\tau} + \tau_i)\mathcal{T}' = \frac{\kappa' - \kappa}{\sigma_z^2}\Big[\kappa\kappa' - \sigma_z^2(\tau_\varepsilon + \tau_i) - \tau_i\bar{\gamma}^{-1}(\kappa + \kappa')\Big].$$

At $\tau_i = 0$, the bracket is negative exactly when $\tau_p\tau_p' < \tau_\varepsilon^2$, or $\tau^{\mathrm{R}} + \tau^{\mathrm{R}\prime} < 1$. Substituting $\kappa' = (1 + \Delta_\kappa)\kappa$ gives the stated threshold. The bracket decreases in τ_i, so the threshold covers every nonrecipient. Finite losses have the same ordering: $\partial[(\mathbb{E}[q_i'] - \mathbb{E}[q_i])/\gamma_i^{-1}]/\partial\tau_i = 1/\mathcal{T}' - 1/\mathcal{T} < 0$, and $\mathbb{E}[q_i']/\mathbb{E}[q_i]$ decreases in τ_i when public precision rises. □

Proposition C.1 (Information rents and expected demand)**.** *In the benchmark economy, fix a stock and suppress its index. Let no investor have a mandate. Define the stock-selection*

position by removing the common factor premium from expected excess payoff:

$$q_i^{\text{sel}} = \gamma_i^{-1}\hat{\tau}_i\left(m_i - \frac{\beta H_\beta}{\mathcal{T}}\right),$$
$$q_i = q_i^{\text{sel}} + \gamma_i^{-1}\hat{\tau}_i\beta\left(\frac{H_\beta}{\mathcal{T}} - g_i^\beta\right).$$

The second identity decomposes full demand into stock selection and a realized factor adjustment. When $\beta = 0$, $q_i^{\text{sel}} = q_i$. Keep the eligible investors, their risk tolerances and the supply $S > 0$ fixed, with $S = 1$ in Section 3 and S equal to the sample institutions' aggregate holdings in Section 7. Let $\tilde{\tau} > 0$, $\bar{q}_i = \mathbb{E}[q_i^{\text{sel}}]$ and $\Pi_i = \mathbb{E}[q_i^{\text{sel}}\tilde{\varepsilon}]$. Then

$$\begin{aligned}\Pi_i &= \gamma_i^{-1}\frac{\tau_i}{\tilde{\tau}}, \qquad \int \Pi_j\, dj = \frac{\kappa}{\tilde{\tau}},\\ \frac{\bar{q}_i}{S} &= \frac{\gamma_i^{-1} + \Pi_i}{\bar{\gamma}^{-1} + \int \Pi_j\, dj} = \frac{\gamma_i^{-1}\hat{\tau}_i}{\mathcal{T}}.\end{aligned} \tag{A5}$$

At fixed $\tau_i > 0$, more public precision lowers the stock-selection rent Π_i. Between two information allocations, expected stock-selection holdings satisfy

$$\frac{\bar{q}_i'}{\bar{q}_i} = \frac{\hat{\tau}_i'}{\hat{\tau}_i}\frac{\mathcal{T}}{\mathcal{T}'} = \frac{\tilde{\tau}' + \tau_i'}{\tilde{\tau} + \tau_i}\frac{\mathcal{T}}{\mathcal{T}'}. \tag{A6}$$

Holding all private precisions fixed, the effect of public precision on expected stock-selection demand is

$$\left.\frac{\partial \bar{q}_i}{\partial \tilde{\tau}}\right|_{\{\tau_j\}} = \frac{\gamma_i^{-1}S\left(\kappa - \bar{\gamma}^{-1}\tau_i\right)}{\mathcal{T}^2}. \tag{A7}$$

It raises holdings below the risk-tolerance-weighted mean private precision $\kappa/\bar{\gamma}^{-1}$ and lowers them above it.

For a rise $\Delta\kappa = \kappa' - \kappa > 0$ with $\Delta\tilde{\tau} = \tilde{\tau}' - \tilde{\tau} \geq 0$, write $\Delta\tau_i = \tau_i' - \tau_i$ and define the accounting allocation that changes private information first,

$$\mathcal{T}^{\text{D}} = \bar{\gamma}^{-1}\tilde{\tau} + \kappa', \qquad q_i^{\text{D}} = \frac{\gamma_i^{-1}S(\tilde{\tau} + \tau_i')}{\mathcal{T}^{\text{D}}}.$$

Its two contributions to the change in expected holdings are

$$\begin{aligned} \bar{q}_i' - \bar{q}_i &= (q_i^{\mathrm{D}} - \bar{q}_i) + (\bar{q}_i' - q_i^{\mathrm{D}}), \\ q_i^{\mathrm{D}} - \bar{q}_i &= \frac{\gamma_i^{-1} S\left[\mathcal{T}\,\Delta\tau_i - \hat{\tau}_i \Delta\kappa\right]}{\mathcal{T}\,\mathcal{T}^{\mathrm{D}}}, \\ \bar{q}_i' - q_i^{\mathrm{D}} &= \frac{\gamma_i^{-1} S \Delta\tilde{\tau}\left(\kappa' - \bar{\gamma}^{-1}\tau_i'\right)}{\mathcal{T}^{\mathrm{D}}\mathcal{T}'}. \end{aligned} \tag{A8}$$

For an investor receiving no new private information, the first contribution is strictly negative. When $\Delta\tilde{\tau} > 0$, the second has the sign of $\kappa' - \bar{\gamma}^{-1}\tau_i$.

Holding public precision fixed in the intermediate allocation assigns the interaction between the two precision changes to learning from prices. In the conditional experiment of Appendix G, the common factor premium and each investor's expected factor contribution stay at their baseline values. Changes in full expected holdings then equal changes in $\bar{q}_i$; ratios in the proposition refer to stock-selection holdings.

Proof. Benchmark clearing gives $\mathbb{E}[m_i] = (S + \beta H_\beta)/\mathcal{T}$, hence $\bar{q}_i = \gamma_i^{-1}\hat{\tau}_i S/\mathcal{T}$. In q_i^{sel}, only $\gamma_i^{-1}\tau_i\tilde{\varepsilon}$ covaries with residual payoff news, giving $\Pi_i = \gamma_i^{-1}\tau_i/\tilde{\tau}$. Dividing the numerator and denominator of $\bar{q}_i/S$ by $\tilde{\tau}$ proves equation (A5); taking its ratio across allocations proves equation (A6).

Differentiating these expressions gives $\partial\Pi_i/\partial\tilde{\tau} = -\gamma_i^{-1}\tau_i/\tilde{\tau}^2$ and equation (A7), using $\mathcal{T} - \bar{\gamma}^{-1}\hat{\tau}_i = \kappa - \bar{\gamma}^{-1}\tau_i$. Subtracting expected holdings over common denominators, with $\mathcal{T}^{\mathrm{D}} = \mathcal{T} + \Delta\kappa$ and $\mathcal{T}' = \mathcal{T}^{\mathrm{D}} + \bar{\gamma}^{-1}\Delta\tilde{\tau}$, gives equation (A8). Its positive denominators give the stated signs. □

For full positions, the realized factor adjustment covaries with residual payoff news, giving $\mathbb{E}[q_i\tilde{\varepsilon}] = \Pi_i(1 - h_i)$ in Theorem 1. This covariance can persist with a fixed mean factor contribution across counterfactual allocations. The unattenuated rent uses q_i^{sel}; the information estimator uses full positions and corrects their exposure attenuation.

Define prior-based risk-bearing capacity relative to noise trades as

$$D(n) = \frac{\bar{\gamma}^{-1}\sqrt{\tau_\varepsilon(n)}}{\sigma_z(n)}. \tag{A9}$$

It measures the shares holders trading on the prior would buy against a one-standard-deviation mispricing, per standard deviation of noise trades.

Lemma C.1 (The slope and aggregate information)**.** *Fix τ_ε, σ_z and $\bar{\gamma}^{-1}$, suppressing the stock index. The price-response coefficient satisfies*

$$\mathcal{T}_p = \kappa + \bar{\gamma}^{-1}\tau_\varepsilon d_p, \qquad \frac{\partial \mathcal{T}_p}{\partial \kappa} = 1 - D^2 d_p^2.$$

At $\kappa > 0$, this derivative is negative exactly when $\kappa < \bar{\gamma}^{-1}\tau_\varepsilon d_p(1-d_p)$. The derivative increases to one as κ grows, so any initial decline eventually reverses.

Corollary C.3 (A finite rise in aggregate information)**.** *Under the conditions of Proposition 4, let $\kappa > 0$ rise to $\kappa' = (1+\Delta_\kappa)\kappa$, with $\Delta_\kappa > 0$, and define $\Delta_d = 1 - d_p'/d_p$. Then*

$$d_p' = \frac{d_p}{1 + \Delta_\kappa(1-d_p)}, \qquad \Delta\mathcal{T}_p = \Delta_\kappa\kappa\Big(1 - D^2 d_p d_p'\Big).$$

The magnitude of aggregate demand elasticity at a fixed positive price and quantity falls if and only if

$$\kappa < \frac{\bar{\gamma}^{-1}\tau_\varepsilon d_p(1-d_p)}{1 + \Delta_\kappa(1-d_p)}.$$

This condition tends to the local condition as $\Delta_\kappa \to 0$ and is stricter for a finite increase.

Corollary C.4 (Direct effect and learning from prices)**.** *Under the conditions of Corollary C.3, fix price learning at its initial value, holding τ_p and d_p constant. New private information raises both $\mathcal{T}$ and $\mathcal{T}_p$ by $\Delta_\kappa\kappa$. Learning from prices adds $\bar{\gamma}^{-1}\tau_p[(1+\Delta_\kappa)^2 - 1]$ to $\mathcal{T}$ and subtracts $\Delta_d(\mathcal{T}_p - \kappa)$ from $\mathcal{T}_p$. For $\beta(n) = \phi(n) = 0$, aggregate expected demand at a given expected excess payoff is proportional to $\mathcal{T}$.*

Proof of Lemma C.1, Proposition 4, and Corollaries C.3–C.4. Write $d_p = (1 + \bar{\gamma}^{-1}\kappa/\sigma_z^2)^{-1}$. Differentiating $\mathcal{T}_p$ gives $1 - D^2 d_p^2$; multiplying by κ gives $\kappa - \bar{\gamma}^{-1}\tau_\varepsilon d_p(1-d_p)$. Since d_p decreases to zero, the derivative increases to one. Where $\beta(n) = \phi(n) = 0$, the slope is $-R_f\mathcal{T}_p$, so its elasticity at a fixed positive price and quantity has the same proportional change as $\mathcal{T}_p$, proving Proposition 4.

Substituting κ' into d_p gives the displayed d_p'. Subtracting the two coefficients then gives

$$\Delta\mathcal{T}_p = \Delta_\kappa\kappa - \bar{\gamma}^{-1}\tau_\varepsilon(d_p - d_p') = \Delta_\kappa\left[\kappa - \frac{\bar{\gamma}^{-1}\tau_\varepsilon d_p(1-d_p)}{1 + \Delta_\kappa(1-d_p)}\right].$$

The sign gives the finite condition. Holding τ_p and d_p fixed isolates the direct term $\Delta_\kappa\kappa$ in both coefficients. The remaining terms follow from $\tau_p' = (1+\Delta_\kappa)^2\tau_p$ and $d_p - d_p' = \Delta_d d_p$. Finally, summing individual demand coefficients $\gamma_i^{-1}\hat{\tau}_i$ gives $\mathcal{T}$. □

Corollary C.5 (The gain in price informativeness)**.** *Let $\kappa > 0$ rise to $(1+\Delta_\kappa)\kappa$ with σ_z and τ_ε fixed. Then $\tau_p' = (1+\Delta_\kappa)^2\tau_p$, so the ratio of the price precision to the payoff precision, $\tau_p/\tau_\varepsilon = \tau^{\mathrm{R}}/(1-\tau^{\mathrm{R}})$, rises by the factor $(1+\Delta_\kappa)^2$ whatever the stock, and*

$$\tau^{\mathrm{R}\prime} = \frac{(1+\Delta_\kappa)^2\tau^{\mathrm{R}}}{1-\tau^{\mathrm{R}}+(1+\Delta_\kappa)^2\tau^{\mathrm{R}}},$$
$$\tau^{\mathrm{R}\prime} - \tau^{\mathrm{R}} = \frac{\left[(1+\Delta_\kappa)^2-1\right]\tau^{\mathrm{R}}(1-\tau^{\mathrm{R}})}{1-\tau^{\mathrm{R}}+(1+\Delta_\kappa)^2\tau^{\mathrm{R}}}.$$

For a given Δ_κ, the gain rises with τ^{R} up to $1/(2+\Delta_\kappa)$ and falls beyond. At that level the gain is $\Delta_\kappa/(2+\Delta_\kappa)$ and $\tau^{\mathrm{R}\prime} = (1+\Delta_\kappa)/(2+\Delta_\kappa)$, so $\tau^{\mathrm{R}} + \tau^{\mathrm{R}\prime} = 1$, the boundary of Corollary C.2. Locally, $d\tau^{\mathrm{R}} = 2\tau^{\mathrm{R}}(1-\tau^{\mathrm{R}})\, d\ln\kappa$.

Proof. Substituting $\tau_p' = (1+\Delta_\kappa)^2\tau_p$ into $\tau^{\mathrm{R}\prime}$ gives both formulas. Write $a = 1+\Delta_\kappa > 1$ and $x = \tau^{\mathrm{R}}$. The gain's derivative has the sign of $1-2x-(a^2-1)x^2$, whose unique root in $(0,1)$ is $1/(a+1)$. Substitution gives the peak and its endpoints. Differentiating $\tau^{\mathrm{R}} = \kappa^2/(\kappa^2+\sigma_z^2\tau_\varepsilon)$ gives the local result. □

Corollary C.6 (Exact zero and public invariance)**.** *If $k_i(n) = 0$, then $\mathbb{E}[q_i(n)\tilde{\varepsilon}(n)] = 0$, the factor and mandate adjustments included. The zero is exact in any equilibrium if $K_i = 0$, and in the benchmark economy if the investor has data on other stocks (the finite-stock discussion in Appendix C). Any variable measurable with respect to public information, times any deterministic coefficient, is uncorrelated with $\tilde{\varepsilon}(n)$, so removing a public component from a holding leaves the moment of Theorem 1 unchanged.*

For the information identity, write $\hat{\varepsilon}_p(n) = \tau_p(n)\eta_p(n)/\tilde{\tau}(n)$, $\tilde{\varepsilon}(n) = \varepsilon(n) - \hat{\varepsilon}_p(n)$ and $\tilde{m}(n) = \mu(n) + \hat{\varepsilon}_p(n) - R_f p(n)$. The residual variance is $1/\tilde{\tau}(n)$. Normality makes it independent of prices; in the continuum benchmark it is also independent of individual signal noises and other stocks' residual surprises.

Proof of Theorem 1 and Corollary C.6. Normal updating gives $\hat{\tau}_i m_i = \hat{\tau}_i\tilde{m} + \tau_i\tilde{\varepsilon} + \tau_i\bar{u}_i$. Substituting into demand yields

$$q_i(n) = \gamma_i^{-1}\hat{\tau}_i(n)\tilde{m}(n) + \gamma_i^{-1}\tau_i(n)\tilde{\varepsilon}(n) + \gamma_i^{-1}\tau_i(n)\bar{u}_i(n) - \gamma_i^{-1}\hat{\tau}_i(n)v_n^\top G_i^{-1}B^\top\hat{\Omega}_i^{-1}m_i.$$

Public demand and signal noise are orthogonal to $\tilde{\varepsilon}(n)$. Only stock n contributes to

$$\mathbb{E}[B^\top\hat{\Omega}_i^{-1}m_i\tilde{\varepsilon}(n)] = v_n\tau_i(n)/\tilde{\tau}(n).$$

Thus

$$\mathbb{E}[q_i(n)\tilde{\varepsilon}(n)] = \gamma_i^{-1}\frac{\tau_i(n)}{\tilde{\tau}(n)}[1-h_i(n)] = \gamma_i^{-1}k_i(n)[1-\tau^{\mathrm{R}}(n)][1-h_i(n)].$$

The attenuation vanishes when $v_n = 0$ and satisfies $1 - h_i(n) = [M_i^{-1}]_{nn}/\hat{\tau}_i(n) > 0$. For $\delta_i^{\phi} \leq 0$, positive-definite G_i gives $h_i(n) \geq 0$; Corollary C.1's bound gives $h_i(n) = O(1/N)$. Dividing by $[1-\tau^{\mathrm{R}}(n)][1-h_i(n)]$ and summing proves the identity. At $k_i(n) = 0$, the moment is zero. For any public variable $V(n)$, $\mathbb{E}[V(n)\tilde{\varepsilon}(n)] = 0$ by iterated expectations, proving public invariance. This also gives the exact zero at $K_i = 0$ when conditioning on the full finite-stock price vector. □

The correction $1/[1-h_i(n)]$ in Theorem 1 uses the investor's own precisions. The following feasible correction uses public precisions.

Corollary C.7 (A feasible exposure correction)**.** *Under the conditions of Theorem 1, let* $\delta_i^{\phi} \leq 0$. *The uncorrected sum* $\sum_n \mathbb{E}[q_i(n)\tilde{\varepsilon}(n)]/[1-\tau^{\mathrm{R}}(n)]$ *lies between* $[1-\max_n h_i(n)]\,\gamma_i^{-1}K_i$ *and* $\gamma_i^{-1}K_i$. *With*

$$\hat{h}(n) = \tilde{\tau}(n)\, v_n^{\top}\Big[\sum_{n'}\tilde{\tau}(n')v_{n'}v_{n'}^{\top}\Big]^{-1}v_n,$$

which is $\tilde{\tau}(n)\beta(n)^2/\sum_{n'}\tilde{\tau}(n')\beta(n')^2$ *without a mandate, in place of* $h_i(n)$*, the corrected sum misses* $\gamma_i^{-1}K_i$ *by a relative error of order* $\max_{n'} k_i(n')/N + 1/N^2$.

Proof. The uncorrected sum is $\gamma_i^{-1}\sum_n k_i(n)[1-h_i(n)]$, giving the bounds. Write $\hat{G} = \sum_{n'}\tilde{\tau}(n')v_{n'}v_{n'}^{\top}$. Then

$$h_i(n) - \hat{h}(n) = \tau_i(n)v_n^{\top}G_i^{-1}v_n + \tilde{\tau}(n)v_n^{\top}(G_i^{-1} - \hat{G}^{-1})v_n.$$

Pervasiveness bounds both inverses by $O(1/N)$, so the first term is $O(k_i(n)/N)$. In the second, use $G_i^{-1} - \hat{G}^{-1} = -G_i^{-1}(G_i - \hat{G})\hat{G}^{-1}$. The difference $G_i - \hat{G}$ is the fixed matrix $\mathrm{diag}(1, -\gamma_i^2/\delta_i^{\phi})$ (one without a mandate) plus $\sum_{n'}\tau_i(n')v_{n'}v_{n'}^{\top}$. Their contributions are $O(N^{-2})$ and $O(\sum_{n'}k_i(n')/N^2)$, respectively. Since $\hat{h}(n) = O(1/N)$, each corrected term $\gamma_i^{-1}k_i(n)[1-h_i(n)]/[1-\hat{h}(n)]$, and their sum, has relative error $O(\max_{n'} k_i(n')/N + 1/N^2)$. □

For a hard exposure target, see Online Appendix Proposition OA3.1.

Let $F \sim N(0, I)$ have fixed dimension K_F as N grows, with loading matrix β whose rows are $\beta(n)^{\top}$. Loadings are bounded and pervasive within each investor's universe:

$$\lambda_{\min}\left(N_i^{-1}\sum_{n\in\mathcal{N}_i}\beta(n)\beta(n)^{\top}\right)$$

is bounded away from zero, also with weights bounded above and away from zero.

Proposition C.2 (Multiple factors)**.** *With no mandates, Lemma 1 holds and factor beliefs remain their prior. Set* $G_i = I + \beta^\top \hat{\Omega}_i^{-1}\beta$*. Demand and factor exposure become*

$$q_i(n) = \gamma_i^{-1}\hat{\tau}_i(n)[m_i(n) - \beta(n)^\top g_i^\beta],$$
$$g_i^\beta = G_i^{-1}\beta^\top \hat{\Omega}_i^{-1} m_i, \qquad \beta^\top q_i = \gamma_i^{-1} g_i^\beta.$$

Benchmark prices replace the factor term by $\beta(n)^\top H_\beta(n)$*, with* $H_\beta(n) = \int \gamma_i^{-1}\hat{\tau}_i(n)\bar{g}_i^\beta \, di$*;* $\kappa(n)$ *and* $\tau_p(n)$ *are unchanged. Theorem 1 and Corollary C.6 apply with*

$$h_i(n) = \hat{\tau}_i(n)\beta(n)^\top G_i^{-1}\beta(n) \in [0, 1),$$

of order $1/N$*. The feasible correction is*

$$\hat{h}(n) = \tilde{\tau}(n)\beta(n)^\top \Big[\sum_{n'} \tilde{\tau}(n')\beta(n')\beta(n')^\top\Big]^{-1}\beta(n),$$

with Corollary C.7's error bound. In Proposition 2, comparable stocks satisfy

$$\hat{\tau}_i(n')\beta(n') = \hat{\tau}_i(n)\beta(n).$$

Propositions 3 and 4 apply where $\beta(n) = 0$*.*

Proof. Signals do not load on F. Woodbury gives $M_i^{-1} = \hat{\Omega}_i^{-1} - \hat{\Omega}_i^{-1}\beta G_i^{-1}\beta^\top \hat{\Omega}_i^{-1}$, proving demand and exposure. Substituting mean adjustments in benchmark clearing leaves the aggregate signal coefficient κ unchanged; the same Ξ argument gives unique mean adjustments. The moment uses $\mathbb{E}[g_i^\beta \tilde{\varepsilon}(n)] = G_i^{-1}\beta(n)\tau_i(n)/\tilde{\tau}(n)$. Positive-definite M_i and G_i imply $0 \le h_i(n) < 1$, while pervasiveness gives $O(1/N)$. Using public precisions and dropping I gives the feasible correction. The slope calculation applies with vector loadings, and zero loadings give scalar demand for the crowding and elasticity results. □

D Measuring Information: Estimation and Properties

D.1 Estimation Specification

Earnings forecasts. I forecast the change in operating earnings from the same fiscal quarter a year earlier, per dollar of book equity at the holdings date. Earnings are announced in the following quarter. I use the latest fiscal period when multiple reports map to a holdings quarter and winsorize the change at that quarter's 1st and 99th percentiles. The earnings surprise in Section 2.4 sums four quarterly changes and scales by their past volatility.

The baseline forecast uses stock characteristics, analyst-consensus revisions for the same fiscal period, log one plus analyst coverage, and forecast dispersion. Revisions compare quarterly forecast snapshots. The augmented forecast adds price, price–characteristic interactions, and twelve-industry indicators; its error is the residual earnings surprise in Theorem 1. The variance comparison captures this entire added block; attribution to price alone requires industry effects to contribute no separate reduction in forecast uncertainty.

Variances and weights. Models of squared forecast errors give payoff precision $\hat{\tau}_\varepsilon(n)$ and public precision $\hat{\tau}_\varepsilon(n) + \hat{\tau}_p(n)$. The baseline variance model is pooled; the augmented model is fitted within quarter. I assume a payoff-proxy scale common to stocks and quarters, which scales information estimates equally and cancels from the variance ratio.

Price informativeness is $\hat{\tau}^{\mathrm{R}}(n) = \hat{\tau}_p(n)/[\hat{\tau}_\varepsilon(n) + \hat{\tau}_p(n)]$. Within each size-quintile and market-to-book-tercile bin, I anchor the median weight $1/[1 - \hat{\tau}^{\mathrm{R}}(n)]$ to the degrees-of-freedom-adjusted ratio of baseline to augmented mean squared errors. Fitted variances supply within-bin variation. I bound weights between one and 100, and public precision between one hundredth and one hundred times its quarterly median. The lower weight bound sets $\hat{\tau}^{\mathrm{R}}(n) = 0$ for 786 of the 1,933 counterfactual stocks in 2024Q4.

Exposure and position corrections. The exposure correction of Corollary C.7 weights squared estimated market betas, based on 60 months of monthly returns, by public precision:

$$\hat{h}(n) = \frac{[\hat{\tau}_\varepsilon(n) + \hat{\tau}_p(n)]\beta(n)^2}{\sum_{n' \in \mathcal{N}_{it}} [\hat{\tau}_\varepsilon(n') + \hat{\tau}_p(n')]\beta(n')^2}.$$

I cap $\hat{h}(n)$ at 0.5. Position controls Z include all forecasting regressors and precision- and informativeness-weight-scaled price and characteristic terms over the same investor-quarter universe. Their projection is $P = Z(Z^\top Z)^\dagger Z^\top$, where $\dagger$ denotes the Moore–Penrose inverse; the leverage divisor is $\max\{1 - P_{nn}, 0.05\}$.

For an observed mandate, add its attribute to the loading matrix subject to rank and pervasiveness, using a basis when the attribute is redundant. If characteristics span ϕ, its scaled public component also enters the position controls. Proposition C.2 supplies the loadings and weighted Gram matrix for multiple factors.

Forecast-coefficient errors enter the estimate through weighted sums of residual positions; the controls remove components spanned by the unscaled and scaled regressors. A common proportional change in both variance laws cancels from the informativeness ratio, and a common scalar rescaling of public precision leaves the exposure projection unchanged. Institutions share earnings surprises and fitted forecasts, so standard errors for the group

statistics in Section 4 use their quarterly time series.

Persistence and size. Following Carhart (1997), Online Appendix Table OA4 uses twelve-quarter formation averages and follow-up windows separated by a four-quarter gap. Within each quarter and group, I residualize standardized information on the log market value and log stock count of positions entering the estimate, then sort on and follow the residual. Separate sorts on trailing log size and log breadth follow standardized information.

D.2 Statistical Properties

Proposition 5 and Corollary 2 use full positions, known forecasting objects, nonbinding estimation bounds and a common payoff-proxy scale normalized to one. Information is allocated before signals and confined to $\mathcal{N}_{it}$. Maintain Assumption C.1, $\delta_i^{\phi} \leq 0$, uniformly bounded $[1-\tau^{\mathrm{R}}(n)]^{-1}$, and $\max_n P_{nn} = O(N_{it}^{-1})$, where $N_{it} = |\mathcal{N}_{it}|$. Exposure loadings satisfy pervasiveness on this universe. All moments condition on public information and the allocation.

Proof of Proposition 5 and Corollary 2. Book-equity positions times earnings surprises per dollar of book equity equal ownership shares times dollar earnings surprises. In these model units, controls and precision corrections are conditionally fixed, and $(\tilde{\varepsilon}(n), \bar{u}_i(n))$ are independent across stocks.

Let $a = (a_{nn'})$ map direct private-information trades into residual positions:

$$a = (I - P)\Big[I - \operatorname{diag}(\hat{\tau}_i) B G_i^{-1} B^{\top}\Big].$$

Bounded, pervasive loadings give $O(N_{it}^{-1})$ entries for the exposure matrix $\operatorname{diag}(\hat{\tau}_i) B G_i^{-1} B^{\top}$. The projection identities $|P_{nn'}| \leq \sqrt{P_{nn} P_{n'n'}}$ and $\sum_{n'} P_{nn'}^2 = P_{nn}$ then give $a_{nn} = 1 + O(N_{it}^{-1})$, $a_{nn'} = O(N_{it}^{-1})$ for $n' \neq n$, and $\hat{h}(n) = O(N_{it}^{-1})$.

With $c_n = \{[1-\hat{h}(n)][1-P_{nn}]\}^{-1}$ and $\tilde{m}(n) = \mu(n) + \mathbb{E}[\varepsilon(n) \mid p] - R_f p(n)$, define adjusted residual public demand and the leading stock contribution as

$$e_{it}^{\mathrm{mis}}(n) = c_n \gamma_{it}^{-1} \Big[a \operatorname{diag}(\hat{\tau}_i) \tilde{m}\Big]_n,$$
$$Y_{it}(n) = \frac{\Big\{\gamma_{it}^{-1} \tau_{it}(n)[\tilde{\varepsilon}(n) + \bar{u}_i(n)] + e_{it}^{\mathrm{mis}}(n)\Big\} \tilde{\varepsilon}(n)}{1 - \tau^{\mathrm{R}}(n)}.$$

The $Y_{it}(n)$ are conditionally independent with mean $\gamma_{it}^{-1} k_{it}(n)$. The exact own-stock private term has multiplier $c_n a_{nn} = 1 + O(N_{it}^{-1})$; off-diagonal terms have zero mean. Hence $\mathbb{E}[\hat{\theta}_{it}] - \theta_{it} = O(\theta_{it}/N_{it})$.

Using $\mathrm{Var}[\tilde{\varepsilon}(n)] = 1/\tilde{\tau}(n)$ and $\tau_i = k_i\tau_\varepsilon$, conditional normality gives $V_{it}(n) = \mathrm{Var}[Y_{it}(n)]$ as

$$V_{it}(n) = 2\gamma_{it}^{-2}k_{it}(n)^2 + \frac{\gamma_{it}^{-2}k_{it}(n)}{1-\tau^{\mathrm{R}}(n)} + \frac{\mathbb{E}\left[e_{it}^{\mathrm{mis}}(n)^2\right]}{\tau_\varepsilon(n)\left[1-\tau^{\mathrm{R}}(n)\right]}.$$

Define $V_{it} = \sum_n V_{it}(n)$ and

$$\mathrm{HHI}_{it} = \sum_n \left[\frac{k_{it}(n)}{K_{it}}\right]^2, \qquad \bar{w}_{it} = \sum_n \frac{k_{it}(n)/K_{it}}{1-\tau^{\mathrm{R}}(n)}, \qquad \sigma^2_{\mathrm{mis},it} = \sum_n \frac{\mathbb{E}\left[e_{it}^{\mathrm{mis}}(n)^2\right]}{\tau_\varepsilon(n)\left[1-\tau^{\mathrm{R}}(n)\right]}.$$

Dividing V_{it} by θ_{it}^2 gives the decomposition in Proposition 5.

Collect the centered cross-stock terms and own-stock approximation error in $\mathcal{R}_{it}$:

$$\hat{\theta}_{it} - \mathbb{E}[\hat{\theta}_{it}] = \sum_n \{Y_{it}(n) - \mathbb{E}[Y_{it}(n)]\} + \mathcal{R}_{it}.$$

Cross-stock products $\tilde{\varepsilon}(n)[\tilde{\varepsilon}(n') + \bar{u}_i(n')]$, $n' \neq n$, have coefficients of order $\gamma_{it}^{-1}\tau_{it}(n')/N_{it}$. Bounded counts and precisions and Gaussian independence give

$$\mathbb{E}[\mathcal{R}_{it}^2] = O\left(\gamma_{it}^{-2}K_{it}/N_{it}\right).$$

These products are uncorrelated with each $Y_{it}(n)$; the diagonal approximation changes the private-component variance by a relative $O(N_{it}^{-1})$. Since $V_{it} \geq \gamma_{it}^{-2}K_{it}$, $\mathrm{Var}(\hat{\theta}_{it}) = V_{it}[1 + O(N_{it}^{-1})]$ and $\mathcal{R}_{it}/\sqrt{V_{it}} \to 0$ in probability. When $K_{it} = 0$, the remainder and bias are zero.

Bounded counts imply $K_{it} \leq CN_{it}$. For $K_{it} > 0$, the bias and variance bounds give

$$\frac{|\mathbb{E}[\hat{\theta}_{it}] - \theta_{it}|}{\sqrt{V_{it}}} = O\left(\frac{\sqrt{K_{it}}}{N_{it}}\right) = O(N_{it}^{-1/2}) \to 0.$$

Each centered $Y_{it}(n)$ is a Gaussian polynomial of degree at most two, with third absolute moment bounded by a universal constant times $V_{it}(n)^{3/2}$. The Lyapunov ratio is therefore bounded by a constant times $[\max_n V_{it}(n)/V_{it}]^{1/2}$. Under the no-dominating-stock condition in Corollary 2, this tends to zero; the negligible remainder and bias give the normal limit centered at θ_{it}.

If $\mathrm{HHI}_{it} \to 0$, $K_{it} \to \infty$ and $\sigma^2_{\mathrm{mis},it}/\theta_{it}^2 \to 0$, bounded informativeness weights imply $V_{it}/\theta_{it}^2 \to 0$. The bias bound and Chebyshev's inequality then give $\hat{\theta}_{it}/\theta_{it} \to 1$ in probability. □

The empirical noise scale squares to $\hat{V}_{it} = \sum_b \hat{\theta}_{it,b}^2$, where $\hat{\theta}_{it,b}$ sums contributions within

each of twelve Fama–French industries. Its relation to sampling variance is

$$\mathbb{E}\Big[\hat{V}_{it}\Big] - \mathrm{Var}\Big(\hat{\theta}_{it}\Big) = \sum_{b} \Big(\mathbb{E}\hat{\theta}_{it,b}\Big)^2 - \sum_{b \neq c} \mathrm{Cov}\Big(\hat{\theta}_{it,b}, \hat{\theta}_{it,c}\Big).$$

With independent industries, the covariance term vanishes. The excess is at least $(\mathbb{E}[\hat{\theta}_{it}])^2/12$. Cauchy–Schwarz bounds $|\hat{\theta}_{it}|/\sqrt{\hat{V}_{it}}$ by $\sqrt{12}$.

Lemma D.1 (Scale invariance)**.** *Fix forecasting objects, controls, the lagged-holdings set and industries. (i) Scaling every position by a positive constant scales $\hat{\theta}_{it}$ and each $\hat{\theta}_{it,b}$ by that constant, and $\hat{V}_{it}$ by its square. Standardized information $\hat{\theta}_{it}/\sqrt{\hat{V}_{it}}$ is unchanged. (ii) For investors with $\delta_i^{\phi} = 0$ and the same data and signals, scaling risk tolerance has this effect. Under Assumption F.1, it also follows from scaling lagged AUM at equal local relative risk aversion in the reference quarter; information per dollar of lagged AUM is then unchanged.*

Proof. Residualization $(I - P)Q_{it}$ and equation (22) are linear in positions; $\hat{V}_{it}$ is quadratic. Positive scaling preserves position signs and the lagged-holdings set. Without mandates, $M_i = \hat{\Omega}_i + \beta\beta^\top$ and m_i are independent of γ_i, so $q_i = \gamma_i^{-1} M_i^{-1} m_i$ scales with risk tolerance. Assumption F.1 gives the reference-quarter wealth scaling. □

Lemma D.2 (Censoring)**.** *Let $q_i^+(n) = \max\{q_i(n), 0\}$ be the long position that 13F reports. Then*

$$\mathbb{E}\Big[q_i^+(n)\,\tilde{\varepsilon}(n)\Big] = \Pr\Big(q_i(n) > 0\Big)\,\mathbb{E}\Big[q_i(n)\,\tilde{\varepsilon}(n)\Big].$$

On stocks fixed before the quarter's signals, each expected information contribution is attenuated by its holding probability, up to exposure and projection corrections. The expected estimate from long positions therefore bounds θ_{it} from below to this order.

Proof. For jointly normal X, Y and absolutely continuous g, Stein's lemma gives

$$\mathrm{Cov}(g(X), Y) = \mathbb{E}[g'(X)]\mathrm{Cov}(X, Y).$$

Set $g(x) = \max\{x, 0\}$ and use $\mathbb{E}[\tilde{\varepsilon}(n)] = 0$. Apply the identity to each long position entering residualization. Cross-stock covariances vanish without exposure management and are of its order otherwise. The own term has coefficient $1 - P_{nn}$; its leverage correction leaves the holding probability $\Pr(q_i(n) > 0)$. □

E The Production Function: Identification

With known forecasting objects, long holdings measure information expressed through long positions. Lemma D.2 gives, up to the exposure approximation,

$$\mathbb{E}[\hat{\theta}_{it}] = \gamma_{it}^{-1} K_{it}^{+}, \qquad K_{it}^{+} = \sum_{n \in \mathcal{N}_{it}} \Pr(q_{it}(n) > 0)\, k_{it}(n).$$

Assumption E.1 (Stable investor scale)**.** Over the production sample, risk tolerance and investor-specific factors converting research output to measured information are constant. Their product with persistent research efficiency is a positive investor scale.

For the positive observations used in equation (23), write

$$\log \hat{\theta}_{it} = \log c_i + \log A_{y(t)} + \alpha \log L_{i,t-1}^{\mathrm{DS}} + X_{i,t-1}^{\top} b + u_{it},$$

where investor effects absorb $\log c_i$, which combines efficiency, risk tolerance, expression through long positions and the payoff-proxy scale. The disturbance u_{it} combines sampling error, productivity disturbances and unmeasured input changes on the log scale.

Let Z_{it} collect the log DS input, controls and year indicators. Identification requires

$$\mathbb{E}\Big[\widetilde{Z}_{it}\, u_{it} \mid \text{estimation sample}\Big] = 0,$$

where a tilde removes investor means. The within-regressor matrix must have full column rank after omitting the 2011 indicator, requiring within-investor employment variation beyond the common annual path and observations connecting the annual comparisons. This moment condition applies to the positive-outcome sample and its combined log disturbance. I estimate the coefficients jointly by within-investor least squares.

Sample and timing. Active status uses each investor's mean quarterly active share over available quarters from 2008Q1 through 2024Q4. The cutoff of 0.553 is the 20th percentile of these averages in the broader sample of 2,033 asset managers used for the classification. I hold this cutoff fixed when applying the production-sample restrictions.

I apply the screens in Section 5.1 and workforce definitions in Online Appendix OA1.2, matching employment at exactly the previous calendar quarter. Unavailable coverage contributes zero observed positive quarters to the four-of-eight screen. Each investor needs at least two eligible observations. The dependent variable uses logs of raw quarterly information.

Inference. I use investor-clustered CR1 standard errors and the t_{G-1} reference distribution, where G is the number of investors. For a vector c selecting annual coefficients, the log contrast has variance $c^\top \widehat{V} c$, using the full coefficient covariance matrix. Figure 4 exponentiates the endpoints of 90 percent log-contrast intervals and normalizes productivity in 2011 to one.

The estimated DS labor elasticity is 0.143 (standard error 0.081). The coefficients on log combined employment of investment analysts and portfolio managers and log employment outside that group and DS are 0.252 (0.156) and 0.230 (0.198), respectively. The productivity ratio for 2024 relative to 2011 is 3.28, with a 90 percent confidence interval of $[2.29, 4.69]$.

The finite-investor extension applies under stable effective risk tolerance and the benchmark information-moment interpretation in Appendix H.3.

F Demand Estimation: Details

Units and the first difference. Let $Q_{is}(n) = \mathrm{BE}_s(n) q_{is}(n)$ be investor i's position in book-equity dollars, where $q_{is}(n)$ is her fraction of outstanding shares. Price is market-to-book, $\mathrm{MB}_s(n) = p_s(n)/\mathrm{BE}_s(n)$ (Koijen et al., 2024). In the schedules below, payoff and private-signal precisions use payoffs per dollar of book equity. Both dates of a quarterly comparison use the same lagged AUM and reference risk tolerance.

Assumption F.1 (Reference tolerance in a quarterly comparison)**.** For quarters $t-1$ and t, hold reference risk tolerance fixed and write

$$\gamma_i^{-1} = \frac{\mathrm{AUM}_{i,t-1}}{\gamma_i^{\mathrm{R}}}, \qquad \gamma_i^{\mathrm{R}} = \mathrm{AUM}_{i,t-1}\gamma_i.$$

The local relative-risk-aversion scale γ_i^{R} may vary across comparisons as assets change.

For $s = t-1, t$, the level schedules are

$$\frac{Q_{is}(n)}{\mathrm{AUM}_{i,t-1}} = -\frac{R_f}{\gamma_i^{\mathrm{R}}}[\tau_{is}(n) + \tau_{\varepsilon,s}(n) d_{p,s}(n)]\mathrm{MB}_s(n) + C_{is}(n) + u_{is}(n),$$
$$u_{is}(n) = \frac{\tau_{is}(n)}{\gamma_i^{\mathrm{R}}}\eta_{is}(n).$$

Here C_{is} collects expected-payoff and exposure-management terms, and u_{is} is demand from the full private signal. Empirical residual demand can also include stock-specific tastes.

I value both quarters' shares at lagged book equity per split-adjusted share. With $\rho_t(n)$

the ratio of this value in $t-1$ to its value in t,

$$\Delta Q_{it}(n) = \frac{\rho_t(n) Q_{it}(n) - Q_{i,t-1}(n)}{\text{AUM}_{i,t-1}}, \qquad \Delta \text{MB}_t(n) = \text{MB}_{t-1}(n) r_t(n),$$

where r_t is the ex-dividend return. The earnings innovation accrues to book equity, so the price change excludes dividends. A nontrading holder has zero flow under issuance, splits and dividends.

Separately, estimation holds the relative demand slope fixed in lagged book-equity units:

$$\rho_t(n)^2 [\tau_{it}(n) + \tau_{\varepsilon,t}(n) d_{p,t}(n)] = \tau_{i,t-1}(n) + \tau_{\varepsilon,t-1}(n) d_{p,t-1}(n).$$

The two powers of ρ_t convert quantity and price. Expected-payoff and latent-demand terms require only the quantity conversion.

Proposition F.1 (Exact first difference under a fixed slope)**.** *Suppose both quarter-end positions lie on the level schedules above. Under Assumption F.1 and the fixed-slope restriction,*

$$\begin{aligned} \Delta Q_{it}(n) = & -\frac{R_f}{\gamma_i^{\text{R}}} [\tau_{i,t-1}(n) + \tau_{\varepsilon,t-1}(n) d_{p,t-1}(n)] \Delta \text{MB}_t(n) \\ & + \rho_t(n) C_{it}(n) - C_{i,t-1}(n) + \Delta u_{it}(n), \\ \Delta u_{it}(n) = & \rho_t(n) u_{it}(n) - u_{i,t-1}(n). \end{aligned}$$

Proof. Substitute the level schedules into ΔQ_{it} and use $\text{MB}_t = \rho_t(\text{MB}_{t-1} + \Delta \text{MB}_t)$. The fixed-slope restriction cancels the terms in MB_{t-1}, leaving the stated price change and differences in C and u. □

Substitution conditional on information. In the share units of Section 3, let $v_n = (\beta(n), \phi(n))^\top$ and let G_i be the matrix in Lemma 2. Without a mandate, keep only the factor component. Proposition 2 gives

$$\begin{aligned} E_i(n,m) \equiv & \frac{\partial q_i(n)}{\partial p(m)} \\ = & -R_f \gamma_i^{-1} \Big[\mathbf{1}\{n=m\} - \hat{\tau}_i(n) v_n^\top G_i^{-1} v_m \Big] \\ & \times \Big[\tau_i(m) + \tau_\varepsilon(m) d_p(m) \Big]. \end{aligned} \tag{A10}$$

For each investor, the exposure adjustment depends on the stock through its precision-weighted loadings, $\hat{\tau}_i(n) v_n$ (Haddad et al., 2026). Matching two distinct stocks on these

precision-weighted loadings therefore gives

$$E_i(n,n) - E_i(n',n) = -R_f\gamma_i^{-1}[\tau_i(n) + \tau_\varepsilon(n)d_p(n)],$$

the relative slope in equation (16). Since $\hat{\tau}_i = \tau_\varepsilon + \tau_p + \tau_i$, matching combines exposures with information from all three sources. Private precision and the discount also shift the relative slope, as in Haddad et al. (2026, equation (29) and Appendix A.5).

At fixed book equity and AUM, conversion to the demand units gives

$$\frac{\partial[Q_i(n)/\text{AUM}_i]}{\partial \text{MB}(m)} = \frac{\text{BE}(n)\text{BE}(m)}{\text{AUM}_i} E_i(n,m).$$

The corresponding row loadings are proportional to $\text{BE}(n)\hat{\tau}_i(n)v_n$. The factor controls interact exposures with public and measured private information, with coefficients free by institution-quarter.

Estimation transformations. I winsorize flows and log ρ separately and use winsorized lagged market-to-book. Missing returns are set to zero and returns are bounded below at -0.9. Discrepancies from the raw accounting identities enter the residual in the projected moments below.

Information units. Let $\text{AUM}_i^{\text{est}}$ and N_i denote the value and number of held positions within the sample used to estimate information, and AUM_i the full demand denominator. In original prediction units, the private-information slope implied by Appendix G is

$$-R_f c \frac{\bar{k}_i}{N_i} \frac{\text{AUM}_i^{\text{est}}}{\text{AUM}_i} \tau_\varepsilon^{\text{lag}}(n).$$

Demand uses the same winsorized per-position measure and lagged payoff precision, then centers and standardizes $\bar{k}_i/N_i$. The fitted interaction approximates the AUM ratio by a common scale across investors; its coefficient is the IV information gradient under this approximation. The counterfactual uses each investor's actual $\text{AUM}_i^{\text{est}}$.

Control variables. Twenty controls have coefficients free by institution-quarter: an intercept; five characteristics (log book equity, operating profitability, investment, dividends to book equity and market beta) at lagged levels and in ρ-adjusted changes; the standardized within-quarter change in the public payoff forecast excluding price information; four factor-loading interactions with public precision, payoff precision, the payoff-news loading and payoff precision times the change in relative price signal precision; and four columns for

the change in ℓ, that change times payoff precision, the change in the uninformative-price indicator and the discount. The discount is one in this linear specification, duplicating the intercept.

For every slope-index term except private information, expected-payoff controls enter separately at both dates. Each multiplies that term's variable by the date's payoff precision and expected payoff divided by R_f; the current control also includes ρ_t. Both dates use the lagged information prediction. These controls have separate coefficients common across institutions and quarters and serve as their own instruments. Private-information expected-payoff terms remain in the residual.

Predicting private information. The target divides the raw quarterly information estimate by same-quarter AUM^{est}. I winsorize it within quarter at the 5th and 95th percentiles and predict its within-quarter deviation by weighted least squares. Regressors are seven type indicators, lagged log AUM^{est} (current size if the lag is missing), a historical score and a missing-score indicator. At date s, the score is the mean winsorized target through $s-4$, using at least four observations, divided by its Newey–West standard error with four lags.

Weights equal the number of demand observations. Five folds of institutions give each investor coefficients estimated on the other folds. The prediction adds the fitted deviation to the equal-weighted average of weighted quarterly target means, with a floor of five percent of that common mean. Coefficients and the common mean use the full panel. Demand in quarter t uses the $t-1$ prediction; its historical score stops at $t-5$, excluding nearby signal noise and the $t-4$ estimate that shares an earnings figure with t through seasonal differencing. I divide by $N_{i,t-1}$, winsorize within quarter at the 1st and 99th percentiles across investor–stock observations, and standardize over those observations.

Measuring relative price signal precision. Forecasting gives $\hat{\tau}_p = \hat{\tau}_\varepsilon \hat{\tau}^{\text{R}}/(1-\hat{\tau}^{\text{R}})$. Measured aggregate information $\hat{\kappa}$ is payoff precision times the sum of holders' terms in equation (22). On stock-quarters with informative prices and positive $\hat{\kappa}$, the identity $\kappa = \sqrt{\tau_p}\sigma_z$ implies a noise-trade scale. I regress its log on log turnover and log dollar volume per dollar of book equity, with quarter effects, and use fitted $\hat{\sigma}_z$ throughout the panel. The measured ratio $\sqrt{\hat{\tau}_p}/\hat{\sigma}_z$ is zero for uninformative prices. Multiplying it by aggregate risk tolerance gives the dimensionless argument of the discount.

For informative prices, ℓ is the log ratio minus its quarterly median across investor–stock observations; the between-quarter term is that median minus the pooled median. Every observation has equal weight. The reference slope sets both terms to zero. The between-quarter coefficient is 0.01 (s.e. 0.01) in Table 3's units.

The GIV. Within each stock-quarter, I project flows on type-by-trading-activity indicators, the holder variables governing price and expected-payoff terms, and predicted payoff-news loadings at both dates. Trading activity groups investors into terciles of their median residual-flow variance across quarters. Projection weights are inverse residual variances estimated leaving the current quarter out. Holder variables include log age and predicted information per dollar per position.

For a position dated t, the payoff-news loading is

$$\bar{k}_{j,t-1}\tau_\varepsilon(n)\hat{\psi}^{\mathrm{G}}_{j,t-1}(n)\hat{\psi}^{\mathrm{W}}_{t-1}(n)\frac{\mathrm{AUM}^{\mathrm{est}}_{jt}}{\mathrm{AUM}_{jt}}.$$

The first fitted allocation distributes information across size-and-growth groups using type and age; the second uses each stock's share of aggregate information within its group. Both allocations are lagged. For the previous position, the construction shifts back one quarter, including $\bar{k}_{j,t-2}$ and $\mathrm{AUM}^{\mathrm{est}}_{j,t-1}/\mathrm{AUM}_{j,t-1}$. These own-date AUM ratios convert the empirical loadings separately from the common lagged-AUM normalization of the level schedules. The slope assigns information evenly across positions; the GIV uses the fitted allocation of information.

I exclude investor i from the projection and sum other holders' residuals with lagged-AUM weights, as in equation (26). Rows with leverage one have undefined leave-one-out residuals and are omitted from instrumented moments. Decompose private-signal demand as

$$u_{is}(n) = \frac{\tau_{is}(n)}{\gamma^{\mathrm{R}}_i}\varepsilon_s(n) + u^{\perp}_{is}(n), \qquad \Delta u^{\perp}_{it}(n) = \rho_t(n)u^{\perp}_{it}(n) - u^{\perp}_{i,t-1}(n),$$

where $u^{\perp}$ contains signal noise and tastes. Conditional on the projection design and weights, exclusion requires

$$\mathbb{E}\Big[\Delta u^{\perp}_{it}(n) \mid \Delta u^{\perp}_{jt}(n), \varepsilon_t(n), \varepsilon_{t-1}(n), \psi_{it}(n), X_{it}(n)\Big] = 0, \qquad j \neq i, \tag{A11}$$

where ψ_{it} stacks the slope-index variables.

Proposition F.2 (The GIV)**.** *Maintain equation* (A11) *jointly conditional on the projection design and weights, with each holder-specific component having zero conditional mean given both payoff innovations. Under the same conditioning, require the GIV to be uncorrelated with nonpredetermined controls and other holders' remaining characteristic-based demand to be orthogonal to investor i's latent demand given ψ_{it} and X_{it}.*

Suppose Assumption F.1 holds, the projection spans true payoff-news loadings at both dates, and finitely many competitive holders generate a nonzero aggregate idiosyncratic de-

mand component. Then:

(i) The GIV is constructed from observables before estimating the demand coefficients, so moments (27) are linear in those coefficients.

(ii) $\mathbb{E}[Z_{it}(n)\Delta u_{it}(n) \mid \psi_{it}(n), X_{it}(n)] = 0.$

(iii) $\mathrm{Cov}(Z_{it}(n), \Delta\mathrm{MB}_t(n)) > 0$ *if other holders' lagged-AUM-weighted price covariances exceed the component removed by projection.*

Proof. Let P_{-i} be the weighted projection on other holders' design, a_{-i} their lagged AUM vector and r_{-i} their holder-specific shocks plus remaining characteristic-based demand. The design spans modeled price and expected-payoff responses for every coefficient value and both payoff-news loading vectors. Thus $Z_i = a_{-i}^\top(I - P_{-i})r_{-i}$, a combination of holder-specific disturbances determined by the observed design and weights. Control projections are also fixed by the observed variables. The conditional restrictions make this combination orthogonal to Δu_i, proving (i)–(ii). For (iii),

$$\mathrm{Cov}(Z_i, \Delta\mathrm{MB}) = a_{-i}^\top \mathrm{Cov}(r_{-i}, \Delta\mathrm{MB}) - a_{-i}^\top P_{-i}\mathrm{Cov}(r_{-i}, \Delta\mathrm{MB}),$$

conditional on the design and weights. The relevance condition makes this difference positive. □

Holding net issuance and outside demand fixed, market clearing gives

$$\Delta\mathrm{MB}_t(n) \approx \frac{\sum_j \mathrm{AUM}_{j,t-1}\Delta u_{jt}(n)}{R_f \mathcal{T}_{p,t}(n)} + \text{other terms}, \tag{A12}$$

where $-R_f\mathcal{T}_{p,t}$ is the aggregate institutional slope and the other terms collect controls and fixed supply. Other investors hold private signals fixed and infer payoff news from price movements whose source is uncertain (Haddad et al., 2026, Appendix B.6.1). Appendix H gives the signal-extraction conditions and the restrictions under which effective tolerances incorporate investors' own price impact.

Exclusion after controls. Let X stack the institution-quarter and common controls, Z^ψ the interacted GIVs, and $D_l = \psi_l\tau_\varepsilon\Delta\mathrm{MB}$ the price regressors. Remove redundant control columns and residualize Z^ψ and D on X with equal weight on every estimation row, giving $\widetilde{Z}$ and $\widetilde{D}$. Identification additionally imposes

$$\mathbb{E}[\widetilde{Z}e] = 0, \qquad \mathrm{rank}\,\mathbb{E}[\widetilde{Z}\widetilde{D}^\top] = \dim(D).$$

These projected restrictions apply directly to the full residual: slope changes, remaining expected-payoff and exposure terms, normalized risk-bearing heterogeneity, loading errors and preprocessing discrepancies. A persistent taste enters as $(\rho_t - 1)$ times its level. Full rank gives a unique slope-index coefficient vector. Inverse-variance weights enter only instrument construction.

Inference. I approximate the covariance in equation (A.7) of Chaudhary et al. (2026). Conditional on the design, write the shock component as $Z_i = \sum_{j \neq i} b_{ij} v_j$, with independent zero-mean holder–stock–quarter shocks of variance σ_j^2. Before control projections, suppressing interactions and row weights,

$$\mathrm{Var}(Z_i) = \sum_{j \neq i} b_{ij}^2 \sigma_j^2, \qquad \mathrm{Cov}(Z_i v_i, Z_j v_j) = b_{ij} b_{ji} \sigma_i^2 \sigma_j^2.$$

The second term accounts for each row's shock entering the other's GIV. The reported approximation substitutes lagged-AUM weights before projection for b_{ij} in both terms and uses holder-quarter residual variances estimated leaving the quarter out. The full coefficient covariance also depends on subsequent control projections; intervals condition on information inputs and the independent-row approximation.

Elasticities and profiles. Position elasticity is fitted slope times market-to-book divided by book-equity holdings per dollar of AUM. I average with book-equity weights within investor-quarter and report medians across investor-quarters. The notes to Table 3 give the common multiplier used to report price coefficients and standard errors.

Figure 5 evaluates a synthetic investor–stock position. Information percentiles use investors with positive estimated information and at least one informative holding, counting each investor once per quarter. The quarter-specific winsorization and pooled standardization remain those of the regression. The axis spans the 5th–95th percentiles of informativeness across distinct informative stock-quarters.

Reference prior precision, fitted noise-trading standard deviation and market-to-book are the medians across distinct informative stock-quarters: $\tau_{\varepsilon,0} = 366.405$, $\sigma_{z,0} = 210.179$ and $\mathrm{MB}_0 = 2.35272$. Age is the investor-quarter median of 37 quarters, and type is fixed at investment adviser, the modal category among eligible investor-quarters. Reference normalized holdings, $Q_0/\mathrm{AUM}_0 = 0.000374284$, are the median across eligible investor-quarters of each investor-quarter's median informative book-equity holding per dollar of lagged AUM.

At hypothetical informativeness x, I set $\tau_p = \tau_{\varepsilon,0} x/(1-x)$ and calculate

$$\ell(x) = \frac{1}{2}\log\left(\tau_{\varepsilon,0}\frac{x}{1-x}\right) - \log\sigma_{z,0} - m_{\text{pool}}, \qquad \bar{\ell} = 0,$$

where $m_{\text{pool}} = -3.30270$ is the original pooled median log price information-to-noise ratio. At each information percentile, I evaluate the slope index in equation (25) using the unscaled fitted coefficients and multiply it by $\tau_{\varepsilon,0}\text{MB}_0/(Q_0/\text{AUM}_0) = 2.30319 \times 10^6$ to obtain the signed elasticity. The profiles describe the conditional relative demand response at fixed reference price and quantity, without solving a new equilibrium.

The discount implied by the estimates. Let $\bar{d}$ denote the discount at the median informative stock. With risk tolerance common across stocks in a quarter, equation (14) gives $d_p(\ell) = 1/[1 + e^{\ell}(1-\bar{d})/\bar{d}]$. Let $\bar{\zeta}$ be the AUM-weighted uninformed slope index at $\ell = 0$.

Lemma F.1 (The discount implied by relative price signal precision)**.** *If ζ_ℓ equals the derivative at $\ell = 0$ of this average slope index, then $\bar{d} = 1 + \zeta_\ell/\bar{\zeta}$.*

Proof. The average uninformed slope index is $\bar{\zeta}\, d_p(\ell)/\bar{d}$. Its derivative at zero is $-\bar{\zeta}(1-\bar{d})$ because $d_p'(\ell) = -d_p(\ell)[1 - d_p(\ell)]$. Equating this derivative to ζ_ℓ gives the result. □

The uninformed average evaluates private information at zero; the reference institution has mean information per position, investment-adviser type and log age zero. Applying the lemma to unrounded estimates gives $\bar{d} = 0.39$. Its 90 percent confidence set, $[0.31, 0.49]$, inverts a test using Table 3's covariance.

G The Counterfactual: Inputs and Solution

Table A4 lists the experiment's inputs.

Information and its placement. Adopters are asset managers with DS employees in 2024Q3 and time-averaged active share at or above 0.55319, the fixed active-manager cutoff used in production estimation. For every institution, the per-position index is $\theta_i^{\text{CF}} = (\bar{k}_i/N_i)\text{AUM}_i^{\text{est}}$, where N_i and $\text{AUM}_i^{\text{est}}$ are the number and value of positions entering the information estimate. The prediction $\bar{k}_i$ uses original information-per-dollar units; $\bar{k}_i/N_i$ uses the demand estimation's 2024Q4 winsorization at the 1st and 99th percentiles.

Table A4: Inputs of the Counterfactual

Input	Value	Source
Panel (a) Estimated		
Demand Slopes	By type, age and information	Table 3
Private-Information Component	0.16	Appendix G
Price-Inference Discount	0.39	Section 6
Annual DS Productivity	2011–2024, normalized in 2011	Figure 4
Labor Elasticities	0.14 (DS), 0.25 (FR), 0.23 (other)	Section 5
Panel (b) Measured		
Information per Dollar	Institution level, 2024Q3	Section 6
Payoff Precision	Stock level, 2024Q3	Section 4
Price Informativeness	Stock level, 2024Q4	Section 4
Institutional Holdings	13F and CRSP, 2024Q4	Section 2
Adopters	172 institutions, 10.7% of holdings	Section 7
Panel (c) Fixed Inputs and Assumptions		
Historical Level Increment	$g = 1 - 1/R = 0.695$	Section 5
Technology Step	$1 + \delta g$, $\delta = 1, 2, 3$	Section 7
Placement of New Information	Proportional to current information	Section 7
Non-13F Investors	No information, price-inelastic	Appendix G
Variance of Noise Trades	Solved from price informativeness	Appendix G

Notes. The table reports inputs for the 2024Q4 experiment in Section 7.1. The information and risk-tolerance scales are chosen to match the price-learning discount and the sum of estimated demand slopes. Private-Information Component is $\kappa/\mathcal{T}_p$: aggregate private precision weighted by risk tolerance divided by the aggregate price-response coefficient with learning from prices. The advance repeats the estimated 2011–2024 increase in annual information productivity. It raises each adopter's private information by 69.5 percent at fixed employment and risk tolerance. The per-position information input multiplies predicted information per dollar per position, winsorized within quarter and expressed before standardization, by the value of positions used in its estimate. Private precision weighted by investor risk tolerance is proportional to this index times lagged payoff precision. The 2024Q3 Payoff Precision row reports this lagged input; current 2024Q4 precision determines posterior uncertainty and price precision. Appendix G explains the mapping and treatment of missing lags. Prices with measured zero informativeness remain uninformative. Measured information, production inputs and proportional information gains are fixed across demand-coefficient draws.

Let $\tau_\varepsilon^{\text{lag}}(n)$ be 2024Q3 payoff precision, replaced by 2024Q4 precision if missing or non-positive. With common scale $c > 0$, every initially held stock receives

$$\gamma_i^{-1}\tau_i^{\text{CF}}(n) = c\theta_i^{\text{CF}}\tau_\varepsilon^{\text{lag}}(n), \qquad \gamma_i^{-1}k_i^{\text{CF}}(n) = c\theta_i^{\text{CF}}\frac{\tau_\varepsilon^{\text{lag}}(n)}{\tau_\varepsilon(n)}.$$

Posterior uncertainty uses current, 2024Q4 payoff precision $\tau_\varepsilon(n)$. The total data count is therefore $\gamma_i^{-1}K_i^{\text{CF}} = c\theta_i^{\text{CF}}\sum_{n\in\mathcal{N}_i}\tau_\varepsilon^{\text{lag}}(n)/\tau_\varepsilon(n)$. The assignment is equal per position in lagged-prior units. Equation (28) multiplies each adopter's assigned precision by $1 + 0.6949643\delta$, holding both priors, employment, investor efficiency and risk tolerance fixed. Other institutions' private information stays unchanged.

Matching demand estimates and market clearing. Each stock's institutional supply $S(n)$ equals its 2024Q4 holdings. Outside holders have zero private information and fixed demand; institutions trade competitively with zero mandate penalties. The mean stock-selection position in Proposition C.1 is

$$\bar{q}_i(n) = \mathbb{E}[q_i^{\text{sel}}(n)] = \frac{\gamma_i^{-1}\hat{\tau}_i(n)S(n)}{\mathcal{T}(n)}.$$

I hold the mean factor contribution $\gamma_i^{-1}\hat{\tau}_i(n)\beta(n)[H_\beta(n)/\mathcal{T}(n) - \bar{g}_i^\beta]$ and common factor premium $\beta(n)H_\beta(n)/\mathcal{T}(n)$ at their initial values. I adjust the bracketed payoff term so that the mean factor contribution to holdings stays fixed as posterior precision changes. Thus $\Delta\mathbb{E}[q_i] = \Delta\bar{q}_i$ for full expected holdings, and elasticity changes describe stock-selection demand. The unattenuated rent formula applies to stock-selection positions; the information estimator corrects the full-position moment for exposure attenuation.

This condition fixes cross-stock factor rebalancing. The unrestricted model offsets 98 percent of added factor exposure for the median institution. Chaudhry and Davis (2026) estimate cross-stock substitution at 0.1 to 2.5 percent of the frictionless mean-variance benchmark.

Risk tolerance is 2024Q3 AUM times fitted type-age tolerance, with current AUM replacing an unusable lag for 110 institutions. These effective tolerances stay fixed under Assumption H.1. One scale matches the sums of model and estimated slopes; the information scale matches the inferred discount. Across stocks, the model-to-estimated slope ratio has median 0.8536 and 10th–90th percentiles 0.2401–1.2310. Among informative stocks, the median discount is 0.34 and its holder-weighted mean is 0.45. Aggregate private precision is 0.16 of the coefficient governing trading with price learning, with a conditional 90 percent interval of $[0.12, 0.20]$.

Beliefs and positions. Mapped information sums to $\kappa(n)$. I infer noise variance $\sigma_z^2(n) = \kappa(n)^2/\tau_p(n)$ and hold it fixed. Zero measured price precision implies infinite noise variance and stays zero after the advance. Proposition 1 gives $\tau_p'(n) = (1 + \Delta_\kappa(n))^2\tau_p(n)$ for proportional aggregate information gain $\Delta_\kappa(n)$. Updated posterior precisions determine $\mathcal{T}'$, discounts and slopes.

Ownership decomposition. Index initial and final private information by $a = 0, 1$ and public precision $\tilde{\tau}_b(n) = \tau_\varepsilon(n) + \tau_{p,b}(n)$ by $b = 0, 1$. The four stock-selection allocations use the same clearing formula:

$$q_i^{ab}(n) = \frac{\gamma_i^{-1} S(n)[\tilde{\tau}_b(n) + \tau_{ia}(n)]}{\bar{\gamma}^{-1}\tilde{\tau}_b(n) + \kappa_a(n)}. \tag{A13}$$

The initial and final allocations are $\bar{q}_{i0} = q_i^{00}$ and $\bar{q}_{i1} = q_i^{11}$. Private information changes first through q_i^{10}; public information changes first through q_i^{01}. At fixed private information, public learning raises model holdings below the risk-tolerance-weighted average private precision $\kappa_a/\bar{\gamma}^{-1}$ and lowers them above it (Proposition C.1).

For observed book-equity positions $q_i^{\text{obs}}(n)$ and $S(n) = \sum_j q_j^{\text{obs}}(n) > 0$, map any model allocation x to

$$\mathcal{H}_n(x)_i = S(n)\frac{[q_i^{\text{obs}}(n) + x_i - \bar{q}_{i0}(n)]_+}{\sum_j [q_j^{\text{obs}}(n) + x_j - \bar{q}_{j0}(n)]_+}, \qquad [y]_+ = \max\{y, 0\}. \tag{A14}$$

Every x sums to $S(n)$, so the denominator is at least $S(n)$. This rule anchors all evaluations to the same observed and model positions, preserves each stock's institutional holdings and gives $\mathcal{H}_n(\bar{q}_0) = q^{\text{obs}}$. At $\delta = 1$, clipping affects 10.54 percent of rows with 0.0419 percent of initial book-equity positions; the negative mass removed is 0.0671 percent of that total.

Let $Q^{\text{D}} = \mathcal{H}(q^{10})$ and $q^{\text{CF}} = \mathcal{H}(q^{11})$. The reported decomposition is

$$q_i^{\text{CF}} - q_i^{\text{obs}} = \underbrace{Q_i^{\text{D}} - q_i^{\text{obs}}}_{\text{private information}} + \underbrace{q_i^{\text{CF}} - Q_i^{\text{D}}}_{\text{learning from prices}}. \tag{A15}$$

The reverse order uses $Q^{\text{P}} = \mathcal{H}(q^{01})$: learning contributes $Q^{\text{P}} - q^{\text{obs}}$, then private information contributes $q^{\text{CF}} - Q^{\text{P}}$. Shapley averages each contribution across the two orders.

Table 7 sums each contribution over a group's investors and stocks and divides by its initial book-equity positions, multiplied by 100. Online Appendix Table OA5 compares the orders and their average at $\delta = 1, 2, 3$. All contributions include clipping and rescaling and add to the same group change, separately in each coefficient draw. Percentile endpoints summarize each component separately.

[Insert Online Appendix Table OA5 here]

Counterfactual prices. Holding the factor premium fixed gives the model price change and empirical mapping

$$R_f \Delta \mathbb{E}[p] = \frac{S}{\mathcal{T}}\left(1 - \frac{\mathcal{T}}{\mathcal{T}'}\right), \qquad \left(\frac{\Delta p}{p}\right)^{\text{reported}} = \left(1 - \frac{\mathcal{T}}{\mathcal{T}'}\right)\frac{d_p S}{R_f \widehat{\mathcal{T}}_p p}. \tag{A16}$$

Here p is market-to-book and S is institutional book-equity holdings. The estimated slope gives elasticity magnitude $R_f \widehat{\mathcal{T}}_p p/S$ at initial prices and holdings. After the anticipated advance, its fixed-belief counterpart is $(R_f \widehat{\mathcal{T}}_p p/S) d_p^{-1} \mathcal{T}'/\mathcal{T}$, which clears the proportional demand shift $\mathcal{T}'/\mathcal{T} - 1$. Schedule-elasticity changes in Tables 7 and 10 equal proportional changes in $\mathcal{T}_p$, evaluated at initial prices and holdings.

Coverage, information shares and crowding out. For adopters $\mathcal{A}$ and initial holders $\mathcal{I}(n)$, let $\kappa_{\mathcal{A}}(n) = \sum_{i \in \mathcal{A} \cap \mathcal{I}(n)} \gamma_i^{-1} \tau_i(n)$ be adopters' risk-tolerance-weighted private precision. The sorting variables are

$$\text{Coverage}(n) = \frac{\kappa_{\mathcal{A}}(n)}{\mathcal{T}(n)}, \qquad \text{Information share}(n) = \frac{\kappa_{\mathcal{A}}(n)}{\kappa(n)}.$$

Coverage divides by total posterior precision; information share divides by private precision alone. Table 6 and Figure 7(a) use approximately equal-count coverage quintiles. Panel (b) takes medians of non-adopters' percentage changes in total book-equity positions within quintiles of winsorized $\bar{k}_i/N_i$. Each institution enters once; both totals cover the same initially held stocks.

Elasticity contributions. In Table 7(b), the direct change in magnitude is $100(\kappa' - \kappa)/\mathcal{T}_p$ and the total is $100(\mathcal{T}_p'/\mathcal{T}_p - 1)$. The panel takes separate medians across informative stocks and defines learning as their difference. Figure 8 averages each stock's direct, learning and total changes within fixed equal-count bins of initial information share or informativeness. Contributions add within each stock and bin.

Conditional uncertainty. I draw 200 joint demand-coefficient vectors using Appendix F's covariance, conditional on measured information, production coefficients, the gain rule, recipients and allocation rule. Each draw supplies its discount and calibration. All 200 have a negative information gradient and a discount in $(0, 1)$ and enter the 90 percent intervals. Stock and investor bins stay fixed at the point estimate across draws and advance sizes.

Statistics are evaluated within each draw before taking the 5th and 95th percentiles, including total-effect bands in Figure 8. Table 10's shares of declining slope magnitudes average the indicator within each draw, equally across stocks or with book-equity weights.

H Finite Investors and Price Impact

Under the execution costs below, price impact reduces effective risk tolerance. With finitely many investors, prices contain part of each investor's own signal, so the holdings moment measures information gained beyond the price. Here J counts investors and N counts stocks.

H.1 Portfolio choice with price impact

Consider one stock with no factor or mandate. Let p be its public reference price, $m_i = \mathbb{E}[f \mid \mathcal{I}_i] - R_f p$ expected excess payoff, and $1/\hat{\tau}_i$ posterior payoff variance. A marginal execution-price schedule $p + \lambda_i^K x$, with $\lambda_i^K \geq 0$, adds date-1 cost $\lambda_i^K q_i^2/2$ to payment at the quote. This execution cost is separate from the public quote's response to noise trades, $1/(R_f \mathcal{T}_p)$ in Proposition 1.

Proposition H.1 (Price impact and effective risk tolerance)**.** *Given posterior moments and $\lambda_i^K \geq 0$, the unique optimal position is*

$$q_i^\lambda = \frac{m_i}{\gamma_i/\hat{\tau}_i + R_f \lambda_i^K} = \left(\gamma_i^{\text{eff}}\right)^{-1} \hat{\tau}_i m_i, \qquad \gamma_i^{\text{eff}} = \gamma_i + R_f \lambda_i^K \hat{\tau}_i. \tag{A17}$$

Proof. The normalized objective $m_i q_i - (\gamma_i/\hat{\tau}_i + R_f \lambda_i^K) q_i^2/2$ is strictly concave. Its first-order condition gives the result. At fixed beliefs, higher impact reduces the magnitude of long and short positions. □

Portfolio exposures. Write $\Sigma_i = \hat{\Omega}_i + \beta\beta^\top$ for the full posterior payoff covariance and $q_i^\top \Lambda_i^K q_i/2$ for date-1 execution cost, with Λ_i^K symmetric and positive semidefinite. The normalized first-order condition is

$$\left(\gamma_i \Sigma_i + R_f \Lambda_i^K - \frac{\delta_i^\phi}{\gamma_i} \phi\phi^\top\right) q_i = m_i.$$

If the execution-cost matrix satisfies

$$R_f \Lambda_i^K = (\gamma_i^{\text{eff}} - \gamma_i)\Sigma_i, \qquad \delta_i^{\phi,\text{eff}} = \delta_i^\phi \frac{\gamma_i^{\text{eff}}}{\gamma_i}, \tag{A18}$$

the benchmark condition holds with risk aversion γ_i^{eff} and mandate coefficient $\delta_i^{\phi,\text{eff}}$. Costs are proportional to the full posterior covariance, and the mandate adjustment preserves its coefficient relative to expected payoff. Other cost matrices generally change factor management. In Appendix G, this rule applies to the stock-selection position in Proposition C.1, using payoff net of the baseline factor premium and costs proportional to posterior idiosyncratic variance. The baseline mean factor contribution is added back to holdings.

H.2 Learning from prices with finite investors

In the one-stock, no-factor, no-mandate setting, J investors have signals $\eta_j = \varepsilon + u_j$. The normal errors u_j are mutually independent and independent of ε and z, with $\text{Var}(u_j) = 1/\tau_j$. Holdings cover entire institutions, whose sizes enter γ_j^{-1}. Suppose an affine public price reveals

$$y = \sum_{j=1}^{J} a_j^p \eta_j + z, \qquad A^p = \sum_j a_j^p, \qquad V^p = \sigma_z^2 + \sum_j \frac{(a_j^p)^2}{\tau_j}, \qquad \tau_p = \frac{(A^p)^2}{V^p}. \tag{A19}$$

The coefficients a_j^p are deterministic, $\sigma_z^2 > 0$, and $A^p \neq 0$. Investors without a signal have $a_j^p = 0$ and contribute no signal-noise term. Inference conditions on this conjectured equilibrium price function, including each signal's covariance with price.

Let $\tilde{\tau} = \text{Var}(\varepsilon \mid p)^{-1} = \tau_\varepsilon + \tau_p$ and $\hat{\tau}_i = \text{Var}(\varepsilon \mid \eta_i, p)^{-1}$ denote the public and private posterior precisions. Execution coefficients and these precisions are deterministic in the conditional Gaussian calculation.

Lemma H.1 (Information beyond the price with finite investors)**.** *Under the stated conjecture for the equilibrium price function and the portfolio choice in Proposition H.1,*

$$\tau_i^{\text{inc}} = \hat{\tau}_i - \tilde{\tau} \geq 0, \qquad \mathbb{E}\Big[q_i^\lambda \big(\varepsilon - \mathbb{E}[\varepsilon \mid p]\big)\Big] = \big(\gamma_i^{\text{eff}}\big)^{-1} \frac{\tau_i^{\text{inc}}}{\tilde{\tau}}. \tag{A20}$$

The moment is zero when the private signal adds no information beyond the price.

Proof. Write $m_p = \mathbb{E}[\varepsilon \mid p]$, $m_i^\varepsilon = \mathbb{E}[\varepsilon \mid \eta_i, p]$ and $r = \varepsilon - m_p$. Optimal demand is $q_i^\lambda = (\gamma_i^{\text{eff}})^{-1}\hat{\tau}_i(\mu + m_i^\varepsilon - R_f p)$. Its public component is orthogonal to r. The projection identity gives $\mathbb{E}[(m_i^\varepsilon - m_p)r] = \mathbb{E}[(m_i^\varepsilon - m_p)^2]$. Conditional Gaussian variances are constant, and the law of total variance gives this last expression as $1/\tilde{\tau} - 1/\hat{\tau}_i \geq 0$. Multiplying by $(\gamma_i^{\text{eff}})^{-1}\hat{\tau}_i$ proves both claims. □

Setting $\lambda_i^K = 0$ gives the price-taking result; positive costs reduce the moment at fixed beliefs. With or without costs, positions based only on public information have zero covariance with the public forecast error.

For $\tau_i > 0$, investor i computes incremental precision by removing her observed signal from the price statistic: $y_{-i} = y - a_i^p \eta_i$. Conditional on ε, y_{-i} and η_i have independent errors. Consequently,

$$\tau_{p,-i} = \frac{(A^p - a_i^p)^2}{V^p - (a_i^p)^2/\tau_i}, \qquad \hat{\tau}_i = \tau_\varepsilon + \tau_i + \tau_{p,-i}, \qquad \tau_i^{\text{inc}} = \tau_i + \tau_{p,-i} - \tau_p. \tag{A21}$$

The adjustment $\tau_{p,-i} - \tau_p$ can have either sign; incremental precision remains nonnegative.

For convergence to the continuum information identity, let $A^p \to \kappa > 0$, let σ_z^2 stay bounded away from zero, and let $\sum_j (a_j^p)^2/\tau_j = o(\sigma_z^2)$ along a sequence of economies. If $a_i^p \to 0$, then $\tau_{p,-i} - \tau_p \to 0$ and $\tau_i^{\text{inc}} - \tau_i \to 0$. Relative recovery follows when $|\tau_{p,-i} - \tau_p| = o(\tau_i)$, including when τ_i is bounded away from zero under these limits. The benchmark equilibrium price function additionally requires convergence of the aggregate signal coefficient to $\kappa = \int (\gamma_j^{\text{eff}})^{-1} \tau_j \, dj$.

H.3 Implications for estimation and counterfactuals

Assumption H.1 (Stable effective risk tolerance)**.** Investor i has constant preference and effective risk aversion, γ_i and $\gamma_i^{\text{eff}} \geq \gamma_i$. In the scalar stock problem, execution costs satisfy

$$R_f \lambda_{it}^K(n) \hat{\tau}_{it}(n) = \gamma_i^{\text{eff}} - \gamma_i \tag{A22}$$

for the relevant stocks, periods and counterfactual information states. The full portfolio problem satisfies equation (A18), including its mandate adjustment, in each state. The counterfactual applies the condition to stock-selection positions, holding the mean factor contribution and common factor premium at their initial values.

Execution costs therefore fall with posterior variance enough to keep effective tolerance constant. Gârleanu and Pedersen (2013, Assumption 1) motivate covariance-proportional costs through liquidity suppliers' compensation for inventory risk. Here the covariance is the investor's posterior covariance, with stable risk-bearing capacity and holding horizons.

Under Assumption H.1 and benchmark inference or the large-market limits above, the applications use effective tolerance in place of preference tolerance. The full-position information target is $(\gamma_i^{\text{eff}})^{-1} K_{it}$, subject to the exposure and estimator approximations; long positions measure expressed information. Appendices E and F give the identification conditions for production and demand. Appendix G fixes fitted effective tolerances and AUM while information changes, aggregating $(\gamma_i^{\text{eff}})^{-1} \tau_i(n)$ into $\kappa(n)$ and updating expected stock-selection holdings as $(\gamma_i^{\text{eff}})^{-1} \hat{\tau}_i S/\mathcal{T}$. These applications condition on the stipulated execution costs and inference rule.